

IDENTIFICATION

|               |                                          |
|---------------|------------------------------------------|
| Product Code: | DEC-12-ZR1A-D                            |
| Product Name: | DIAL-MS Assembler Program<br>Description |
| Date Created: | July 1, 1970                             |
| Maintainer:   | Software Services                        |



LAP6-DIAL is an editor, filing system and assembler for use with the PDP-12 computer. The editor and filing portions are derived from the basic LINC program LAP6<sup>1</sup> by Mary Allen Wilkes of Washington University. The assembly portion is derived from several programs used for the PDP-8 computer including PAL-D<sup>2</sup>.

The Digital Equipment Corporation wishes to express to the author, Mary Allen Wilkes (Clark), and the Computer Research Laboratory of Washington University, St. Louis, Missouri, its appreciation for the development set forth in LAP6 as well as its thanks for permission to use parts of the LAP6 program.

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<sup>1</sup>M. A. Wilkes, LAP6 Handbook, Computer Research Laboratory Tech. Rep. No. 2, Washington University, St. Louis, May 1, 1967.

<sup>2</sup>PAL-D Assembler Programmer's Reference Manual DEC-D8-ASAA-D.



## 1.0 PROGRAM OVERVIEW

The DIAL Assembler is a bilingual assembler which can assemble program which have been edited by the DIAL editor or entered by PIP.

Additional features to the assembler are:

1. Ability to save and load parts of the symbol table
2. Conditional assembly of various lines
3. Listing between line numbers
4. A "quick list" entry point which removes comments and line numbers and replaces tabs by blanks
5. Listing on disk or tape
6. Chaining between source files.

This assembler is based on PAL-D which, in turn, was based on MACRO-8. Most of the routines in the DIAL Assembler will be found and described in the PAL-D or MACRO-8 system description.

## 2.0 ENVIRONMENT

The Assembler leaves its binary output in the Working Area on tape 1 (blocks 370-427), and the header block in 447. The Assembler itself (including necessary scratch blocks) resides in blocks 324-345 of tape 0.

## 2.1 LOADER MAP

|           |                                                                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0000-0177 | Page 0: Common literals and pointers                                                                                                                                    |
| 0200-0377 | Bank Check Routine; Symbol Search Routine                                                                                                                               |
| 0400-0577 | Symbol Table Getter Routine                                                                                                                                             |
| 0600-0777 | Routine to get a symbol from Symbol Table (or add on).<br>Ring buffering routines                                                                                       |
| 1000-1177 | General Recursive Expression Evaluator                                                                                                                                  |
| 1200-1377 | Recursive Address Evaluator                                                                                                                                             |
| 1400-1577 | Central Processing Loop, Main Start of Assembler<br>after Initialization                                                                                                |
| 1600-1777 | Text Processor, Current Line Tester, plus Minor<br>Support Routines                                                                                                     |
| 2000-2177 | An Input Character Scanner Routine, the output<br>buffer routine and tab routine                                                                                        |
| 2200-2377 | Number Conversions, Type Checks                                                                                                                                         |
| 2400-2577 | Symbol Sort for Output, Teletype Routine                                                                                                                                |
| 2600-2777 | Symbol Output and Value Routine, Octal Print<br>Routine, Field Change Routine, and Work Area                                                                            |
| 3000-3177 | Output List Routine, 8K Pseudo Op Initialization<br>(destroyed and never used) Work Area                                                                                |
| 3200-3377 | Error Message Processor, Type Evaluator, System<br>Symbol Table Swapper, Fairly Low Level Character<br>Getter, and Routine to Skip if Not Pass 2 or Not<br>in List Mode |
| 3400-3577 | Input Character Convertor to Internal Form, Plus<br>Page Checkers, Other Little Routines                                                                                |
| 3600-3777 | Internal to External Character Convertor,<br>Symbol Table Type Getters and Setters<br><br>Low Level Character Getter, Cmmment Processor,<br>List and No List Processors |
| 4000-4177 | General Tape I/O Call Routines, Initialization<br>for Tape Routines, Other Short Routines                                                                               |
| 4200-4377 | More Tape I/O Routines, Pseudo Op Processor,<br>and Loader Core Map                                                                                                     |
| 4400-4777 | Input Buffer and Swapped In and Out Work Area<br>for SAVSYM and LODSYM                                                                                                  |
| 5000-5377 | Output Buffer, plus Initialization for Entire<br>Assembler after it is first loaded in                                                                                  |

|              |                                                                  |
|--------------|------------------------------------------------------------------|
| 54000-5777   | Pseudo-Op Processor                                              |
| 60000-6377   | Line Printer Checker and Basic Line Printer Routine. Ring buffer |
| 64000-6777   | Listape Buffer                                                   |
| 70000-7777   | DIAL-MS routines                                                 |
| 100000-16777 | User defined symbols                                             |
| 170000-17377 | System Mode Symbols (L-mode or P-mode)                           |
| 174000-17577 | Pseudo Ops and Special Character Table                           |

### 3.0 THE PSEUDO OP PROCESSOR

After the desired pseudo op is found in the table by "EXPR", it then Jumps to an ISZ chain. The ISZ chain is a group of sequential ISZs of location "OPL". After falling through the ISZs, OPL contains the desired pseudo op number; the routine then JMSs to it with the desired operation number in the AC.

### 4.0 SYMBOL TABLE ORGANIZATION

A symbol table entry is composed of four words. The first three words are the name and the last is the value. If the symbol is a pseudo-op, then the last word contains a pointer to the routine which will process the pseudo op and the rest of the statement.

The name is composed of six characters. These characters are usually the 26 alphabetic letters plus 10 numbers plus a null character if less than six letters long, permitting 45 octal possible characters for the left or the right half of

a word. If six bit notation were used, 23 possible positions on the character would be wasted; therefore, six bit is not used. The first character of a name must be a letter, thus leaving only 32 octal combinations for the first character. If the left characters of a word are multiplied by 45, there are two free bits in the first word of a name ( $32*45 < 2000$  octal). These two left bits are used as follows:

|     |                   |
|-----|-------------------|
| 00: | Special Character |
| 01: | Undefined         |
| 10: | Defined           |
| 11: | Pseudo operation  |

## 5.0 SUBROUTINES

### 1. Initialization

|                          |                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| NAME:                    | <u>SETORG</u>                                                                                                |
| FUNCTION:                | Called at the beginning of each pass to initialize counters, pointers, and tables. Sets origin to SEGMENT 2. |
| INPUT:                   | Clear AC.                                                                                                    |
| OUTPUT:                  | Clear AC.                                                                                                    |
| MAJOR REGISTERS CHANGED: | AADR, APAGE, EVAL, ABANK                                                                                     |

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| NAME:                    | <u>RESET</u>                                                    |
| FUNCTION:                | Called when a new page is to be started to initialize pointers. |
| INPUT:                   | Clear AC.                                                       |
| OUTPUT:                  | Clear AC.                                                       |
| MAJOR REGISTERS CHANGED: | APAGE.                                                          |

## 2. Input

NAME: GETIN

FUNCTION: Gets an 8-bit character from GETCHR. Saves it in ITEM. Also scans off comments if in QUICK LIST mode.

MAJOR REGISTERS CHANGED: ITEM

## 3. Input Conversion

NAME: BIT6

FUNCTION: Translates the 8-bit ASCII character in ITEM to its 6-bit internal code.

INPUT: Clear AC

OUTPUT: 6-bit internal code in AC

MAJOR REGISTERS CHANGED: None

## 4. Output

NAME: PUNONE

FUNCTION: Outputs the contents of EVAL (the generated binary).

1. PASS 1 - no operation
2. Save word in correct core location. If correct block not in core, it calls NOTBC1 to load it in.

INPUT: Clear AC when EVAL contains assembled binary.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CUNADD.

Output (cont.)

NAME: OPS

FUNCTION: This is the octal print routine. It converts the binary word in the AC to 4 octal characters (8-bit ASCII code) and calls TYPO to output them..

INPUT: Value to be output in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: None.

NAME: EMPTY

FUNCTION: Outputs the contents of the symbolic buffer via TYPO during optional PASS 2. It then calls TYCAR to output a carriage return/line feed.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CHARAC - the buffer counter/pointer.

NAME: TYCAR

FUNCTION: Outputs a carriage return and a line feed in 8-bit ASCII code via two calls to TYPO.

INPUT: Clear AC.

Output (cont.)

OUTPUT: Clear AC

MAJOR REGISTERS CHANGED: None.

NAME: UNTRAN

FUNCTION: Converts the 6-bit internal code character in the AC to 8-bit ASCII and then calls TYPO to output it.

INPUT: Internal code character in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: TYCAR - symbol character counter.

NAME: OX

FUNCTION: Used to decode a word of a symbol table entry by dividing by  $45_8$  and then calling UNTRAN twice to output the two characters that are now in 6-bit internal code.

INPUT: Word to be decoded in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: LWC, HIC, MPI

Output (cont.)

|                          |                                                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME:                    | <u>OSYM</u>                                                                                                                                        |
| FUNCTION:                | Outputs the symbol in TEM1A-TEM1A+2. It initializes the symbol character counter, TYCAR, and then calls OX once for each word of the symbol.       |
| INPUT:                   | Clear AC.                                                                                                                                          |
| OUTPUT:                  | Clear AC.                                                                                                                                          |
| MAJOR REGISTERS CHANGED: | TYCAR                                                                                                                                              |
| NAME:                    | <u>OSANDV</u>                                                                                                                                      |
| FUNCTION:                | Outputs a symbol via OSYM and its value via OPS. It calls SKIP2 to output spaces for format and TYCAR for a terminating carriage return/line feed. |
| INPUT:                   | Clear AC.                                                                                                                                          |
| OUTPUT:                  | Clear AC.                                                                                                                                          |
| MAJOR REGISTERS CHANGED: | None.                                                                                                                                              |
| NAME:                    | <u>SKIP2</u>                                                                                                                                       |
| FUNCTION:                | Outputs spaces (8-bit ASCII) for formatting purposes via TYPO.                                                                                     |
| INPUT:                   | 2's complement of number of spaces in AC.                                                                                                          |

Output (cont.)

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: None.

NAME: ALPHA

FUNCTION: Called at the end of PASS 2 to output an alphanumeric ordered symbol table print of all the defined symbols and their values. It uses the following "lower level" output routines to accomplish this: LT2, TYPE, TYCAR and OSANDV. It also uses

- MOVE - the triple precision moving routine,
- TRIPLE - the triple precision comparison routine.
- and GETYPE - to determine the type of the symbol currently being ordered.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: SADR, auto-index  $16_8$ , TEM1-TEM3, TEM1A-TEM1A+2, and VAL.

NAME: ONEREG

FUNCTION: This routine is entered with the AC containing the value that is to be loaded into the location now pointed

Output (cont.)

at by AADR (the current address counter) when the assembled program is to be run. This value is output via PUNONE.

INPUT:

Value to be output in AC.

OUTPUT:

Clear AC.

MAJOR REGISTERS CHANGED:

EVAL  
AADR - incremented by 1.  
TYPINST+page number - this contains the highest instruction address +1 and was changed if the value just output was the highest instruction so far on this page.

NAME:

ERROR

FUNCTION:

Called to output error message (8-bit ASCII) via TYPO corresponding to error number in the AC. The error address is either relative to a symbolic tag or absolute depending on whether or not a tag has been encountered since the last origin setting.

INPUT:

Error number in AC.

OUTPUT:

Clear AC.

MAJOR REGISTERS CHANGED:

None.

## 5. Pseudo-Instruction Processors

NAME: PAGEC

FUNCTION: Handles the PAGE pseudo-op. Calls EXPR to see if a page number follows PAGE. If so, the origin is set to the beginning of that page. If no number is specified, the origin is set to the beginning of the next page. PUNORG is called to punch the new origin.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: AADR, EVAL

NAME: PERIOD

FUNCTION: Handles the "." pseudo instruction. Combines the value of the current address counter, AADR, with the value already accumulated according to the last operator received. If in LMODE, AADR is ANDed with 1777.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CON (indirectly).

## Pseudo-Instruction Processors (cont.)

NAME: H

FUNCTION: Handles the "!" pseudo instruction. The indirect indicator register is set to  $400_8$  and the scanning for the rest of the expression resumes at A+1.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: H+3

NAME: ORGIC

FUNCTION: Handles the "\*" pseudo instruction. EXPR is called to get the address of the new origin which follows the \*. PUNORG is called to punch the new origin.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: AADR

NAME: STRING

FUNCTION: This handles the TEXT pseudo-instruction. It uses calls to GETIN

## Pseudo-Instruction Processors (cont.)

to input the beginning delimiter, the string of characters, and the terminating delimiter in 6-bit DIAL code. It then uses calls to ONEREG to output two string characters per register in 6-bit DIAL code.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: VAL, EVAL, ITEM.

NAME: FIELDM

FUNCTION: Handles the FIELD pseudo-instruction. EXPR is called to pick up the bank designation which follows FIELD. This memory bank designator is output (pass 2 only) via TYP02 in a format acceptable to the binary loader.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: None.

NAME: DECIM

FUNCTION: Handles the DECIMAL pseudo-instruction. Calls the subroutine DECIMS, which by

## Pseudo-Instruction Processors (cont.)

setting a switch in the MT10 sub-routine, effects all subsequent numbers to be considered in the decimal radix.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: MTSW - the switch register.

NAME: OCT

FUNCTION: Handles the OCTAL pseudo-instruction. Calls the subroutine OCTS, which by setting a switch in the MT10 sub-routine, effects all subsequent numbers to be considered in the octal radix.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: MTSW - the switch register.

## 6. Symbol Table Routines

NAME: SEARCH

FUNCTION: Searches the symbol table for the symbol found in TEM1, TEM2, and TEM3.

INPUT: Clear AC.

Symbol Table Routines (cont.)

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: SADR, Autoindex  $16_8$ , INST, VAL,  
VADR, ANY, TYPE

NAME: GETYPE

FUNCTION: Extracts the type from the left hand  
2 bits of the 1st word and the left  
hand bit of the second word of the  
current symbol and places it in TYPE.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: TYPE.

NAME: TYPADD

FUNCTION: Change or add the type of the symbol  
pointed at by SADR.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: TYPE, 1st 2 words of symbol table  
entry.

## Symbol Table Routines (cont.)

|                          |                                                                                                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME:                    | <u>CURREN</u>                                                                                                                                                                   |
| FUNCTION:                | Handles the special character ", ".<br>It defines the symbol preceding the<br>comma as equal to the current<br>address counter.                                                 |
| INPUT:                   | Clear AC.                                                                                                                                                                       |
| OUTPUT:                  | Clear AC.                                                                                                                                                                       |
| MAJOR REGISTERS CHANGED: | TEMLA-TEMLA+2, VADR, VAL1A                                                                                                                                                      |
| NAME:                    | <u>ENTS</u>                                                                                                                                                                     |
| FUNCTION:                | Enter a symbol in the symbol table<br>with type equal to the parameter<br>following the call. The symbol to<br>be entered is in TEM1-TEM3 and the<br>value in VAL.              |
| INPUT:                   | Clear AC.                                                                                                                                                                       |
| OUTPUT:                  | Clear AC.                                                                                                                                                                       |
| MAJOR REGISTERS CHANGED: | SEND, SADR, VADR, TYPE.                                                                                                                                                         |
| NAME:                    | <u>ERR7</u>                                                                                                                                                                     |
| FUNCTION:                | This routine is called by CURREN to<br>determine whether the symbol to be<br>defined has already been defined and,<br>if so, whether the new definition<br>agrees with the old. |

## Symbol Table Routines (cont.)

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: None.

NAME: SAVE

FUNCTION: Called by SEARCH to save the pointers (SADR, TYPE, VADR) of the previous symbol, in case the current symbol is a definition character ( , or =) or a macro escape character.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: REPUN, RESTOR-1, VADR1

NAME: RESTOR

FUNCTION: Restores the registers saved by SAVE. CURREN, the ", " processing routine, is one place where RESTOR is called to allow definition of the symbol preceding the ", ".

INPUT: Clear AC.

OUTPUT: Clear AC.

## Symbol Table Routines (cont.)

MAJOR REGISTERS CHANGED:      SADR, TYPE, VADR

NAME:                            LDT

FUNCTION:                        Used in the assembling of a symbol.  
If the character is alphanumeric,  
exit is made with it in the AC after  
having incremented IN and called IDX  
for the next character. If non-  
alphanumeric, CHARX is cleared to  
effect immediate exit from LDT in  
subsequent calls.

INPUT:                           Clear AC.

OUTPUT:                          Alphanumeric character of 0 in AC.

MAJOR REGISTERS CHANGED:      NUPAGE (for temporary storage),  
IN -- alphanumeric character  
counter, CHARX.

NAME:                            AlW

FUNCTION:                        Used in the assembling of a symbol.  
It makes two calls to LDT and adds  
the output of the 2nd to 45<sub>8</sub> times  
the 1st. This gains type bits in the  
1st two words of the symbol table  
entry.

INPUT:                           Clear AC.

Symbol Table Routines (cont.)

|                          |                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OUTPUT:                  | Word of symbol in AC.                                                                                                                                                                    |
| MAJOR REGISTERS CHANGED: | NUPAGE (for temporary storage), VAL.                                                                                                                                                     |
| NAME:                    | <u>AAS</u>                                                                                                                                                                               |
| FUNCTION:                | Assembles a 3 word symbol by calling<br>AlW three times. By calls to LDT,<br>it causes all letter/digits beyond 6<br>to be ignored.                                                      |
| INPUT:                   | Clear AC.                                                                                                                                                                                |
| OUTPUT:                  | Clear AC.                                                                                                                                                                                |
| MAJOR REGISTERS CHANGED: | TEM1, TEM2, TEM3.                                                                                                                                                                        |
| NAME:                    | <u>ANY</u>                                                                                                                                                                               |
| FUNCTION:                | Control comes here from the SEARCH<br>routine. If the type of the symbol<br>found is defined or undefined, ANY is<br>indexed to indicate something valid<br>was assembled for this line. |
| INPUT:                   | Clear AC.                                                                                                                                                                                |
| OUTPUT:                  | Clear AC.                                                                                                                                                                                |
| MAJOR REGISTERS CHANGED: | ANY                                                                                                                                                                                      |

Symbol Table Routines (cont.)

|                          |                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAME:                    | <u>GETASY</u>                                                                                                                                                              |
| FUNCTION:                | Calls AAS to get a symbol and then SEARCH to see if it is already in the symbol table. If it is not in the table, it is entered as undefined by calling ENTS.              |
| INPUT:                   | Non-zero AC when called by GETSYM; otherwise clear AC.                                                                                                                     |
| OUTPUT:                  | Non-zero AC if symbol undefined and pass is 1; otherwise clear AC.                                                                                                         |
| MAJOR REGISTERS CHANGED: | IN, VAL, TRUBL.                                                                                                                                                            |
| NAME:                    | <u>TRIPLE</u>                                                                                                                                                              |
| FUNCTION:                | Triple precision comparison routine used by ALPHA to order its output. It exits with the link on if the symbol in WORK-WORK+2 is greater than the symbol in TEM1 - TEM1+2. |
| INPUT:                   | Clear AC.                                                                                                                                                                  |
| OUTPUT:                  | Clear AC; link on or off.                                                                                                                                                  |
| MAJOR REGISTERS CHANGED: | MOVE (for temporary storage)                                                                                                                                               |

## Symbol Table Routines (cont.)

NAME:

GETADR

FUNCTION:

Called by EXPR when a defined or undefined symbol is terminated by a space. Its job is to get the address portion of this expression. Because it calls EXPR to accomplish this, it must save its own exit point for possible recursion. It will call CONC to generate a link when the address is an out-of-page reference.

INPUT:

Clear AC.

OUTPUT:

Clear AC.

MAJOR REGISTERS CHANGED:

ADR, NUPAGE (for temporary storage).

## 7. Numeric Routines

NAME:

DECIMS

FUNCTION:

Sets a switch in MT10 to effect decimal radix for all subsequent numbers.

INPUT:

Clear AC.

OUTPUT:

Clear AC.

MAJOR REGISTERS CHANGED:

MTSW (the switch address).

Numeric Routines (cont.)

NAME: OCTS

FUNCTION: Sets a switch in MT10 to effect octal radix for all subsequent numbers.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: MTSW (the switch address).

NAME: MT10

FUNCTION: Uses calls to MTRL (which shifts LWC left 1 place) and calls to MTAD (which adds the contents of TIC to the pseudo accumulator, LWC) to combine the number in MTDG, according to the currently declared radix, with the previously accumulated value, LWC.

INPUT: Clear AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: TIC.

NAME: DPCV

FUNCTION: Numeric input and conversion routine. It calls IDX for input and MT10 for conversion to the currently declared radix. A non-numeric character will stop processing.

## Numeric Routines (cont.)

INPUT: Clear AC.

OUTPUT: AC non-zero.

MAJOR REGISTERS CHANGED: ANY, HIC, LWC, RTDIG, SIGN, DPN, MTDG.

## 8. Miscellaneous Routines

NAME: RADD

FUNCTION: Add the value in the AC to the accumulated value, CON, and store the result in CON.

INPUT: Value to be added to CON in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CON.

NAME: RSUB

FUNCTION: Subtract the value in the AC from the accumulated value, CON, and store the result in CON.

INPUT: Value to be subtracted from CON in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CON.

Miscellaneous Routines (cont.)

NAME: RAND

FUNCTION: AND the value in the AC with the accumulated value, CON, and store the result in CON.

INPUT: Value to be ANDed with CON in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CON.

NAME: RIOR

FUNCTION: OR the value in the AC with the accumulated value, CON, and store the result in CON.

INPUT: Value to be OR'd with CON in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: CON, VAL (for temporary storage).

NAME: UNDERR

FUNCTION: Channels the error printout number, which is in the AC, to ERROR if the current pass is not 1. It suppresses error printouts during pass 1.

Miscellaneous Routines (cont.)

INPUT: Error printout number in AC.

OUTPUT: Pass 1 - error printout number in  
AC. Not Pass 1 - clear AC.

MAJOR REGISTERS CHANGED: None.

NAME: PUSH2

FUNCTION: Enters the contents of the AC into  
the push down list.

INPUT: Entry for push down list in AC.

OUTPUT: Clear AC.

MAJOR REGISTERS CHANGED: POINT, PACK (for temporary storage).

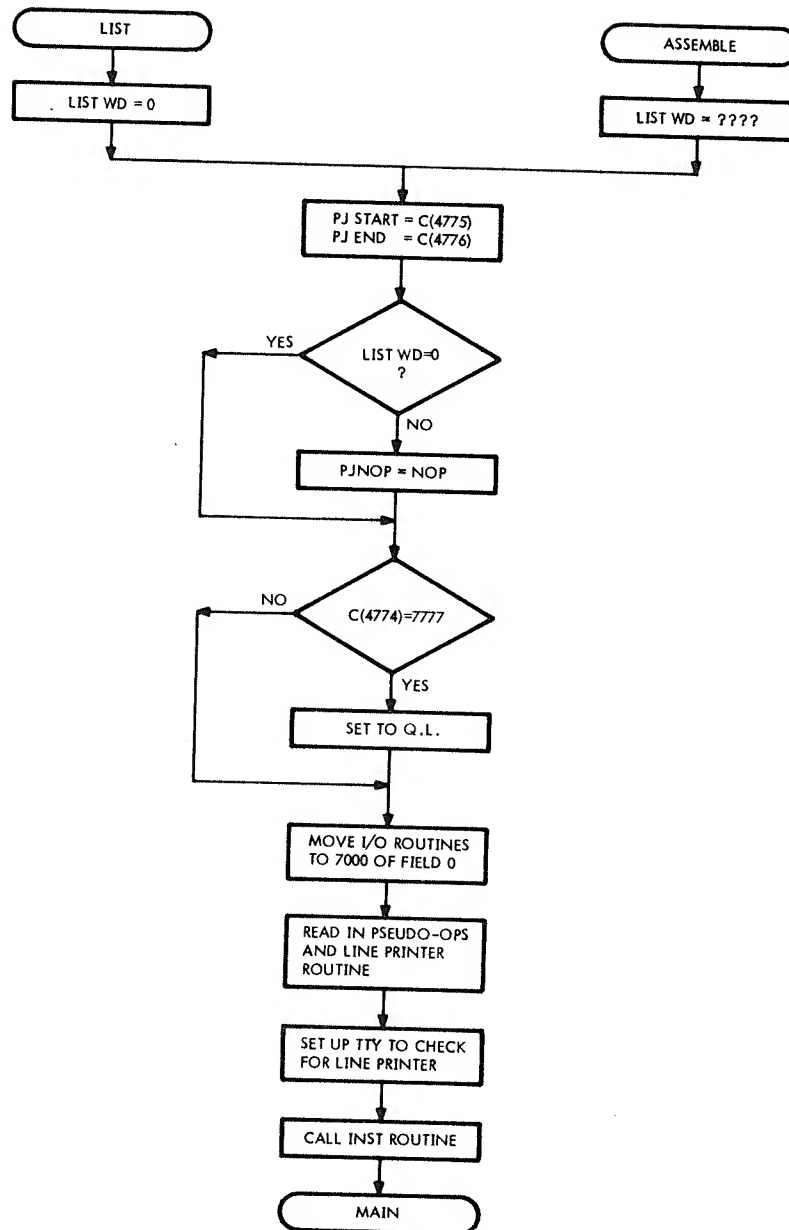
NAME: POPUP1

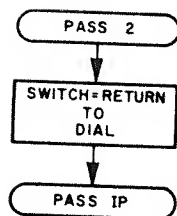
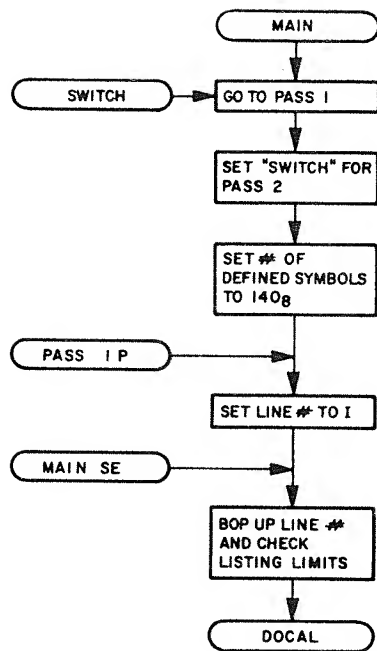
FUNCTION: Retrieves the last entry from push  
down list.

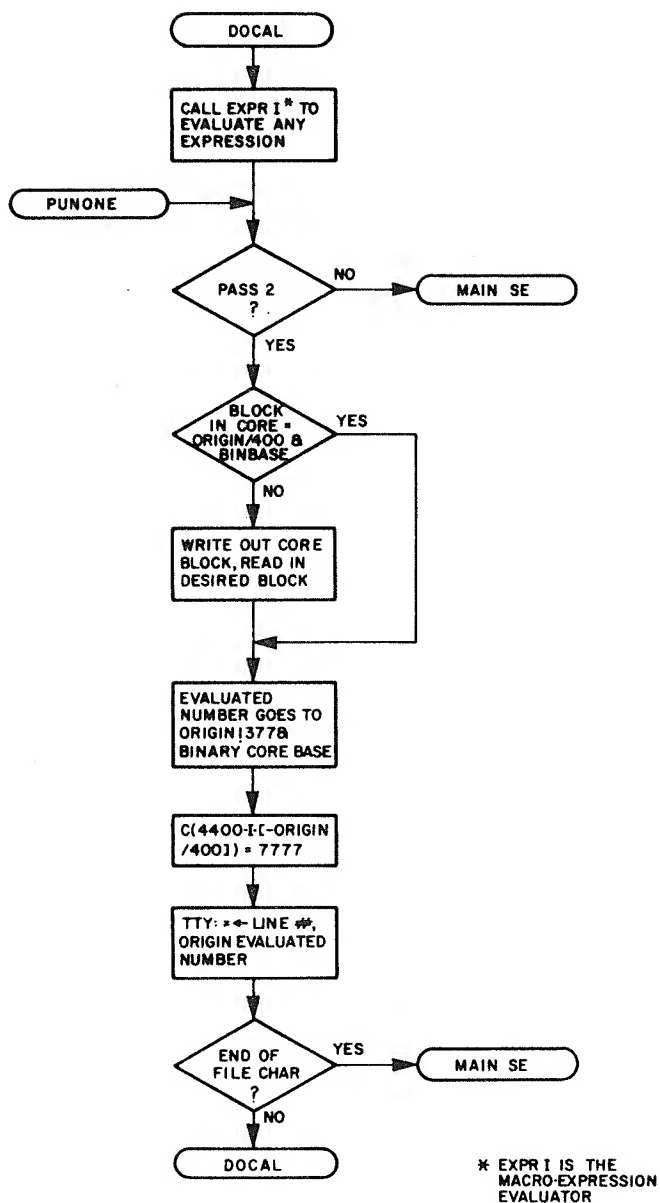
INPUT: Clear AC.

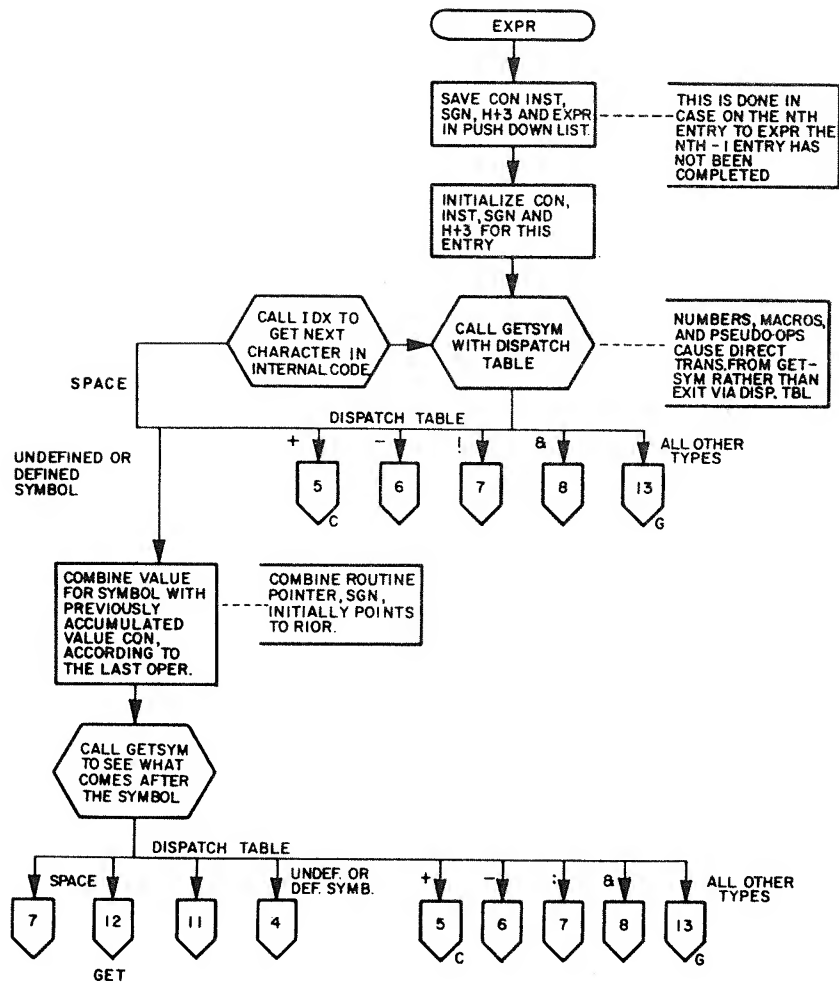
OUTPUT: Entry in AC.

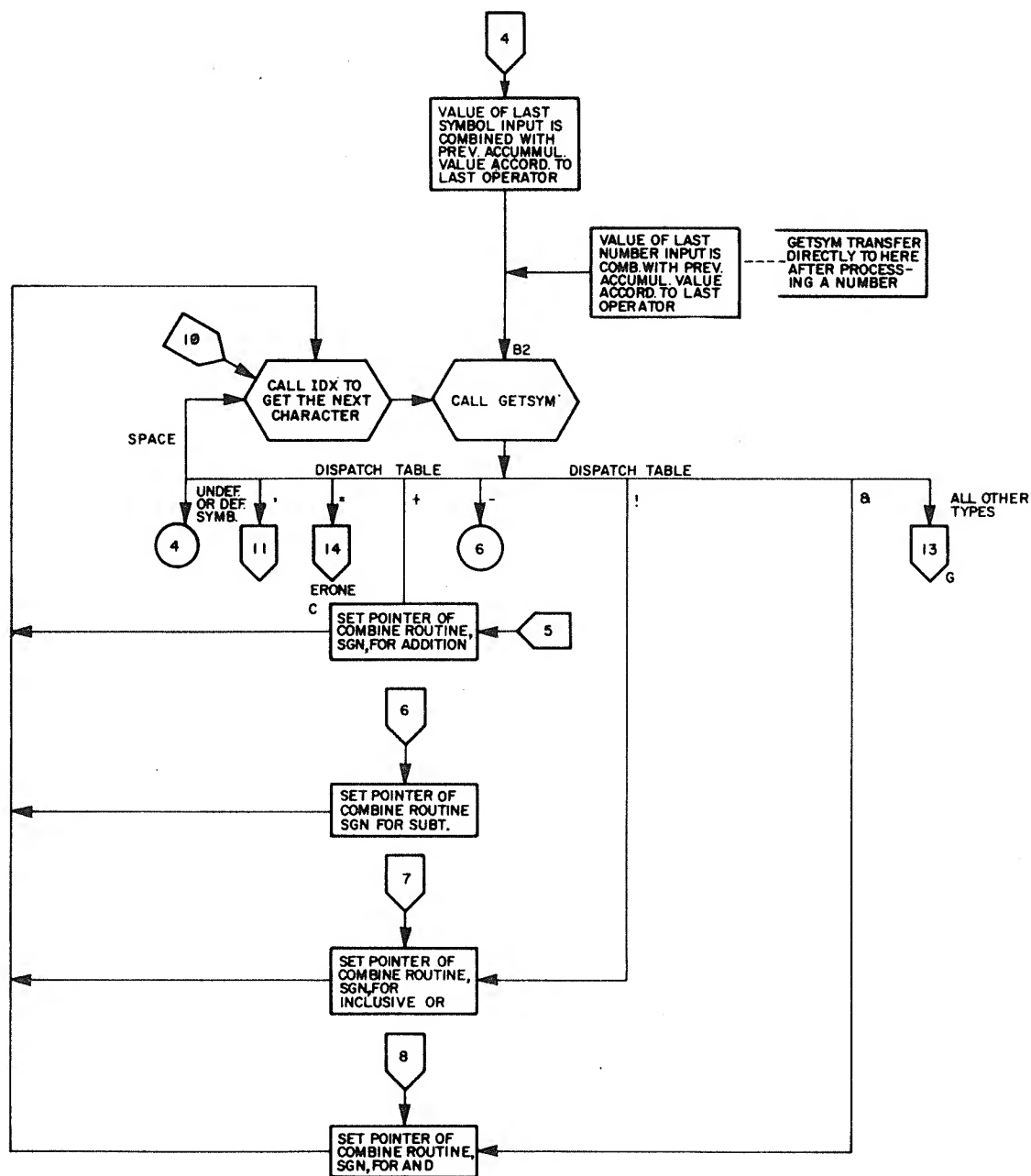
MAJOR REGISTERS CHANGED: POINT.

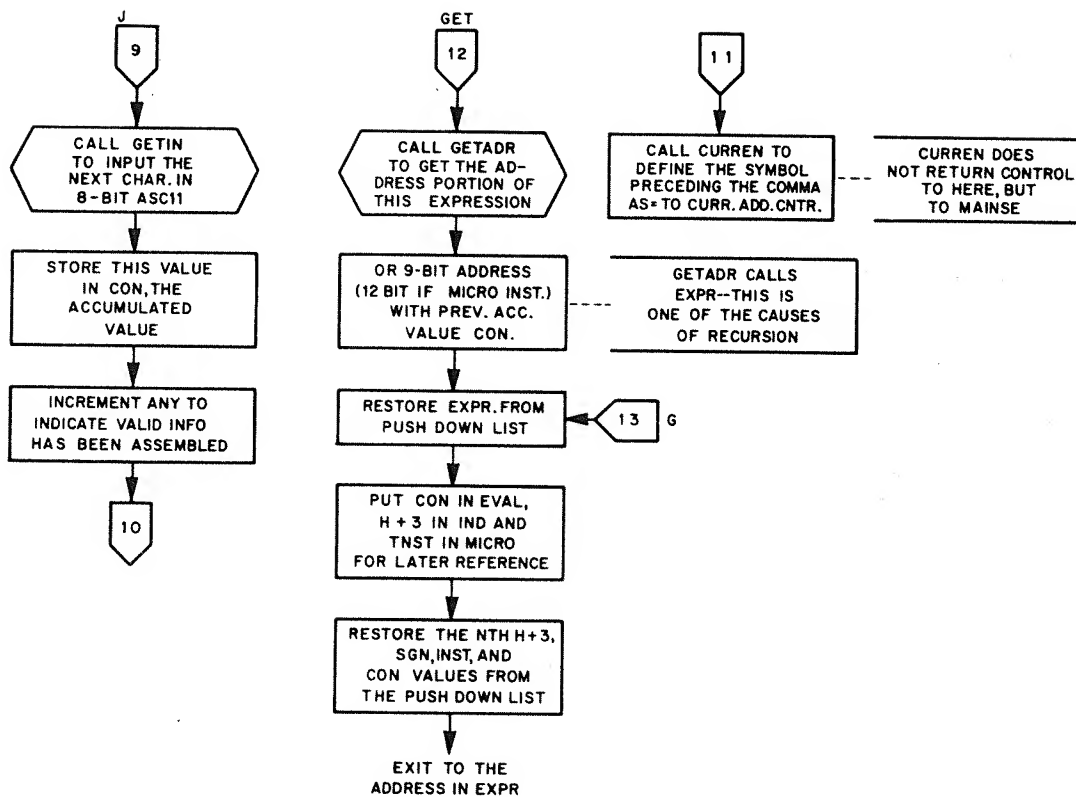


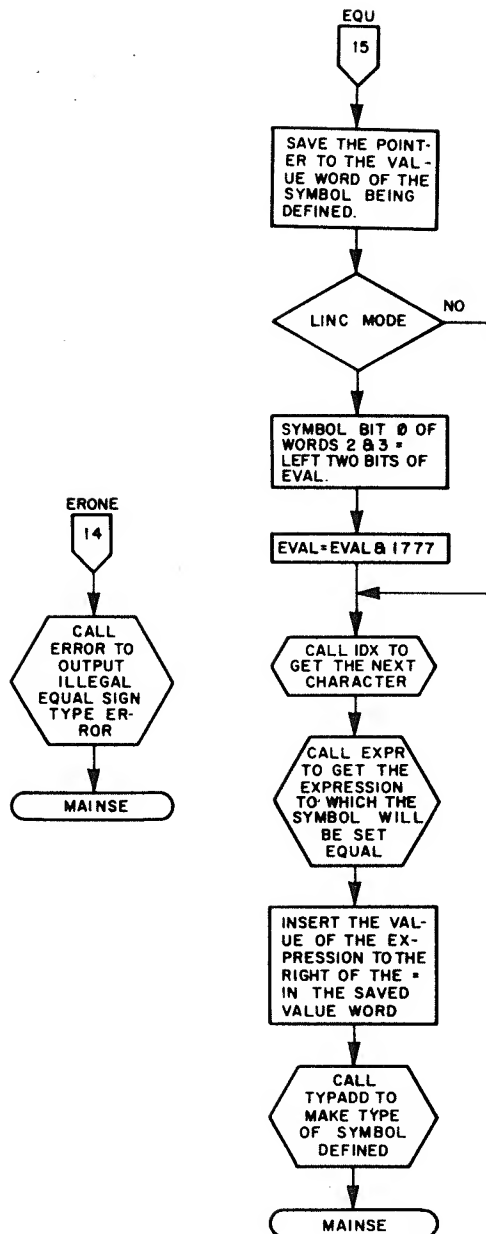


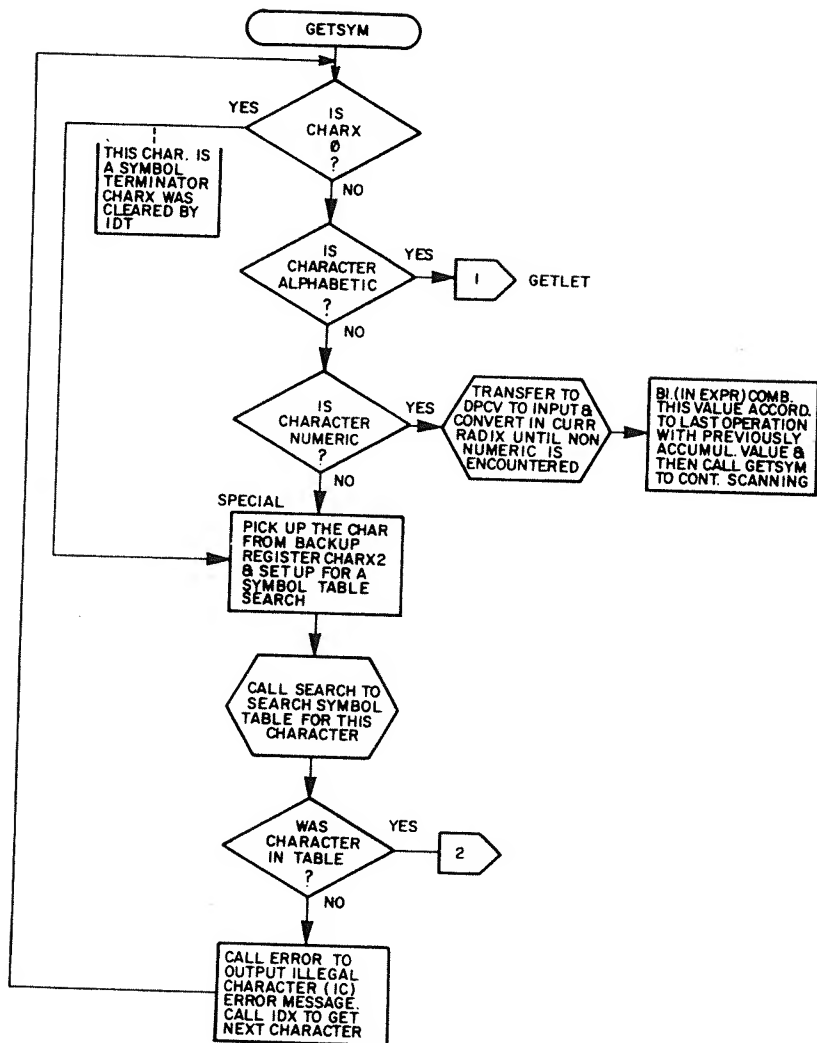


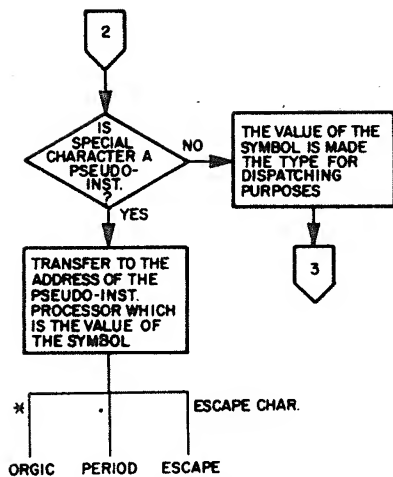






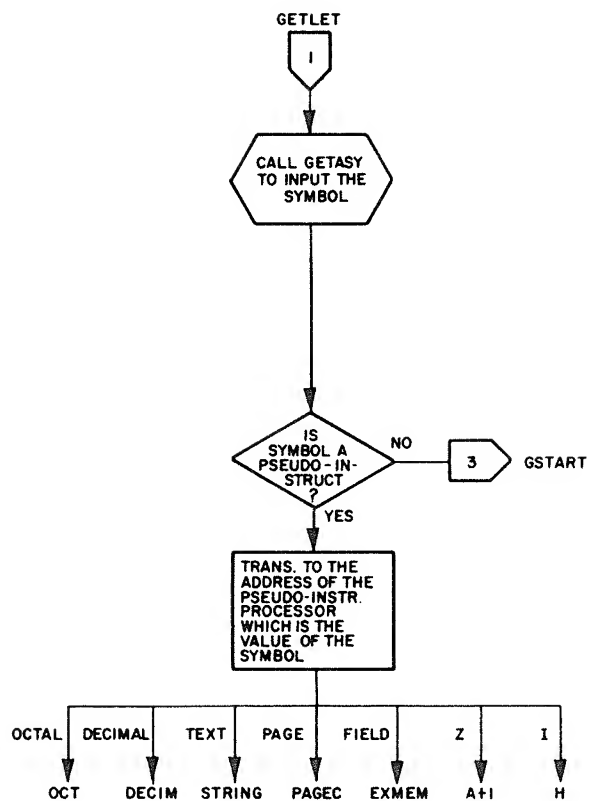


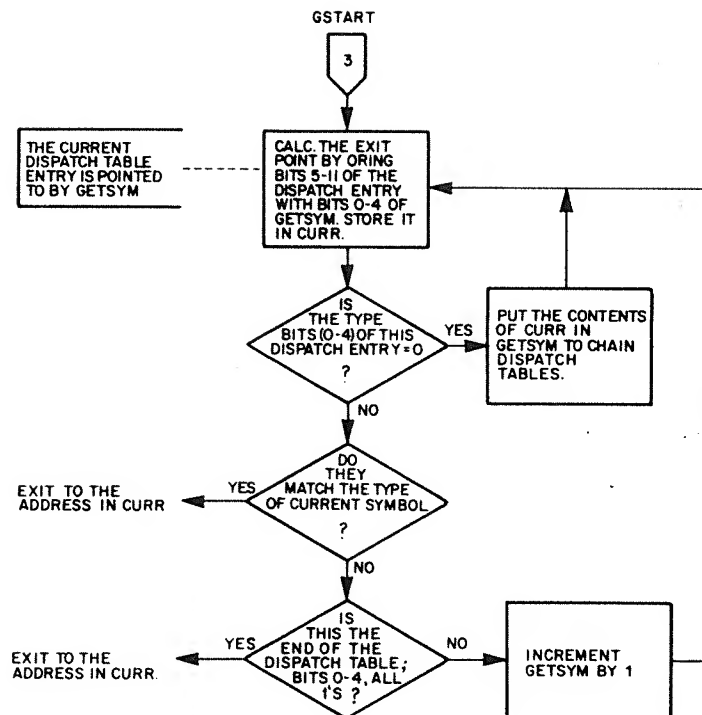




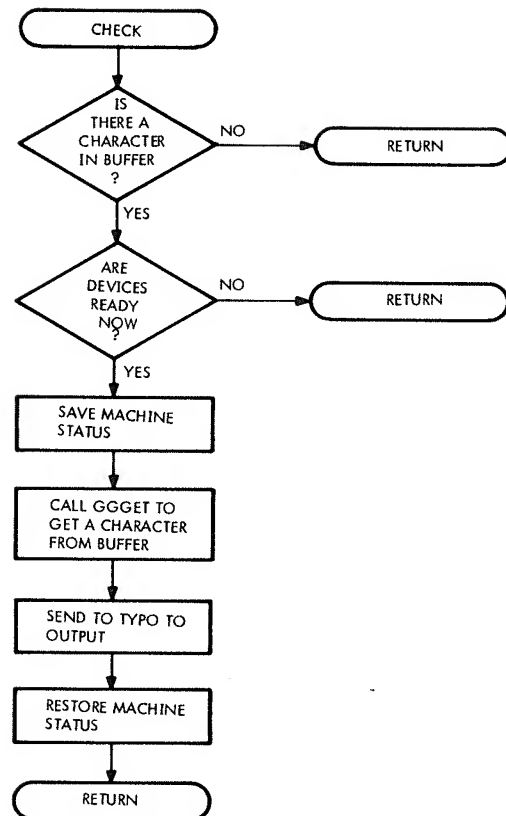
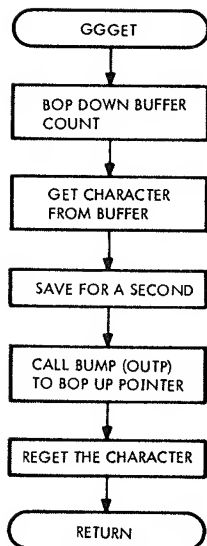
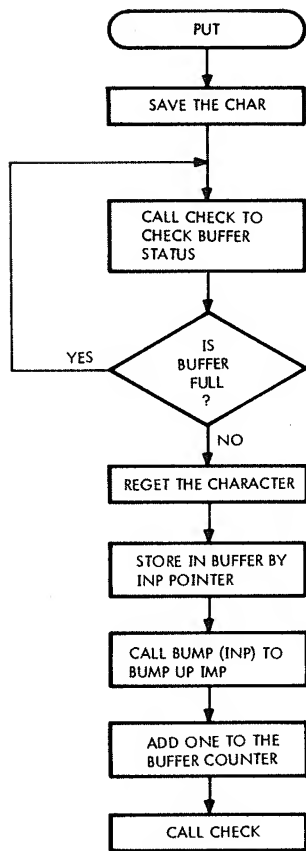
possibilities:

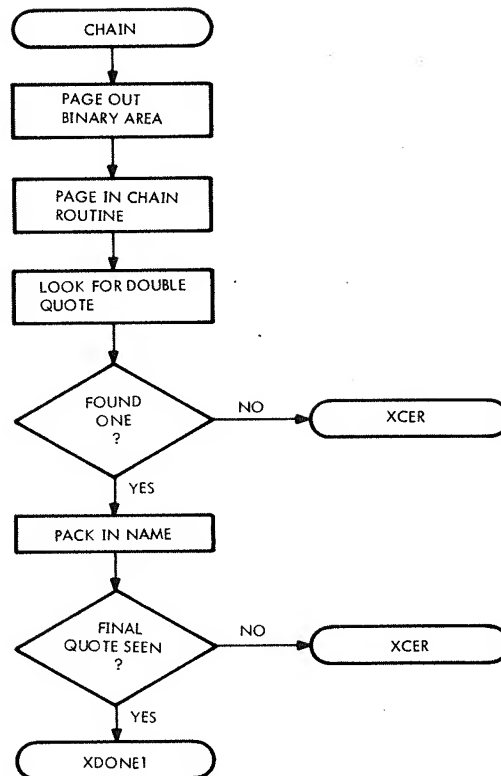
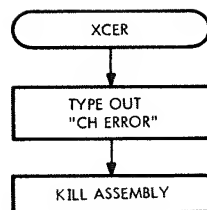
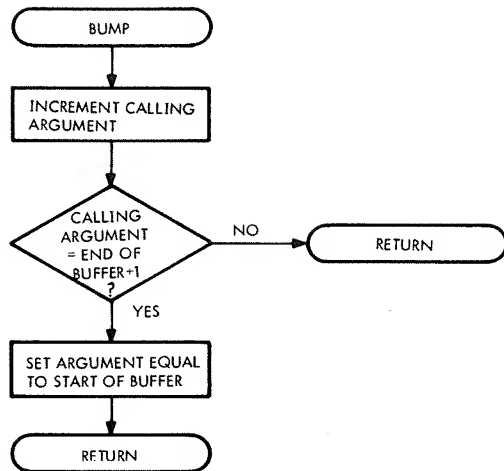
~  
: SPACE

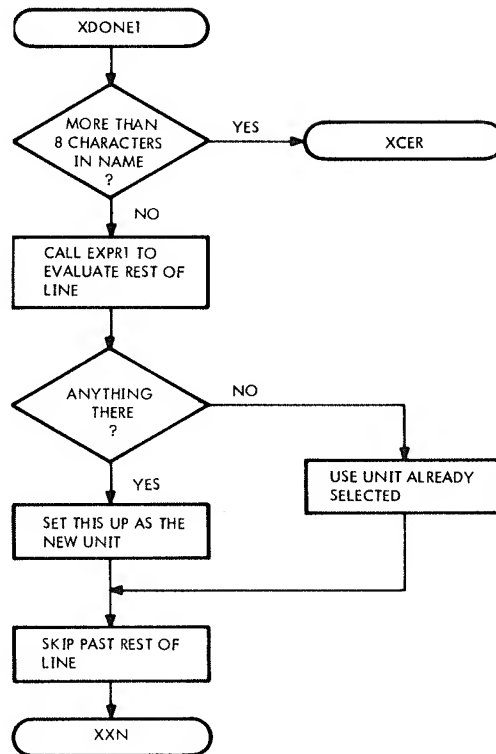
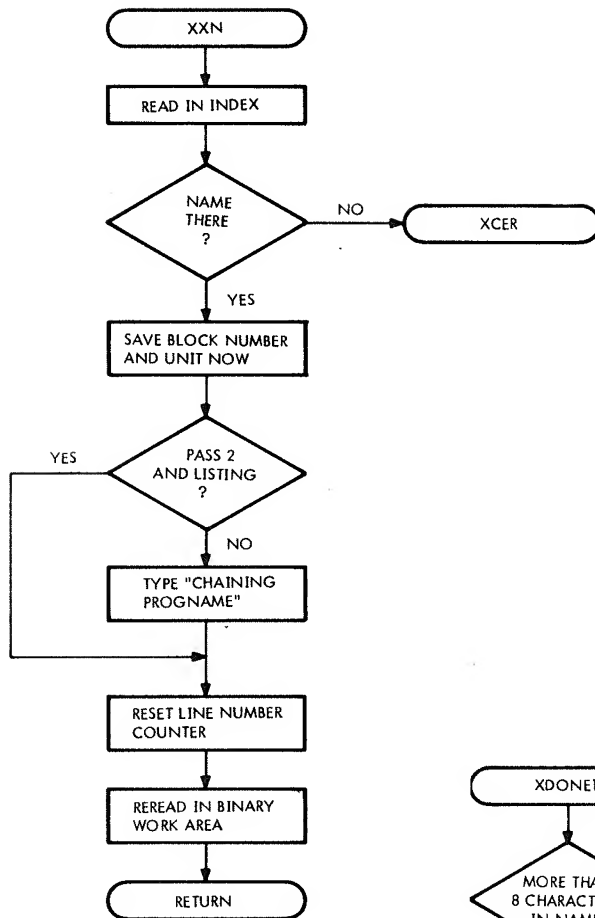




(DECOUT)







0005  
0006  
0007  
0010  
0011  
0012  
0013  
0014  
0015  
0016  
0017  
0020  
0021  
0022  
0023  
0024  
0025

# DIAL ASSEMBLER - - - VERSION 2

/COPYRIGHT 1969, DIGITAL EQUIPMENT CORPORATION, MAYNARD MASSACHUSSETS, 01754  
/PDP-12 DISK DIAL ASSEMBLER

THIS ASSEMBLER REQUIRES THE STANDARD DISK DIAL HARDWARE

LATEST ASSEMBLER <5/14/70>, INCLUDES CHAINING.

EJECT

*internal code (at least that used by MACRO-8)*

|     |       |    |         |
|-----|-------|----|---------|
| A-Z | 1-32  | \$ | 60      |
| 6-9 | 33-44 | %  | 61      |
| .   | 45    | &  | 62      |
| ;   | 46    | '  | 63      |
| <   | 47    | (  | 64      |
| =   | 50    | )  | ignored |
| >   | 51    | *  | 66      |
| ?   | 52    | +  | 67      |
| @   | 53    | ,  | 70      |
| [   | 54    | -  | 71      |
| !   | 55    | .  | 72      |
| "   | 56    | /  | 73      |
| #   | 57    | [  | 74      |
|     |       | \  | 75      |
|     |       | ]  | ignored |

car. ret . 46  
tab 54

```

0026 /
0027 /
0030 /
0031 /0000-0177
0032 /0200-0377
0033 /0400-0577
0034 /0600-0777
0035 /
0036 /
0037 /
0040 /
0041 /
0042 /1000-1177
0043 /1200-1377
0044 /1400-1577
0045 /1600-1777
0046 /2000-2177
0047 /2200-2377
0050 /2400-2577
0051 /2600-2777
0052 /3000-3177
0053 /3200-3377
0054 /
0055 /3400-3577
0056 /
0057 /3600-3777
0060 /
0061 /
0062 /
0063 /THE FOLLOWING LOCATIONS ARE IN
0064 /
0065 /
0066 /4000-4177
0067 /4200-4377
0070 /4400-4777
0071 /5000-5377
0072 /
0073 /
0074 /5400-5777
0075 /
0076 /6000-6777
0077 /7000-7377
0100 /7400-7577
0101 /7600-7777
0102 /
0103 /
0104 /
0105 /*****
0106 /
0107 /
0110 /
0111 /

```

4 K L O A D E R M A P

PAGE 0: COMMON LITERALS AND POINTERS,  
 BANK CHECK ROUTINE; SYMBOL SEARCH ROUTINE  
 SYMBOL TABLE GETTER ROUTINE  
 ROUTINE TO GET A SYMBOL FROM SYMBOL TABLE( OR ADD ON).  
 SDECO4 IS ACTIVE, SDECO8 IS NOT USED,  
 SDECO4 USES 6777-6000 FOR USER SYMBOLS,  
 5400-5777 IS USED FOR USER SYMBOLS AND FOR A SWAPPING BUFFER/  
 IF MORE THEN 300 SYMBOLS . EXTRA SYMBOLS GO OUT ON UNIT 1.  
 GENERAL RECURSIVE EXPRESION EVALUATOR,  
 RECURSIVE ADDRESS EVALUATOR,  
 /CENTRAL PROCESSING LOOP, MAIN START OF ASSEMBLER AFTER INITIALIZATION.  
 /TEXT PROCESSOR, CURRENT LINE TESTER, PLUS MINOR SUPPORT ROUTINES.  
 AN INPUT CHARACTER SCANNER ROUTINE, THE PRINTER BUF ROUTINE, AND TAB ROUTINE,  
 NUMBER CONVERSIONS, TYPE CHECKS,  
 SYMBOL SORT FOR OUTPUT, TTY ROUTINE,  
 SYMBOL OUTPUT AND VALUE ROUTINE, OCTAL PRINT ROUTINE, FIELD CHANGE ROUTINE, AND WORK  
 AREA,  
 OUTPUT LIST ROUTINE, 8K PSEUDO OP INITIALIZATION(DESTROYED AND NEVER USED), WORK AR  
 EA,  
 ERROR MESSAGE PROCESSOR, TYPE EVALUATOR, SYSTEM SYMBOL TABLE SWAPPER,  
 FAIRLY LOW LEVEL CHARACTER GETTER, AND ROUTINE TO SKIP IF NOT PASS2 OR NOT IN LIST M  
 ODE,  
 INPUT CHARACTER CONVERTOR TO INTERNAL FORM, PLUS PAGE CHECKERS, OTHER  
 LITTLE ROUTINES,  
 INTERNAL TO EXTERNAL CHARACTER CONVERTOR, SYMBOL TABLE TYPE GETTERS AND SETTERS,  
 LOW LEVEL CHARACTER GETTER, COMMENT PROCESSOR, LIST AND NOLIST PROCESSORS.

GENERAL TAPE I/O CALL ROUTINES, INITIALIZATION FOR TAPE ROUTINES, PLUS EXTRA LITTLE  
 MORE TAPE I/O ROUTINES, PSEUDO OP PROCESSOR(FOR PAGED IN PSEUDO OPS), AND LOADER CO  
 RE MAP,  
 INPUT BUFFER+SWAPPED IN AND OUT WORK AREA FOR SAVSYM AND LODSYM,  
 OUTPUT BUFFER, PLUS INITIALIZATION FOR ENTIRE ASSEMBLER AFTER IT S FIRST LOADED IN  
 AND THIS ALSO CONTAINS THE SWAPPED IN PSEUDO OP PROCESSOR. NOTE: WHEN THINGS ARE SW  
 APPED IN,  
 THE PREVIOUS CONTENTS OF THE LOCATIONS ARE FIRST SWAPPED OUT TO PRESERVE THEM,  
 USER SYMBOLS IF < 300 SYMBOLS, ELSE IT S BUFFER AREA FOR SWAPPING SYMBOLS FROM TAP  
 1.  
 USER SYMBOLS(FIRST 200 OF THEM ANYWAY),  
 MODE SYMBOLS(EITHER PMODE OR LMODE TABLE, DEPENDING ON MODE,).  
 PSEUDO OP TABLE PLUS SPECIAL CHARACTER DEFINITION TABLE,  
 SUPPOSEDLY UNUSED, BUT WHEN PSEUDO OP TABLE IS LOADED, IT S LENGTH  
 OF 400 WORDS OVERLAYS IT.

END OF 4 K M A P

EJECT

```

0112 /
0113 /
0114 /
0115 /THIS MAP SHOWS ONLY CHANGES TO 4K MAP,
0116 /
0117 /
0120 /
0121 /
0122 /0600-0777
0123 /
      SDEC08 IS NOW ACTIVE AND SDEC04 IS PASSIVE. SDEC08 USES THE UPPER 4K OF CORE
      FOR THE SYMBOL TABLE, THUS IT STORES NO USER SYMBOLS ON UNIT1(EXCEPT IF A SAVSYM IS
      GIVEN).
0124 /5000-5377
0125 /5400-5777
0126 /6000-6177
0127 /6200-6377
0130 /6400-6777
0131 /7000-7377
      PSEUDO OP TABLE IS NO LONGER PAGED IN HERE, SEE 7000-7377,
      OVERLAPPED READ ROUTINES FOR READING WHILE CALCULATING.
      LINE PRINTER CHECKER AND BASIC LINE PRINTER ROUTINE.
      OVERLAPPED TAPE OUTPUT ROUTINES PLUS COMMON READ-WRITE ROUTINES.
      INPUT BUFFER NO .2
      RESIDENT PSEUDO OP PROCESSOR, NOTE THAT IT CAN RESIDE IN TWO SEPERATE PAGES(FOR 4 OR
      8K),
0132 /7400-7777
0133 /
0134 /
0135 /
0136 /
0137 /
      INPUT BUFFER NO.1 FOR TAPE.
0140 /10000-16777
0141 /17000-17377
0142 /17400-17577
      F I E L D 1 R E F E R E N C E S
0143 /
0144 /
0145 /
0146 /
0147 /
0150 /
0151 /
0152 /
      USER DEFINED SYMBOLS.
      SYSTEM MODE SYMBOLS(LMODE OR PMODE),
      PSEUDO OPS AND SPECIAL CHARACTER TABLE.
      E N D O F 8 K M A P
      EJECT

```

```

0153 /          T H E   P S E U D O   O P   P R O C E S S O R
0154 /
0155 /
0156 /
0157 /
0160 /
      THE PSEUDO OP PROCESSOR IS A 400 LOCATION ROUTINE WHICH IS PAGED IN ON THE 4K MODE AND RESID
0161 ENT IN THE 8K MODE.
0162 /AFTER THE DESIRED PSEUDO OP IS FOUND IN THE TABLE BY "EXPR" IT THEN JMPS TO AN ISZ CHAIN
      /THE ISZ CHAIN IS A BUNCH OF SEQUENTIAL ISZ S OF LOCATION "OP1", AFTER FALLING THROUGH THE ISZ S OP1 CONTAI
      NS
0163 / THE DESIRED PSEUDO OP NUMBER, THE ROUTINE THEN PAGES IN THE PSEUDO OP PROCESSOR(IF 4K, ELSE IT S RESIDENT)
      , THEN JMS
0164 / TO IT WITH THE DESIRED OPERATION NUMBER IN THE AC, WHEN FINISHED THE PSEUDO OP PROCESSOR RETURNS AND IT S
      SWAPPED BAC
0165 /K OUT, IN THE 8K VERSION IT S RESIDENT AND IS NOT SWAPPED IN OR OUT.
0166 /
0167 /
0170 /
0171 /
0172 /
0173 EJECT

```

-

```

0174 /
0175 /
0176 /
0177 /
0200 /A SYMBOL TABLE ENTRY IS COMPOSED OF 4 WORDS, THE FIRST THREE ARE THE NAME AND THE LAST IS THE VALUE
0201 /IF THE SYMBOL IS A PSEUDO OP THEN THE LAST WORD CONTAINS A POINTER TO THE ROUTINE WHICH WILL PROCESS
0202 /THE PSEUDO OP AND THE REST OF THE STATEMENT.
0203 /
0204 /
0205 /THE NAME IS COMPOSED OF SIX CHARACTERS.
0206 /THESE CHARACTERS ARE USUALLY THE 26 LETTERS PLUS 10 NUMBERS PLUS A NULL CHARACTER IF LESS THEN 6 LETTERS LONG.
0207 /THIS GIVES YOU 45 OCTAL POSSIBLE CHARACTERS FOR THE LEFT OR THE RIGHT HALF OF A WORD.
0210 /IF YOU USE SIXBIT NOTATION, YOU LL WASTE 23 POSSIBLE POSITIONS ON THE CHARACTER. THEREFORE, SIXBIT IS NOT USED.
0211 /NOW, SINCE THE FIRST CHARACTER OF A NAME MUST BE A LETTER, THIS LEAVES ONLY 32 OCTAL
0212 /COMBINATIONS FOR THE FIRST CHARACTER. THUS IF WE MULTIPLY THE LEFT
0213 /CHARACTERS OF A WORD BY 45, WE WILL HAVE 2 FREE BITS IN THE FIRST WORD OF A NAME./
0214 /((32*45+45<2000 OCTAL), THESE TWO LEFT BITS ARE USED AS FOLLOWS:
0215 /
0216 /
0217 /
0220 /
0221 /
0222 /
0223 /
0224 /
0225 /
0226 /
0227 /

```

|     |                   |
|-----|-------------------|
| 00: | SPECIAL CHARACTER |
| 01: | UNDEFINED         |
| 10: | DEFINED           |
| 11: | PSEUDO OPERATION  |

```

EJECT

```

```

0230 /
0231 /
0232 /
0233 /
0234 /
0235 /
0236 /
0237 /
0240 /EVERY TIME YOU ASSEMBLE AND CHANGE THE MAJOR SYMBOL LOCATIONS, UP THIS BY 1.
0241 /
0242 /
0243 /
0244 UNITAS=4777 /PLACE TO GET UNIT FROM
0245 E7=0
0246 D6=300
0247 D7=370
0250 D8=410
0251 D9=446
0252 PTHREE=7325 /CLA CLL CML IAC RAL
0253 /GENERATES A +3 IN AC WHEN EXECUTED
0254 MTHWO=7344 /CLA CLL CMA RAL
0255 /GENERATES A -2 IN AC WHEN EXECUTED,
0256 MTHREE=7346 /CLA CLL CMA RTL
0257 /GENERATES A -3 IN AC WHEN EXECUTED,
0260 /
0261 /
0262 /
0263 /
0264 /
0265 /
0266 /TERMC=00 /SET TERMC TO 44 IF DOLLAR SIGN IS THE END OF FILE CHAR,
0267 /
0270 /
0271 /
0272 /
0273 EJECT

```

```

0274 /PDP-12 ASSEMBLER PAGE 0
0275 PMODE
0276 *1
0277 0001 0710 DECOU, PUT
0300 0002 0000 ERTOT, 0
0301 0003 0000 LISTWD, 0
0302 0004 2353 CNTLPI, CNTRL
0303 0005 0000 MICROB, 0
0304 0006 7777 EQRETI, 7777
0305 0007 0000 TEMP, 0
0306 /VARIABLES
0307 *10
0310 0010 2400 BINPTR, 2000+D7+10
0311 0011 0000 BINBLK, 0
0312 0012 0000 CURLIN, 0
0313 0013 7740 M40, -40
0314 0014 0000 TYPEA, 0
0315 0015 0011 P11, 11
0316 /16 AND 17 ARE USED
0317 /FOR AUTO-INDEXING,
0320 *20
0321 0020 0000 ANY, 0 /WAS THIS AN INSTRUCTION
0322 0021 0000 TYPE, 0 /HOLDS TYPE OF CURRENT SYMBOL
0323 0022 0000 VADR, 0 /VALUE OF CURRENT SYMBOL
0324 0023 0000 SADR, 0 /POINTER OF FIRST WORD OF CURRENT SYMBOL
0325 0024 0000 VAL, 0 /VALUE OF CURRENT SYMBOL
0326 0025 0000 VALK, 0
0327 0026 0000 SCURR, 0 /CURRENT SYMBOL NUMBER
0330 0027 0141 SEND, 141 /POINTER TO FIRST FREE REGISTER AFTER SYMBOL TABLE
0331 0030 0000 AADR, 0 /ACTUAL CURRENT ADDRESS
0332 0031 0000 APAGE, 0 /ORIGIN OF CURRENT PAGE - ACTUAL
0333 0032 0141 PERMA, 141 /FIRST REGISTER AFTER PERMANENT SYMBOL TABLE
0334 0033 0000 EVAL, 0 /HOLDS ADDRESS PORTION DURING CALCULATION
0335 0034 0000 SGN, 0
0336 0035 0000 PASS, 0 /7777 IF PASS 2
0337 0036 0000 FLOWD, 0 /7777 IF EXTENDED MEMORY
0340 0037 0000 MODE, 0 /7777 IF POP-8; 0 IF LINC
0341 0040 0000 TEM1A, 0
0342 0041 0000 0
0343 0042 0000 0
0344 0043 0000 CURR, 0
0345 0044 0000 CHARX, 0
0346 0045 0000 CHARX2, 0
0347 0046 0000 ABANK, 0
0350 0047 0000 POINT, 0 /PUSH DOWN LIST
0351 0050 0000 IN, 0 /TELLS WHETHER SYMBOL WAS ACCEPTED
0352 0051 0000 ITEM, 0
0353 0052 0000 HIC, 0
0354 0053 0000 LWC, 0
0355 0054 0000 MICROL, 0
0356 0055 0000 INST, 0 /- IF SYMBOLIC A MICRO INST.
0357 0056 0000 MICRO, 0
0360 0057 0000 CHARAC, 0
0361 0060 4013 INIT1, INIT
0362 /POINTERS
0363 0061 2054 EMPTY1, EMPTY
0364 0062 2661 SOPS, OPS
0365 0063 3663 RESET1, RESETL
0366 0064 3400 BIT6A, BIT6
0367 0065 3355 PASSEK, PASSER
0370 0066 2520 CURSKA, CURSKP
0371 0067 3317 GETCHR, SYSIN
0372 0070 3524 CHEKER, PAGSET

```

|      |      |      |                |
|------|------|------|----------------|
| 0373 | 0071 | 1454 | SYMLST, LSTSYM |
| 0374 | 0072 | 1400 | MAINEX, MAIN   |
| 0375 | 0073 | 1547 | INAS, CJMP1    |
| 0376 | 0074 | 0524 | MOVE1, MOVE8   |
| 0377 | 0075 | 0764 | SKIP, SKIP2    |
| 0400 | 0076 | 3201 | ERR1, ERROR    |
| 0401 | 0077 | 1000 | EXPR1, EXPR    |
| 0402 | 0100 | 2510 | TYCARI, TYCAR  |
| 0403 | 0101 | 1411 | MAIN1, MAINSE  |
| 0404 | 0102 | 1551 | PAGE1, RESET   |
| 0405 | 0103 | 1046 | B21, B2        |
| 0406 | 0104 | 2037 | IDX1, IDX      |
| 0407 | 0105 | 2341 | PUSH1, PUSH2   |
| 0410 | 0106 | 4307 | POPUP1, POPUP1 |
| 0411 | 0107 | 0400 | GETSY1, GETSYM |
| 0412 | 0110 | 3242 | GETAS1, GETASY |
| 0413 | 0111 | 0622 | TYPAD1, TYPAD0 |
| 0414 | 0112 | 0600 | SDEC01, SDEC08 |
| 0415 | 0113 | 4201 | SRITIC, SWRC   |
| 0416 | 0114 | 1503 | SYB, SYMBU     |
| 0417 | 0115 | 4275 | SRD1, SREAD1   |
| 0420 | 0116 | 4051 | RET1, RETURN   |
| 0421 | 0117 | 4101 | MON, MONIT     |
| 0422 | 0120 | 3001 | PUNON1, PUNONE |
| 0423 | 0121 | 7000 | M1000, -1000   |
| 0424 | 0122 | 7735 | M43, -43       |
| 0425 | 0123 | 7745 | M33, -33       |
| 0426 | 0124 | 0400 | P400, 400      |
| 0427 | 0125 | 7733 | M45, -45       |
| 0430 | 0126 | 7711 | M67, -67       |
| 0431 | 0127 | 0076 | P76, 76        |
| 0432 | 0130 | 1706 | LINENO, JBTEST |
| 0433 | 0131 | 3777 | P3777, 3777    |
| 0434 | 0132 | 2132 | SCLEAR, CLEARR |
| 0435 | 0133 | 7731 | M47, -47       |
| 0436 | 0134 | 0240 | P240, 240      |
| 0437 | 0135 | 7774 | M4, -4         |
| 0440 | 0136 | 6000 | P6000, 6000    |
| 0441 | 0137 | 3662 | PM60A, M60A    |
| 0442 | 0140 | 0000 | LOWTMP, 0      |
| 0443 | 0141 | 0740 | PCHECK, CHECK  |
| 0444 | 0142 | 5000 | P5000, 5000    |
| 0445 | 0143 | 0300 | P300, 300      |
| 0446 | 0144 | 1777 | P1777, 1777    |
| 0447 | 0145 | 0177 | P177, 177      |
| 0450 | 0146 | 7600 | P7600, 7600    |
| 0451 | 0147 | 0200 | P200, 200      |
| 0452 | 0150 | 4000 | P4000, 4000    |
| 0453 | 0151 | 7563 | M215, -215     |
| 0454 | 0152 | 2000 | P2000, 2000    |
| 0455 | 0153 | 0212 | P212, 212      |
| 0456 | 0154 | 0215 | P215, 215      |
| 0457 | 0155 | 7400 | P7400, 7400    |
| 0460 | 0156 | 0777 | P777, 777      |
| 0461 | 0157 | 0007 | P7, 7          |
| 0462 | 0160 | 0077 | P77, 77        |
| 0463 | 0161 | 0000 | TEM1, 0        |
| 0464 | 0162 | 0000 | TEM2, 0        |
| 0465 | 0163 | 0000 | TEM3, 0        |
| 0466 | 0164 | 0000 | CON, 0         |
| 0467 | 0165 | 0000 | ADR, 0         |
| 0470 | 0166 | 0000 | IND, 0         |
| 0471 | 0167 | 0000 | ERRCNT, 0      |

/MAY BE USED ONLY FOR IMM, TEMP, NEVER USED IF YOU LEAVE ROUTINE.

/FIRST  
 /SECOND  
 /THIRD WORD OF CURRENT SYMBOL  
 /HOLDS ACCUMULATED VALUE OF WORD  
 /HOLD FULL 12 BIT ADDRESS  
 /400 IF INDIRECT OTHERWISE 0

```

0472 0170 0000 SWITCH, 0
0473 0171 0000 SNUM, 0
0474 0172 0000 EMPTRA, 0
0475 0173 0000 SPCUNT, 0
0476 0174 3102 SY, SYMBUF
0477 0175 0000 TBCONT, 0
0500 0176 2370 BINTAB, 2000+D7
0501 0177 4321 SETINU, UBITS
0502 /
0503 /
0504 /
0505 /
0506 /
0507

```

/LINE CHARACTER SPACE COUNTER,

/POINTS TO THE ROUTINE TO TURN ON THE HEADER BLOCK BITS

EJECT

```

0510      /PDP-12 ASSEMBLER PAGE 1
0511      *200
0512      0200 4477 BANKCH, JMS I EXPR1
0513      0201 1020      TAD ANY
0514      0202 7640      SZA CLA
0515      0203 5220      JMP BANKEM
0516      0204 7040      CMA
0517      0205 1030      TAD AADR
0520      0206 1152      TAD P2000
0521      0207 0136      AND P6000
0522      0210 3030      DCA AADR
0523      0211 4463 BANKHR, JMS I RESET1
0524      0212 4502      JMS I PAGE1
0525      0213 1030      TAD AADR
0526      0214 3033      DCA EVAL
0527      0215 4465      JMS I PASSEK
0530      0216 4466      JMS I CURSKA
0531      0217 5501      JMP I MAIN1
0532      0220 7307 BANKEM, CLA CLL IAC RTL
0533      0221 0033      AND EVAL
0534      0222 3036      DCA FLOWD
0535      0223 1033      TAD EVAL
0536      0224 7012      RTR
0537      0225 7010      RAR
0540      0226 5207      JMP BANKHR-2
0541      /
0542      0227 0000 CURREN, 0
0543      0230 4663      JMS I REST01
0544      0231 2260      ISZ COMMAN
0545      0232 3171      DCA SNUM
0546      0233 4661      JMS I PSTCHK
0547      0234 4662      JMS I MOVEAA
0550      0235 1037      TAD MODE
0551      0236 7004      RAL
0552      0237 7200      CLA
0553      0240 1030      TAD AADR
0554      0241 7420      SNL
0555      0242 0144      AND P1777
0556      0243 6211      6211
0557      0244 3422      DCA I VADR
0560      0245 6201      6201
0561      0246 7430      SZL
0562      0247 5253      JMP PDP8ER
0563      0250 1030      TAD AADR
0564      0251 0136      AND P6000
0565      0252 7112      CLL RTR
0566      0253 1150 PDP8ER, TAD P4000
0567      0254 3256      DCA ,+2
0570      0255 4511      JMS I TYPAD1
0571      0256 0000      0
0572      0257 5664      JMP I MAIN1A
0573      /
0574      0260 7777 COMMAN, -1
0575      0261 3757 PSTCHK, STOCHK
0576      0262 2275 MOVEAA, ERR2
0577      0263 1751 REST01, RESTOR
0600      0264 1416 MAIN1A, MAINSF
0601      /
0602      0265 1736 SAVE1, SAVE
0603      /
0604      0266 0000 SEARCH, 0
0605      0267 4665      JMS I SAVE1
0606      0270 3026      DCA SCURR

```

/\*4

/CHECK FOR TOO MANY COMMAS,  
/YEP, A BADDY

/PLACE SIGN BIT INTO THE LINK

/RESET IRREGULARLY,

*1 scans the symbol table*  
*1 set symbol # = 0*

*Search starts at top  
of symbol table (#7600)  
and works down*

|      |      |      |         |              |
|------|------|------|---------|--------------|
| 0607 | 0271 | 6211 |         | 6211         |
| 0610 | 0272 | 1026 | SEARC1, | TAD SCURR    |
| 0611 | 0273 | 4512 |         | JMS I SDECO1 |
| 0612 | 0274 | 7240 |         | CLA CMA      |
| 0613 | 0275 | 1023 |         | TAD SADR     |
| 0614 | 0276 | 3016 |         | DCA 16       |
| 0615 | 0277 | 1416 |         | TAD I 16     |
| 0616 | 0300 | 0144 |         | AND P1777    |
| 0617 | 0301 | 7041 |         | CIA          |
| 0620 | 0302 | 1161 |         | TAD TEM1     |
| 0621 | 0303 | 7640 |         | SZA CLA      |
| 0622 | 0304 | 5370 |         | JMP IND1     |
| 0623 | 0305 | 1416 |         | TAD I 16     |
| 0624 | 0306 | 0131 |         | AND P3777    |
| 0625 | 0307 | 7041 |         | CIA          |
| 0626 | 0310 | 1162 |         | TAD TEM2     |
| 0627 | 0311 | 7640 |         | SZA CLA      |
| 0630 | 0312 | 5370 |         | JMP IND1     |
| 0631 | 0313 | 1416 |         | TAD I 16     |
| 0632 | 0314 | 0131 |         | AND P3777    |
| 0633 | 0315 | 7041 |         | CIA          |
| 0634 | 0316 | 1163 |         | TAD TEM3     |
| 0635 | 0317 | 7640 |         | SZA CLA      |
| 0636 | 0320 | 5370 |         | JMP IND1     |
| 0637 | 0321 | 1367 | TAD TOP |              |
| 0640 | 0322 | 7041 |         | CIA          |
| 0641 | 0323 | 1023 |         | TAD SADR     |
| 0642 | 0324 | 7700 |         | SMA CLA      |
| 0643 | 0325 | 5334 |         | JMP PSUD     |
| 0644 | 0326 | 1032 |         | TAD PERMA    |
| 0645 | 0327 | 7041 |         | CIA          |
| 0646 | 0330 | 1026 |         | TAD SCURR    |
| 0647 | 0331 | 7510 |         | SPA          |
| 0650 | 0332 | 3055 |         | DCA INST     |
| 0651 | 0333 | 7300 |         | CLA CLL      |
| 0652 | 0334 | 1416 | PSUD,   | TAD I 16     |
| 0653 | 0335 | 3025 |         | DCA VALK     |
| 0654 | 0336 | 1016 |         | TAD 16       |
| 0655 | 0337 | 3022 |         | DCA VADR     |
| 0656 | 0340 | 7346 |         | MTHREE       |
| 0657 | 0341 | 1016 |         | TAD 16       |
| 0660 | 0342 | 3016 |         | DCA 16       |
| 0661 | 0343 | 1416 |         | TAD I 16     |
| 0662 | 0344 | 0150 |         | AND P4000    |
| 0663 | 0345 | 3007 |         | DCA TEMP     |
| 0664 | 0346 | 1416 |         | TAD I 16     |
| 0665 | 0347 | 0150 |         | AND P4000    |
| 0666 | 0350 | 7110 |         | RAR CLL      |
| 0667 | 0351 | 1007 |         | TAD TEMP     |
| 0670 | 0352 | 1025 |         | TAD VALK     |
| 0671 | 0353 | 3024 |         | DCA VAL      |
| 0672 | 0354 | 1037 |         | TAD MODE     |
| 0673 | 0355 | 7700 |         | SMA CLA      |
| 0674 | 0356 | 4764 |         | JMS I IND18  |
| 0675 | 0357 | 4765 |         | JMS I GETYE  |
| 0676 | 0360 | 2266 |         | ISZ SEARCH   |
| 0677 | 0361 | 4766 |         | JMS I IND1C  |
| 0700 | 0362 | 6201 | TAABA,  | 6201         |
| 0701 | 0363 | 5666 |         | JMP I SEARCH |
| 0702 | 0364 | 1560 | IND18,  | LNINST       |
| 0703 | 0365 | 3623 | GETYE,  | GETYE        |
| 0704 | 0366 | 2367 | IND1C,  | ANY          |
| 0705 | 0367 | 7374 | TOP,    | 7374         |

*current symbol number*

*pointer to 15<sup>th</sup> word of current symbol*

*/NO MATCH.*

*/- IF INSTRUCTION*

*get type of current symbol*

*/Return here if nothing assembled for this line  
/Return here if something found  
/line instruction done*

*If 113 loc is below  
74100 then it's a  
perm. symb. If  
above, a pseudo*

*PERMA = 141 =  
141<sup>st</sup> symbol slot;  
⇒ first symbol  
after permanent  
symbols.*

*1/0 = special char*

*01 = undefined*

*10 = defined*

*11 = pseudo-op*

|      |      |      |       |            |
|------|------|------|-------|------------|
| 0706 | 0370 | 2026 | IN01, | ISZ SCURR  |
| 0707 | 0371 | 1026 |       | TAD SCURR  |
| 0710 | 0372 | 7041 |       | CIA        |
| 0711 | 0373 | 1027 |       | TAD SEND   |
| 0712 | 0374 | 7640 |       | SZA CLA    |
| 0713 | 0375 | 5272 |       | JMP SEARC1 |
| 0714 | 0376 | 2020 |       | ISZ ANY    |
| 0715 | 0377 | 5362 |       | JMP TAABA  |
| 0716 |      |      |       | EJECT      |

*increment symbol  
table pointer*

gets in the next item, whatever  
it is - a symbol, #, or character  
then dispatch according to list  
following the call to this routine  
/ get 1st char.

1st char = letter => symbol

```

0717
0720
0721 0400 0000 GETSYM, 0
0722 0401 1044 TAD CHARX
0723 0402 7450 SNA
0724 0403 5216 JMP SPECIAL
0725 0404 1123 TAD M33
0726 0405 7510 SPA
0727 0406 5241 JMP GETLET
0730 0407 1214 TAD M12A
0731 0410 7700 SNA CLA
0732 0411 5216 JMP SPECIAL
0733 0412 4615 JMS I DPCVOP
0734 0413 5634 JMP I B11
0735 0414 7766 M12A, -12
0736 0415 2200 DPCVOP, DPCV
0737 0416 1045 SPECIAL, TAD CHARX2
0740 0417 3163 DCA TEM3
0741 0420 3162 DCA TEM2
0742 0421 3161 DCA TEM1
0743 0422 4633 JMS I SCH1
0744 0423 5235 JMP B11+1
0745 0424 1021 TAD TYPE
0746 0425 1152 TAD P2000
0747 0426 7650 SNA CLA
0750 0427 5424 JMP I VAL
0751 0430 1024 TAD VAL
0752 0431 3021 DCA TYPE
0753 0432 5246 JMP GSTART
0754 0433 0266 SCH1, SEARCH
0755 0434 1041 B11, B1
0756 0435 7325 PTHREE
0757 0436 4476 JMS I ERR1
0760 0437 4504 JMS I IDX1
0761 0440 5201 JMP GETSYM+1
0762 0441 4510 GETLET, JMS I GETAS1
0763 0442 1021 TAD TYPE
0764 0443 1152 TAD P2000
0765 0444 7650 SNA CLA
0766 0445 5424 JMP I VAL
0767 0446 1600 GSTART, TAD I GETSYM
0770 0447 0145 AND P177
0771 0450 3043 DCA CURR
0772 0451 1200 TAD GETSYM
0773 0452 0146 AND P7600
0774 0453 1043 TAD CURR
0775 0454 3043 DCA CURR
0776 0455 1600 TAD I GETSYM
0777 0456 0146 AND P7600
1000 0457 7440 SZA
1001 0460 5264 JMP GTYPOT
1002 0461 1043 TAD CURR
1003 0462 3200 DCA GETSYM
1004 0463 5246 JMP GSTART
1005 0464 7041 GTYPOT, CIA
1006 0465 1021 TAD TYPE
1007 0466 7650 SNA CLA
1010 0467 5443 JMP I CURR
1011 0470 1600 TAD I GETSYM
1012 0471 0146 AND P7600
1013 0472 1147 TAD P2000
1014 0473 7650 SNA CLA
1015 0474 5443 JMP I CURR

```

/DIGIT

handle special characters here

call search routine

if TYPE = 0000 undefined

/GET SYMBOL

/PSEUDO OP,

get ~~branch~~ branch addr.

fix up ptr. to routine  
get char code

(i.e., new 15

if char code = 0  
double ext so we go to new place

match what we have?  
yes - go to correct service routine  
get char code again

also go if this works out

branch  
addr.  
char.  
code  
branch addr. offset

```

1016 0475 2507 ISZ I GETSY1
1017 0476 5246 JMP GSTART
1020 0477 4477 PAGEC, JMS I EXPR1
1021 0500 1020 TAD ANY
1022 0501 7640 SZA CLA
1023 0502 5316 JMP PAGEM
1024 0503 7040 CMA
1025 0504 1030 TAD AADR
1026 0505 1147 TAD P200
1027 0506 0146 AND P7600
1030 0507 3030 DCA AADR
1031 0510 4502 PAGHER, JMS I PAGE1
1032 0511 1030 TAD AADR
1033 0512 3033 DCA EVAL
1034 0513 4465 JMS I PASSEK
1035 0514 4466 JMS I CURSKA
1036 0515 5001 JMP I MAIN1
1037 0516 1033 PAGEM, TAD EVAL /MULTIPLY BY 200
1040 0517 7012 RTR
1041 0520 7012 RTR
1042 0521 7012 RTR
1043 0522 5306 JMP PAGHER-2
1044 0523 7777 M1,-1
1045 0524 0000 MOVEB, 0
1046 0525 1323 TAD M1
1047 0526 3016 DCA 16
1050 0527 7040 CMA
1051 0530 1724 TAD I MOVEB
1052 0531 3017 DCA 17
1053 0532 2324 ISZ MOVEB
1054 0533 1724 TAD I MOVEB
1055 0534 2324 ISZ MOVEB
1056 0535 3345 DCA MOVA
1057 0536 1156 TAD P777
1060 0537 3007 DCA TEMP
1061 0540 1007 MOVB, TAD TEMP
1062 0541 7124 CLL CML RAL
1063 0542 3007 DCA TEMP
1064 0543 7430 SZL
1065 0544 5724 JMP I MOVEB
1066 0545 0000 MOVA, 0
1067 0546 1416 TAD I 16
1070 0547 0007 AND TEMP
1071 0550 0131 AND P3777
1072 0551 6201 6201
1073 0552 3417 DCA I 17
1074 0553 5340 JMP MOVB
1075 /
1076 0554 7610 DOLIST, CLA SKP /LIST ENTRY PSEUDO,
1077 0555 7240 PLIST, CLA CMA /NOLIST PSEUDO,
1100 0556 3324 DCA MOVEB /IN TEMPORARY,
1101 0557 1035 TAD PASS /IS THIS PASS2?
1102 0560 7650 SNA CLA
1103 0561 5367 JMP DOPOUT /NO, IGNORE IT,
1104 0562 1324 TAD MOVEB /0=LIST,7777=NOLIST
1105 0563 3770 DCA I JJSTART /FUDGE JBTEST PARAMS,
1106 0564 1324 TAD MOVEB
1107 0565 7040 CMA
1110 0566 3771 DCA I JJEND /SET UP LINE NUMBER LIMITS,
1111 0567 5503 DOPOUT, JMP I B21 /CLEAN UP SYSTEM,
1112 0570 1733 JJSTART, JSSTART
1113 0571 1734 JJEND, JEND
1114 /

```

*"page" pseudo op*

*single precision move routine*

|      |      |      |              |                                                  |
|------|------|------|--------------|--------------------------------------------------|
| 1115 |      | /    |              |                                                  |
| 1116 |      | /    |              |                                                  |
| 1117 |      | /    |              |                                                  |
| 1120 |      | /    |              |                                                  |
| 1121 |      | /    |              |                                                  |
| 1122 |      | /    |              |                                                  |
| 1123 |      | /    |              |                                                  |
| 1124 | 0572 | 0000 | VRSTR, 0     |                                                  |
| 1125 | 0573 | 4776 | JMS I VRSTRR | /THIS ROUTINE RESTORE 6400 FROM THE SCRATCH AREA |
| 1126 | 0574 | 4331 | VSAVE        | /CALL THE READ ROUTINE                           |
| 1127 | 0575 | 5772 | JMP I VRSTR  | /POINTER TO THE SAVE AREA                        |
| 1130 |      | /    |              | /RETURN TO THE CALLER                            |
| 1131 |      | /    |              |                                                  |
| 1132 |      | /    |              |                                                  |
| 1133 |      | /    |              |                                                  |
| 1134 | 0576 | 7774 | VRSTRR, READ | /POINTER TO THE READ ROUTINE                     |
| 1135 |      | /    |              |                                                  |
| 1136 |      | /    |              |                                                  |
| 1137 |      | /    |              |                                                  |
| 1140 |      | /    |              |                                                  |
| 1141 |      | /    |              |                                                  |
| 1142 |      | /    |              |                                                  |
| 1143 |      | /    |              |                                                  |
| 1144 |      | /    |              |                                                  |
| 1145 |      |      | EJECT        |                                                  |

-

|      |           |                          |                                                    |
|------|-----------|--------------------------|----------------------------------------------------|
| 1146 |           | /PDP-12 ASSEMBLER PAGE 3 |                                                    |
| 1147 |           | *600                     |                                                    |
| 1150 |           | /SYMBOL NUMBER DECODER   |                                                    |
| 1151 | 0600 0000 | SDEC08, 0                | /CONVERT A SYMBOL NUMBER TO AN ADDRESS             |
| 1152 | 0601 3171 | DCA SNUM                 | /SAVE THE NUMBER                                   |
| 1153 | 0602 1171 | TAD SNUM                 |                                                    |
| 1154 | 0603 0136 | AND P6000                |                                                    |
| 1155 | 0604 7640 | SZA CLA                  |                                                    |
| 1156 | 0605 5674 | JMP I ERSY               | /TOO MANY SYMS                                     |
| 1157 | 0606 1171 | TAD SNUM                 |                                                    |
| 1160 | 0607 7106 | CLL RTL                  |                                                    |
| 1161 | 0610 7041 | CIA                      |                                                    |
| 1162 | 0611 1275 | TAD STAB                 | /STAB IS TOP OF SYMBOL TABLE                       |
| 1163 | 0612 3023 | DCA SADR                 | /ADDRESS OF FIRST WORD OF SYMBOL                   |
| 1164 | 0613 7325 | PTHREE                   |                                                    |
| 1165 | 0614 1023 | TAD SADR                 |                                                    |
| 1166 | 0615 3022 | DCA VADR                 | /POINTS TO SYMBOL VALUE                            |
| 1167 | 0616 1360 | TAD COUNT                | /GET THE NUMBER OF CHARACTERS IN THE OUTPUT BUFFER |
| 1170 | 0617 7640 | SZA CLA                  | /IF NON-ZERO, SEE IF ITS OK TO PRINT ONE,          |
| 1171 | 0620 4340 | JMS CHECK                | /BY CALLING THE CHECK ROUTINE.                     |
| 1172 | 0621 5600 | JMP I SDEC08             |                                                    |
| 1173 | 0622 0000 | TYPADD, 0                | <i>changes type of symbol pointed to by SADR</i>   |
| 1174 | 0623 7200 | CLA                      |                                                    |
| 1175 | 0624 1622 | TAD I TYPADD             |                                                    |
| 1176 | 0625 3014 | DCA TYPEA                |                                                    |
| 1177 | 0626 1014 | TAD TYPEA                |                                                    |
| 1200 | 0627 3021 | DCA TYPE                 |                                                    |
| 1201 | 0630 6211 | 6211                     |                                                    |
| 1202 | 0631 1423 | TAD I SADR               |                                                    |
| 1203 | 0632 0144 | AND P1777                |                                                    |
| 1204 | 0633 3423 | DCA I SADR               |                                                    |
| 1205 | 0634 1021 | TAD TYPE                 | <i>change 1st word</i>                             |
| 1206 | 0635 0136 | AND P6000                |                                                    |
| 1207 | 0636 1423 | TAD I SADR               |                                                    |
| 1210 | 0637 3423 | DCA I SADR               |                                                    |
| 1211 | 0640 2023 | ISZ SADR                 |                                                    |
| 1212 | 0641 1423 | TAD I SADR               |                                                    |
| 1213 | 0642 0131 | AND P3777                |                                                    |
| 1214 | 0643 3423 | DCA I SADR               |                                                    |
| 1215 | 0644 1021 | TAD TYPE                 |                                                    |
| 1216 | 0645 7006 | RTL                      |                                                    |
| 1217 | 0646 0150 | AND P4000                | <i>change 2nd word</i>                             |
| 1220 | 0647 1423 | TAD I SADR               |                                                    |
| 1221 | 0650 3423 | DCA I SADR               |                                                    |
| 1222 | 0651 2023 | ISZ SADR                 |                                                    |
| 1223 | 0652 1423 | TAD I SADR               |                                                    |
| 1224 | 0653 0131 | AND P3777                |                                                    |
| 1225 | 0654 3423 | DCA I SADR               |                                                    |
| 1226 | 0655 1021 | TAD TYPE                 |                                                    |
| 1227 | 0656 7006 | RTL                      |                                                    |
| 1230 | 0657 7004 | RAL                      |                                                    |
| 1231 | 0660 0150 | AND P4000                | <i>change 3rd word</i>                             |
| 1232 | 0661 1423 | TAD I SADR               |                                                    |
| 1233 | 0662 3423 | DCA I SADR               | <i>Point back to 1st word</i>                      |
| 1234 | 0663 7344 | MTWO                     |                                                    |
| 1235 | 0664 1023 | TAD SADR                 |                                                    |
| 1236 | 0665 3023 | DCA SADR                 |                                                    |
| 1237 | 0666 1021 | TAD TYPE                 |                                                    |
| 1240 | 0667 0136 | AND P6000                |                                                    |
| 1241 | 0670 3021 | DCA TYPE                 |                                                    |
| 1242 | 0671 6201 | 6201                     |                                                    |
| 1243 | 0672 2222 | ISZ TYPADD               |                                                    |
| 1244 | 0673 5622 | JMP I TYPADD             |                                                    |

|      |      |      |            |         |        |                                                 |
|------|------|------|------------|---------|--------|-------------------------------------------------|
| 1245 | 0674 | 1175 | ERSY,STEXD |         |        |                                                 |
| 1246 | 0675 | 7600 | STAB,      | 7600    |        | /SET TO POINT TO END OF THE SYMBOL TABLE        |
| 1247 |      |      | /          |         |        |                                                 |
| 1250 |      |      | /          |         |        |                                                 |
| 1251 |      |      | /          |         |        |                                                 |
| 1252 |      |      | /          |         |        |                                                 |
| 1253 |      |      | /          |         |        |                                                 |
| 1254 |      |      | /          |         |        |                                                 |
| 1255 |      |      | /          |         |        |                                                 |
| 1256 |      |      | /          |         |        |                                                 |
| 1257 |      |      | /          |         |        |                                                 |
| 1260 |      |      | /          |         |        |                                                 |
| 1261 |      |      | /          |         |        |                                                 |
| 1262 | 0676 | 0000 | GGGET,     | 0       |        | BUFFERED TTY AND LPT ROUTINE                    |
| 1263 | 0677 | 7240 |            | CLA CMA |        | /GETS A CHARACTER FROM THE BUFFER               |
| 1264 | 0700 | 1360 | TAD        | COUNT   |        | /DECREMENT COUNT BY 1                           |
| 1265 | 0701 | 3360 | DCA        | COUNT   |        |                                                 |
| 1266 | 0702 | 1705 | TAD I      | OUTP    |        | /GET THE CHARACTER                              |
| 1267 | 0703 | 3140 | DCA        | LOWTMP  |        | /SAVE FOR A SECOND                              |
| 1270 | 0704 | 4326 | JMS        | BUMP    |        | /BUMP THE RING POINTER NOW                      |
| 1271 | 0705 | 6110 | OUTP,      | FIRST   |        | /POINTS TO THE NEXT GET FROM SPOT IN THE BUFFER |
| 1272 | 0706 | 1140 | TAD        | LOWTMP  |        | /REGET THE CHARACTER                            |
| 1273 | 0707 | 5676 | JMP I      | GGGET   |        | /AND EXIT                                       |
| 1274 |      |      | /          |         |        |                                                 |
| 1275 |      |      | /          |         |        |                                                 |
| 1276 |      |      | /          |         |        |                                                 |
| 1277 |      |      | /          |         |        |                                                 |
| 1300 |      |      | /          |         |        |                                                 |
| 1301 | 0710 | 0000 | PUT,       | 0       |        |                                                 |
| 1302 | 0711 | 3222 |            | DCA     | TYPADD | /SAVE THE CHAR IN ANOTHER TEMP LOC              |
| 1303 | 0712 | 4541 | JMS I      | PCHECK  |        | /CHECK TO STATUS OF THE DEVICES,                |
| 1304 | 0713 | 1360 | TAD        | COUNT   |        | /GET THE COUNT                                  |
| 1305 | 0714 | 1361 | TAD        | PLTST   |        | /COMPARE AGAINST THE MAXIMUM NUMBER ALLOWED     |
| 1306 | 0715 | 7650 | SNA        | CLA     |        |                                                 |
| 1307 | 0716 | 5312 | JMP        | PUT+2   |        | /BUFFER IS FULL, WAIT                           |
| 1310 | 0717 | 1222 | TAD        | TYPADD  |        | /RECALL THE DESIRED CHARACTER                   |
| 1311 | 0720 | 3722 | DCA I      | INP     |        | /STASH IN THE BUFFER                            |
| 1312 | 0721 | 4326 | JMS        | BUMP    |        | /BUMP THE BUFFER POINTER                        |
| 1313 | 0722 | 6110 | INP,       | FIRST   |        | /THE BUFFER POINTER                             |
| 1314 | 0723 | 2360 | ISZ        | COUNT   |        | /BOP UP THE COUNT                               |
| 1315 | 0724 | 4541 | JMS I      | PCHECK  |        | /SEND OUT ANOTHER CHAR IF ALL IS WELL           |
| 1316 | 0725 | 5710 | JMP I      | PUT     |        | /AND EXIT                                       |
| 1317 |      |      | /          |         |        |                                                 |
| 1320 |      |      | /          |         |        |                                                 |
| 1321 |      |      | /          |         |        |                                                 |
| 1322 |      |      | /          |         |        |                                                 |
| 1323 | 0726 | 0000 | BUMP,      | 0       |        | /THIS BUMPS THE RING BUFFER POINTER             |
| 1324 | 0727 | 2726 | ISZ I      | BUMP    |        |                                                 |
| 1325 | 0730 | 1726 | TAD I      | BUMP    |        |                                                 |
| 1326 | 0731 | 1362 | TAD        | PLOVR   |        | /TEST FOR WRAP AROUND                           |
| 1327 | 0732 | 7650 | SNA        | CLA     |        | /?                                              |
| 1330 | 0733 | 1361 | TAD        | PLTST   |        | /GET MINUS END +START                           |
| 1331 | 0734 | 1726 | TAD I      | BUMP    |        | /NOW READJUST POINTER IF NECESSARY              |
| 1332 | 0735 | 3726 | DCA I      | BUMP    |        |                                                 |
| 1333 | 0736 | 2326 | ISZ        | BUMP    |        |                                                 |
| 1334 | 0737 | 5726 | JMP I      | BUMP    |        |                                                 |
| 1335 |      |      | /          |         |        |                                                 |
| 1336 |      |      | /          |         |        |                                                 |
| 1337 |      |      | /          |         |        |                                                 |
| 1340 |      |      | /          |         |        |                                                 |
| 1341 |      |      | /          |         |        |                                                 |
| 1342 | 0740 | 0000 | CHECK,     | 0       |        | /THIS CHECKS THE OUTPUT STATUS                  |
| 1343 | 0741 | 1360 | TAD        | COUNT   |        | /GET THE NUMBER OF CHARS IN THE BUFFER          |

|      |      |      |         |            |  |                                              |
|------|------|------|---------|------------|--|----------------------------------------------|
| 1344 | 0742 | 7650 | SNA CLA |            |  |                                              |
| 1345 | 0743 | 5740 | JMP I   | CHECK      |  | /RETURN IF NO CHARS IN THE BUFFER            |
| 1346 | 0744 | 6041 | TSF     |            |  | /KEYBOARD OK???                              |
| 1347 | 0745 | 5740 | JMP I   | CHECK      |  | /NOPE, RETURN NOW                            |
| 1350 | 0746 | 6661 | LSD     |            |  | /HOW ABOUT THE LINE PRINTER                  |
| 1351 | 0747 | 5740 | JMP I   | CHECK      |  | /IF NO LINE PRINTER THIS GETS ZAPPED         |
| 1352 | 0750 | 6214 | RDF     |            |  | /READ IN THE PRESENT DATA FIELD              |
| 1353 | 0751 | 1353 | TAD     | CCDF       |  | /SAVE IT AWAY                                |
| 1354 | 0752 | 3356 | DCA     | ,+4        |  | /SAVE IT AWAY FOR A SECOND                   |
| 1355 | 0753 | 6201 | CCDF,   | 0          |  | /RESET THE DATA FIELD TO FIELD ZERO          |
| 1356 | 0754 | 4276 | JMS     | GGGET      |  | /GET A CHAR FROM THE BUFFER                  |
| 1357 | 0755 | 4763 | JMS I   | CTYPO      |  | /NOW OUTPUT IT RIGHT AWAY,                   |
| 1360 | 0756 | 0000 | 0000    |            |  | /RETURN CDF GOES HERE                        |
| 1361 | 0757 | 5740 | JMP I   | CHECK      |  | /NOW RETURN TO THE CALLER                    |
| 1362 |      |      | /       |            |  |                                              |
| 1363 |      |      | /       |            |  |                                              |
| 1364 |      |      | /       |            |  |                                              |
| 1365 |      |      | /       |            |  |                                              |
| 1366 |      |      | /       |            |  |                                              |
| 1367 |      |      | /       |            |  |                                              |
| 1370 | 0760 | 0000 | COUNT,  | 0          |  | /NUMBER OF CHARACTERS NOW IN THE BUFFER      |
| 1371 | 0761 | 7510 | PLTST,  | -BAD+FIRST |  |                                              |
| 1372 | 0762 | 1400 | PLOVR,  | -BAD       |  | /-UPPER LIMIT                                |
| 1373 | 0763 | 2527 | CTYPO,  | TYPO       |  | /ADDRESS OF THE ACTUAL PUTTER                |
| 1374 |      |      | /       |            |  |                                              |
| 1375 |      |      | /       |            |  |                                              |
| 1376 |      |      | /       |            |  |                                              |
| 1377 |      |      | /       |            |  |                                              |
| 1400 |      |      | /       |            |  |                                              |
| 1401 |      |      | /       |            |  |                                              |
| 1402 |      |      | /       |            |  |                                              |
| 1403 |      |      | /       |            |  |                                              |
| 1404 | 0764 | 0000 | SKIP2,  | 0          |  |                                              |
| 1405 | 0765 | 3200 | DCA     | SDEC08     |  | /SAVE SKIP COUNT                             |
| 1406 | 0766 | 1134 | TAD     | P240       |  |                                              |
| 1407 | 0767 | 4401 | JMS I   | DECOU      |  |                                              |
| 1410 | 0770 | 2200 | ISZ     | SDEC08     |  |                                              |
| 1411 | 0771 | 5366 | JMP     | , -3       |  |                                              |
| 1412 | 0772 | 5764 | JMP I   | SKIP2      |  |                                              |
| 1413 |      |      | /       |            |  |                                              |
| 1414 |      |      | /       |            |  |                                              |
| 1415 |      |      | /       |            |  |                                              |
| 1416 |      |      | /       |            |  |                                              |
| 1417 |      |      | /       |            |  |                                              |
| 1420 |      |      | /       |            |  |                                              |
| 1421 |      |      | /       |            |  |                                              |
| 1422 |      |      | /       |            |  |                                              |
| 1423 |      |      | /       |            |  |                                              |
| 1424 |      |      | /       |            |  |                                              |
| 1425 |      |      | /       |            |  |                                              |
| 1426 |      |      | /       |            |  |                                              |
| 1427 | 0773 | 0000 | VPRSRV, | 0          |  | /THIS ROUTINE PRESERVES LOCATIONS 6400-67777 |
| 1430 | 0774 | 4777 | JMS I   | VPRSRW     |  | /IN THE SCRATCH AREA, CALL THE WRITE ROUTINE |
| 1431 | 0775 | 4331 | VSAVE   |            |  |                                              |
| 1432 | 0776 | 5773 | JMP I   | VPRSRV     |  | /AND RETURN TO THE CALLER,                   |
| 1433 |      |      | /       |            |  |                                              |
| 1434 |      |      | /       |            |  |                                              |
| 1435 |      |      | /       |            |  |                                              |
| 1436 |      |      | /       |            |  |                                              |
| 1437 | 0777 | 7775 | VPRSRW, | WRITE      |  | /POINTER TO THE WRITE ROUTINE,               |
| 1440 |      |      | /       |            |  |                                              |
| 1441 |      |      | /       |            |  |                                              |
| 1442 |      |      | /       |            |  |                                              |

1443  
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1453  
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/

EJECT

|      |      |      |                          |
|------|------|------|--------------------------|
| 1454 |      |      | *1000                    |
| 1455 |      |      | SCOLON=200               |
| 1456 |      |      | COMMA=600                |
| 1457 |      |      | SYMB=4000                |
| 1460 |      |      | EQUALS=1400              |
| 1461 |      |      | USYMB=2000               |
| 1462 |      |      | EXCLAM=2200              |
| 1463 |      |      | ANDS=3400                |
| 1464 |      |      | PLUS=4600                |
| 1465 |      |      | MINUS=5200               |
| 1466 |      |      | SPACE=5600               |
| 1467 |      |      | ALL=7600                 |
| 1470 |      |      | /PDP-12 ASSEMBLER PAGE 4 |
| 1471 | 1000 | 0000 | EXPR, 0                  |
| 1472 | 1001 | 1164 | TAD CON                  |
| 1473 | 1002 | 4505 | JMS I PUSH1              |
| 1474 | 1003 | 3164 | DCA CON                  |
| 1475 | 1004 | 1055 | TAD INST                 |
| 1476 | 1005 | 4505 | JMS I PUSH1              |
| 1477 | 1006 | 3055 | DCA INST                 |
| 1500 | 1007 | 1054 | TAD MICROL               |
| 1501 | 1010 | 4505 | JMS I PUSH1              |
| 1502 | 1011 | 3054 | DCA MICROL               |
| 1503 | 1012 | 1034 | TAD SGN                  |
| 1504 | 1013 | 4505 | JMS I PUSH1              |
| 1505 | 1014 | 1365 | TAD CRIOR                |
| 1506 | 1015 | 3034 | DCA SGN                  |
| 1507 | 1016 | 1275 | TAD H6                   |
| 1510 | 1017 | 4505 | JMS I PUSH1              |
| 1511 | 1020 | 3275 | DCA H6                   |
| 1512 | 1021 | 1200 | TAD EXPR                 |
| 1513 | 1022 | 4505 | JMS I PUSH1              |
| 1514 | 1023 | 7410 | SKP                      |
| 1515 | 1024 | 4504 | A, JMS I IDX1            |
| 1516 | 1025 | 4507 | JMS I GETSY1             |
| 1517 | 1026 | 2076 | B-1000+USYMB             |
| 1520 | 1027 | 4076 | B-1000+SYMB              |
| 1521 | 1030 | 5624 | A-1000+SPACE             |
| 1522 | 1031 | 4655 | C-1000+PLUS              |
| 1523 | 1032 | 5260 | C+3-1000+MINUS           |
| 1524 | 1033 | 2262 | C+5-1000+EXCLAM          |
| 1525 | 1034 | 0313 | G-1000+SCOLON            |
| 1526 | 1035 | 3464 | C+7-1000+ANDS            |
| 1527 | 1036 | 7713 | G-1000+ALL               |
| 1530 | 1037 | 1024 | TAD VAL                  |
| 1531 | 1040 | 5243 | JMP ,+3                  |
| 1532 | 1041 | 7200 | B1, CLA                  |
| 1533 | 1042 | 1053 | TAD LWC                  |
| 1534 | 1043 | 4434 | JMS I SGN                |
| 1535 | 1044 | 7410 | SKP                      |
| 1536 | 1045 | 4504 | JMS I IDX1               |
| 1537 | 1046 | 4507 | B2, JMS I GETSY1         |
| 1540 | 1047 | 5662 | C+5-1000+SPACE           |
| 1541 | 1050 | 0707 | GET=1-1000+COMMA         |
| 1542 | 1051 | 4037 | B1-2-1000+SYMB           |
| 1543 | 1052 | 2037 | B1-2-1000+USYMB          |
| 1544 | 1053 | 1540 | ERONE-1000+EQUALS        |
| 1545 | 1054 | 0031 | A+5-1000                 |
| 1546 | 1055 | 1366 | C, TAD CRIOR+1           |
| 1547 | 1056 | 3034 | DCA SGN                  |
| 1550 | 1057 | 5245 | JMP B2-1                 |
| 1551 | 1060 | 1367 | TAD CRIOR+2              |
| 1552 | 1061 | 5256 | JMP C+1                  |

*expression processor (recursive)*

*/save accumulated value*

*/save last operand address*

*/OR - set up initially to do an IOR*

*/save indirect bit*

*/save exit*

*/IE error*

*5B.4 Branch ADDR + 7bit char code  
B - 10000 + USYMB*

|      |      |      |                   |      |
|------|------|------|-------------------|------|
| 1553 | 1062 | 1365 | TAD CR10R         | /OR  |
| 1554 | 1063 | 5256 | JMP C+1           |      |
| 1555 | 1064 | 1370 | TAD CR10R+3       | /AND |
| 1556 | 1065 | 5256 | JMP C+1           |      |
| 1557 | 1066 | 1037 | TAD MODE          |      |
| 1560 | 1067 | 7700 | SMA CLA           |      |
| 1561 | 1070 | 5304 | JMP LINK1         |      |
| 1562 | 1071 | 1124 | TAD P400          |      |
| 1563 | 1072 | 1275 | TAD H6            |      |
| 1564 | 1073 | 3275 | DCA H6            |      |
| 1565 | 1074 | 5225 | JMP A+1           |      |
| 1566 | 1075 | 0000 | 0                 |      |
| 1567 | 1076 | 1024 | TAD VAL           |      |
| 1570 | 1077 | 4434 | JMS I SGN         |      |
| 1571 | 1100 | 4507 | JMS I GETSY1      |      |
| 1572 | 1101 | 1542 | EQU-1000+EQUALS   |      |
| 1573 | 1102 | 5710 | GET-1000+SPACE    |      |
| 1574 | 1103 | 0047 | 82+1-1000         |      |
| 1575 | 1104 | 1373 | TAD P10           |      |
| 1576 | 1105 | 1373 | TAD P10           |      |
| 1577 | 1106 | 5272 | JMP H+4           |      |
| 1600 | 1107 | 4774 | JMS I DEFIN1      |      |
| 1601 | 1110 | 4772 | JMS I GETAD1      |      |
| 1602 | 1111 | 1165 | TAD ADR           |      |
| 1603 | 1112 | 4765 | JMS I CR10R       |      |
| 1604 | 1113 | 4506 | JMS I POPU1       |      |
| 1605 | 1114 | 3200 | DCA EXPR          |      |
| 1606 | 1115 | 1164 | TAD CON           |      |
| 1607 | 1116 | 3033 | DCA EVAL          |      |
| 1610 | 1117 | 1275 | TAD H6            |      |
| 1611 | 1120 | 3166 | DCA IND           |      |
| 1612 | 1121 | 4506 | JMS I POPU1       |      |
| 1613 | 1122 | 3275 | DCA H6            |      |
| 1614 | 1123 | 4506 | JMS I POPU1       |      |
| 1615 | 1124 | 3034 | DCA SGN           |      |
| 1616 | 1125 | 1054 | TAD MICROL        |      |
| 1617 | 1126 | 3005 | DCA MICROB        |      |
| 1620 | 1127 | 4506 | JMS I POPU1       |      |
| 1621 | 1130 | 3054 | DCA MICROL        |      |
| 1622 | 1131 | 1055 | TAD INST          |      |
| 1623 | 1132 | 3056 | DCA MICRO         |      |
| 1624 | 1133 | 4506 | JMS I POPU1       |      |
| 1625 | 1134 | 3055 | DCA INST          |      |
| 1626 | 1135 | 4506 | JMS I POPU1       |      |
| 1627 | 1136 | 3164 | DCA CON           |      |
| 1630 | 1137 | 5600 | JMP I EXPR        |      |
| 1631 |      |      |                   |      |
| 1632 | 1140 | 5741 | ERONE, JMP I .+1  |      |
| 1633 | 1141 | 3766 | EQUERR            |      |
| 1634 |      |      |                   |      |
| 1635 | 1142 | 1771 | EQU, TAD I REPUN1 |      |
| 1636 | 1143 | 3364 | DCA CR10R-1       |      |
| 1637 | 1144 | 3006 | DCA EQRETI        |      |
| 1640 | 1145 | 4504 | JMS I IDX1        |      |
| 1641 | 1146 | 4477 | JMS I EXPR1       |      |
| 1642 | 1147 | 7240 | CLA CMA           |      |
| 1643 | 1150 | 3006 | DCA EQRETI        |      |
| 1644 | 1151 | 1364 | TAD CR10R-1       |      |
| 1645 | 1152 | 4512 | JMS I SDECO1      |      |
| 1646 | 1153 | 1033 | TAD EVAL          |      |
| 1647 | 1154 | 6211 | 6211              |      |
| 1650 | 1155 | 3422 | DCA I VADR        |      |
| 1651 | 1156 | 6201 | 6201              |      |

*check what follows a symbol*

*restore exit*

*restore indirect bit*

*restore last operator*

*same current symbol type*

*restore previous symbol type*

*Routine to handle "="*

*SGN holds the address of the routine which handles the operation indicated by the last operator.*

|      |      |      |                  |
|------|------|------|------------------|
| 1652 | 1157 | 4511 | JMS I TYPAD1     |
| 1653 | 1160 | 4000 | SYMB             |
| 1654 | 1161 | 4465 | JMS I PASSEK     |
| 1655 | 1162 | 4466 | JMS I CURSKA     |
| 1656 | 1163 | 5501 | JMP I MAIN1      |
| 1657 | 1164 | 0000 | 0                |
| 1660 | 1165 | 1600 | CR IOR, RIOR     |
| 1661 | 1166 | 4235 | RADD             |
| 1662 | 1167 | 4211 | RSUB             |
| 1663 | 1170 | 3713 | RAND             |
| 1664 | 1171 | 1747 | REPUN1, RESTOR-2 |
| 1665 | 1172 | 1200 | GETAD1, GETADR   |
| 1666 | 1173 | 0010 | PI0, 10          |
| 1667 | 1174 | 0227 | DEFIN1, CURREN   |
| 1670 | 1175 | 1157 | STEXD,TAD P7     |
| 1671 | 1176 | 4476 | JMS I ERR1       |
| 1672 | 1177 | 5517 | JMP I MON        |
| 1673 |      |      | EJECT            |

|      |      |      |                          |
|------|------|------|--------------------------|
| 1674 |      |      | /PDP-12 ASSEMBLER PAGE 5 |
| 1675 |      |      | *1200                    |
| 1676 | 1200 | 0000 | GETADR, 0                |
| 1677 | 1201 | 4504 | JMS I IDX1               |
| 1700 | 1202 | 1200 | TAD GETADR               |
| 1701 | 1203 | 4505 | JMS I PUSH1              |
| 1702 | 1204 | 4477 | JMS I EXPR1              |
| 1703 | 1205 | 1037 | TAD MODE                 |
| 1704 | 1206 | 7700 | SMA CLA                  |
| 1705 | 1207 | 5250 | JMP LNKOR                |
| 1706 | 1210 | 1056 | TAD MICRO                |
| 1707 | 1211 | 7700 | SMA CLA                  |
| 1710 | 1212 | 5222 | JMP CHKLA                |
| 1711 | 1213 | 1164 | TAD CON                  |
| 1712 | 1214 | 7106 | CLL RTL                  |
| 1713 | 1215 | 7006 | RTL                      |
| 1714 | 1216 | 0157 | AND P7                   |
| 1715 | 1217 | 1261 | TAD M6A                  |
| 1716 | 1220 | 7700 | SMA CLA                  |
| 1717 | 1221 | 5257 | JMP LEVEL+1              |
| 1720 | 1222 | 1033 | CHKLA, TAD EVAL          |
| 1721 | 1223 | 0146 | AND P7600                |
| 1722 | 1224 | 7450 | SNA                      |
| 1723 | 1225 | 5242 | JMP AWAY+1               |
| 1724 | 1226 | 7041 | CIA                      |
| 1725 | 1227 | 1031 | TAD APAGE                |
| 1726 | 1230 | 7650 | SNA CLA                  |
| 1727 | 1231 | 5241 | JMP AWAY                 |
| 1730 | 1232 | 7001 | IAC                      |
| 1731 | 1233 | 4662 | JMS I NUPAGE             |
| 1732 | 1234 | 5236 | JMP LEAVE                |
| 1733 | 1235 | 3165 | DCA ADR                  |
| 1734 | 1236 | 4506 | LEAVE, JMS I POPU1       |
| 1735 | 1237 | 3200 | DCA GETADR               |
| 1736 | 1240 | 5600 | JMP I GETADR             |
| 1737 | 1241 | 1147 | AWAY, TAD P200           |
| 1740 | 1242 | 3007 | DCA TEMP                 |
| 1741 | 1243 | 1033 | TAD EVAL                 |
| 1742 | 1244 | 0145 | AND P177                 |
| 1743 | 1245 | 1007 | TAD TEMP                 |
| 1744 | 1246 | 1166 | TAD IND                  |
| 1745 | 1247 | 5235 | JMP LEAVE-1              |
| 1746 | 1250 | 1005 | LNKOR, TAD MICROB        |
| 1747 | 1251 | 7700 | SMA CLA                  |
| 1750 | 1252 | 5264 | JMP CHKL                 |
| 1751 | 1253 | 1033 | TAD EVAL                 |
| 1752 | 1254 | 0144 | AND P1777                |
| 1753 | 1255 | 1166 | TAD IND                  |
| 1754 | 1256 | 3033 | LEVEL, DCA EVAL          |
| 1755 | 1257 | 1033 | TAD EVAL                 |
| 1756 | 1260 | 5235 | JMP LEAVE-1              |
| 1757 | 1261 | 7772 | M6A, -6                  |
| 1760 | 1262 | 2317 | NUPAGE, UNDERR           |
| 1761 | 1263 | 0000 | 0                        |
| 1762 | 1264 | 1014 | CHKL, TAD TYPEA          |
| 1763 | 1265 | 7006 | RTL                      |
| 1764 | 1266 | 0136 | AND P6000                |
| 1765 | 1267 | 7640 | SZA CLA                  |
| 1766 | 1270 | 5253 | JMP LEVEL-3              |
| 1767 | 1271 | 1033 | TAD EVAL                 |
| 1770 | 1272 | 1166 | TAD IND                  |
| 1771 | 1273 | 5256 | JMP LEVEL                |
| 1772 | 1274 | 0000 | LDT, 0                   |

*assembles address portion of word*

*1st expression for address*

*1st inc mode*

*1st page 0*

*1st error*

*1st store exit*

*1st assemble address*

*1st not MRI*

*1st line mode def.*

*1st, 1st mode*  
*1st, 1st mode*

*TAD TYPEA*  
*AND P 1400*  
*SZA CLA*

|      |      |      |                         |
|------|------|------|-------------------------|
| 1773 | 1275 | 1044 | TAD CHARX               |
| 1774 | 1276 | 7450 | SNA                     |
| 1775 | 1277 | 5674 | JMP I LDT               |
| 1776 | 1300 | 3263 | DCA NUPAGE+1            |
| 1777 | 1301 | 1263 | TAD NUPAGE+1            |
| 2000 | 1302 | 1125 | TAD M45                 |
| 2001 | 1303 | 7700 | SMA CLA                 |
| 2002 | 1304 | 5311 | JMP NOTLDT              |
| 2003 | 1305 | 2050 | ISZ IN /LETTER OR DIGIT |
| 2004 | 1306 | 4504 | JMS I IDX1              |
| 2005 | 1307 | 1263 | TAD NUPAGE+1            |
| 2006 | 1310 | 5674 | JMP I LDT               |
| 2007 | 1311 | 3044 | NOTLDT, DCA CHARX       |
| 2010 | 1312 | 5674 | JMP I LDT               |
| 2011 | 1313 | 0000 | A1W, 0                  |
| 2012 | 1314 | 4274 | JMS LDT                 |
| 2013 | 1315 | 3263 | DCA NUPAGE+1            |
| 2014 | 1316 | 1263 | TAD NUPAGE+1            |
| 2015 | 1317 | 7106 | CLL RTL                 |
| 2016 | 1320 | 3024 | DCA VAL /CHR, *4        |
| 2017 | 1321 | 1024 | TAD VAL                 |
| 2020 | 1322 | 7006 | RTL                     |
| 2021 | 1323 | 7004 | RAL /CHR, *32           |
| 2022 | 1324 | 1024 | TAD VAL                 |
| 2023 | 1325 | 1263 | TAD NUPAGE+1 /CHR *1    |
| 2024 | 1326 | 3024 | DCA VAL                 |
| 2025 | 1327 | 4274 | JMS LDT                 |
| 2026 | 1330 | 1024 | TAD VAL                 |
| 2027 | 1331 | 5713 | JMP I A1W               |
| 2030 | 1332 | 0000 | AAS, 0                  |
| 2031 | 1333 | 4313 | JMS A1W                 |
| 2032 | 1334 | 3161 | DCA TEM1                |
| 2033 | 1335 | 4313 | JMS A1W                 |
| 2034 | 1336 | 3162 | DCA TEM2                |
| 2035 | 1337 | 4313 | JMS A1W                 |
| 2036 | 1340 | 3163 | DCA TEM3                |
| 2037 | 1341 | 4274 | JMS LDT                 |
| 2040 | 1342 | 7650 | SNA CLA                 |
| 2041 | 1343 | 5732 | JMP I AAS               |
| 2042 | 1344 | 5341 | JMP -3                  |
| 2043 | 1345 | 4504 | ORGIC, JMS I IDX1       |
| 2044 | 1346 | 4477 | JMS I EXPR1             |
| 2045 | 1347 | 1037 | TAD MODE                |
| 2046 | 1350 | 7700 | SMA CLA                 |
| 2047 | 1351 | 5370 | JMP LINKIC              |
| 2050 | 1352 | 1033 | TAD EVAL                |
| 2051 | 1353 | 3030 | DCA AADR                |
| 2052 | 1354 | 1033 | TAD EVAL                |
| 2053 | 1355 | 0146 | AND P7600               |
| 2054 | 1356 | 7041 | CIA                     |
| 2055 | 1357 | 1031 | TAD APAGE               |
| 2056 | 1360 | 7650 | SNA CLA                 |
| 2057 | 1361 | 5365 | JMP LNRT                |
| 2060 | 1362 | 4502 | JMS I PAGE1             |
| 2061 | 1363 | 1030 | TAD AADR                |
| 2062 | 1364 | 3033 | DCA EVAL                |
| 2063 | 1365 | 4465 | LNRT, JMS I PASSEK      |
| 2064 | 1366 | 4466 | JMS I CURSKA            |
| 2065 | 1367 | 5501 | JMP I MAIN1             |
| 2066 | 1370 | 1033 | LINKIC, TAD EVAL        |
| 2067 | 1371 | 0144 | AND P1777               |
| 2070 | 1372 | 1046 | TAD ABANK               |
| 2071 | 1373 | 3030 | DCA AADR                |

*Handwritten notes:*  
 "Handle" (under 2043)  
 "\*" (under 2042)

2072  
2073  
-

1374 5365

JMP LNRT  
EJECT

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2140 /

END OF ASSEMBLY!!!!!!!!!!!!  
- - - - -

CHAIN "ASSEM2" /GO GET THE SECOND PART NOW,

|      |     |                        |
|------|-----|------------------------|
| 0000 | *20 |                        |
| 0001 |     | PHODE                  |
| 0002 | /   |                        |
| 0003 | /   |                        |
| 0004 | /   |                        |
| 0005 | /   |                        |
| 0006 | /   |                        |
| 0007 | /   |                        |
| 0010 | /   |                        |
| 0011 | /   |                        |
| 0012 | /   |                        |
| 0013 | /   |                        |
| 0014 | /   |                        |
| 0015 | /   |                        |
| 0016 | /   |                        |
| 0017 | /   | BEGINNING OF ASSEMTWO, |
| 0020 | /   |                        |
| 0021 | /   |                        |
| 0022 | /   |                        |
| 0023 | /   |                        |
| 0024 | /   |                        |
| 0025 | /   |                        |
| 0026 | /   |                        |
| 0027 | /   |                        |
| 0030 | /   |                        |

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EJECT

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0043 /
0044 /
0045 /
0046 /
0047 /
0050 /PDP-12 ASSEMBLER PAGE 6
0051 /
0052 *1400
0053 /
0054 1400 5201 MAIN, JMP PASS1
0055 1401 1032 PASS1, TAD PERMA
0056 1402 3027 DCA SEND
0057 1403 3035 DCA PASS
0060 1404 3037 DCA MODE
0061 1405 4320 JMS SETORG
0062 1406 1350 TAD CJMP1+1
0063 1407 3200 PASS1P, DCA MAIN
0064 1410 4735 JMS I OCTS1
0065 1411 4530 MAINSE, JMS I LINENO
0066 1412 7240 CLA CMA
0067 1413 3777 DCA I PTOCHC
0070 1414 4404 JMS I CNTLP1
0071 1415 4653 JMS I CLENUP
0072 1416 4504 MAINSF, JMS I IDX1
0073 1417 1376 TAD M60
0074 1420 3173 DCA SPCUNT
0075 1421 1357 TAD LIST1
0076 1422 3047 DCA POINT
0077 1423 3020 DCA ANY
0100 1424 3166 DCA IND
0101 1425 5775 JMP I PSINTER
0102 1426 4477 DOCAL, JMS I EXPR1
0103 1427 1020 TAD ANY
0104 1430 7650 SNA CLA
0105 1431 5304 JMP MAINSA
0106 1432 1033 TAD EVAL
0107 1433 4236 JMS ONEREG
0110 1434 4461 TACK, JMS I EMPTY1
0111 1435 5211 JMP MAINSE
0112 1436 0000 ONEREG, 0
0113 1437 3033 DCA EVAL
0114 1440 4520 JMS I PUNON1
0115 1441 2030 ISZ AADR
0116 1442 1037 TAD MODE
0117 1443 7700 SMA CLA
0120 1444 5252 JMP LNKREG
0121 1445 1030 TAD AADR
0122 1446 0145 AND P177
0123 1447 7640 SZA CLA
0124 1450 5636 JMP I ONEREG
0125 1451 4502 JMS I PAGE1
0126 1452 5636 LNKREG, JMP I ONEREG
0127 1453 2132 CLENUP, CLEARR
0130 /SYMBOLIC ASSEMBLY SAVE ROUTINE
0131 LSTSYM, 0
0132 1454 0000 TAD LISTWD
0133 1455 1003 SPA CLA
0134 1456 7710 JMP I LSTSYM
0135 1457 5654 TAD ITEM
0136 1460 1051 TAD M43
0137 1461 1122 SNA CLA
0140 1462 7650 JMP I LSTSYM
0141 1463 5654 TAD ITEM
1464 1051

```

/SWITCH

*1 case all but perm. syms.*

/SET TADIX TO OCTAL

/RESET THE COMMA SWITCH,

*main loop, passed thru  
each time a terminator  
found.*

/NOTHING TO BE ASSEMBLED

*1 output one word (bin)*

/END OF PAGE

|      |      |      |                   |
|------|------|------|-------------------|
| 0142 | 1465 | 2356 | ISZ SWOT          |
| 0143 | 1466 | 5274 | JMP DOLEFT        |
| 0144 | 1467 | 1703 | TAD I SYMBU       |
| 0145 | 1470 | 3703 | DCA I SYMBU       |
| 0146 | 1471 | 2303 | ISZ SYMBU         |
| 0147 | 1472 | 2057 | LSRTE, ISZ CHARAC |
| 0150 | 1473 | 5654 | JMP I LSTSYM      |
| 0151 | 1474 | 7106 | DOLEFT, CLL RTL   |
| 0152 | 1475 | 7006 | RTL               |
| 0153 | 1476 | 7006 | RTL               |
| 0154 | 1477 | 3703 | DCA I SYMBU       |
| 0155 | 1500 | 7040 | CMA               |
| 0156 | 1501 | 3356 | DCA SWOT          |
| 0157 | 1502 | 5272 | JMP LSRTE         |
| 0160 | 1503 | 3102 | SYMBU, SYMBUF     |
| 0161 | 1504 | 1035 | MAINSA, TAD PASS  |
| 0162 | 1505 | 7050 | SNA CLA           |
| 0163 | 1506 | 5211 | JMP MAINSE        |
| 0164 | 1507 | 1003 | TAD LISTWO        |
| 0165 | 1510 | 7710 | SPA CLA           |
| 0166 | 1511 | 5211 | JMP MAINSE        |
| 0167 | 1512 | 1167 | TAD ERRCNT        |
| 0170 | 1513 | 7640 | SZA CLA           |
| 0171 | 1514 | 5234 | JMP TACK          |
| 0172 | 1515 | 4465 | JMS I PASSEK      |
| 0173 | 1516 | 4466 | JMS I CURSKA      |
| 0174 | 1517 | 5211 | JMP MAINSE        |
| 0175 | 1520 | 0000 | SETORG, 0         |
| 0176 | 1521 | 1150 | TAD P4000         |
| 0177 | 1522 | 3030 | DCA AADR          |
| 0200 | 1523 | 1150 | TAD P4000         |
| 0201 | 1524 | 3031 | DCA APAGE         |
| 0202 | 1525 | 1150 | TAD P4000         |
| 0203 | 1526 | 3033 | DCA EVAL          |
| 0204 | 1527 | 1150 | TAD P4000         |
| 0205 | 1530 | 3046 | DCA ABANK         |
| 0206 | 1531 | 3002 | DCA ERTOT         |
| 0207 | 1532 | 3036 | DCA FLOWD         |
| 0210 | 1533 | 3167 | DCA ERRCNT        |
| 0211 | 1534 | 5720 | JMP I SETORG      |
| 0212 | 1535 | 2235 | OCTS1, OCTS       |
| 0213 | 1536 | 3002 | PUNONE+1          |
| 0214 | 1537 | 7240 | PASS2, CLA CMA    |
| 0215 | 1540 | 3035 | DCA PASS          |
| 0216 | 1541 | 1121 | TAD M1000         |
| 0217 | 1542 | 3736 | DCA I OCTS1+1     |
| 0220 | 1543 | 4320 | JMS SETORG        |
| 0221 | 1544 | 4460 | JMS I INIT1       |
| 0222 | 1545 | 1347 | TAD CJMP1         |
| 0223 | 1546 | 5207 | JMP PASS1P        |
| 0224 | 1547 | 5201 | CJMP1, JMP PASS1  |
| 0225 | 1550 | 5337 | JMP PASS2         |
| 0226 | 1551 | 0000 | RESET, 0          |
| 0227 | 1552 | 1030 | TAD AADR          |
| 0230 | 1553 | 0146 | AND P7600         |
| 0231 | 1554 | 3031 | DCA APAGE         |
| 0232 | 1555 | 5751 | JMP I RESET       |
| 0233 | 1556 | 0000 | SWOT, 0           |
| 0234 | 1557 | 2731 | LIST1, LISTA      |
| 0235 | 1560 | 0000 | LNINST, 0         |
| 0236 | 1561 | 1055 | TAD INST          |
| 0237 | 1562 | 7700 | SMA CLA           |
| 0240 | 1563 | 5372 | JMP ,+7           |

*/initialize origins*

*set up for pass 2*

*/sic if symbol is perm. or even lower*

|      |      |      |                |        |
|------|------|------|----------------|--------|
| 0241 | 1564 | 1024 | TAD            | VAL    |
| 0242 | 1565 | 0136 | AND            | P6000  |
| 0243 | 1566 | 7650 | SNA            | CLA    |
| 0244 | 1567 | 5372 | JMP            | .+3    |
| 0245 | 1570 | 7040 | CMA            |        |
| 0246 | 1571 | 3054 | DCA            | MICROL |
| 0247 | 1572 | 1025 | TAD            | VALK   |
| 0250 | 1573 | 3024 | DCA            | VAL    |
| 0251 | 1574 | 5760 | JMP            | LNINST |
| 0252 | 1575 | 1426 | PSINTER, DOCAL |        |
| 0253 | 1576 | 7716 | M60,           | -62    |
| 0254 | 1577 | 0260 | PTOCHC, COMMAN |        |
| 0255 |      |      | /              |        |
| 0256 |      |      | /              |        |
| 0257 |      |      | /              |        |
| 0260 |      |      | EJECT          |        |

*Instruction*

*Set value of symbol*

```

0261      /PDP-12 ASSEMBLER PAGE 7
0262      *1600
0263      1600 0000 RIOR, 0
0264      1601 3024      DCA      VAL
0265      1602 1164      TAD      CON
0266      1603 7040      CMA
0267      1604 0024      AND      VAL
0270      1605 1164      TAD      CON
0271      1606 3164      DCA      CON
0272      1607 5600      JMP I    RIOR
0273      /
0274      /
0275      1610 4703 STRING, JMS I  GENINI
0276      1611 7041      CIA
0277      1612 3024      DCA      VAL
0300      1613 4241      JMS      STGET
0301      1614 5273      JMP      STNODUMP
0302      1615 5224      JMP      STINTO
0303      1616 4241 STLOOP, JMS      STGET
0304      1617 5267      JMP      STDUMP
0305      1620 1033      TAD      EVAL
0306      1621 4704      JMS I    ONEREI
0307      1622 4465      JMS I    PASSEK
0310      1623 4500      JMS I    TYCARI
0311      1624 1051 STINTO, TAD      ITEM
0312      1625 0100      AND      P77
0313      1626 7106      CLL      RTL
0314      1627 7006      RTL
0315      1630 7006      RTL
0316      1631 3033      DCA      EVAL
0317      1632 4241      JMS      STGET
0320      1633 5267      JMP      STDUMP
0321      1634 1051      TAD      ITEM
0322      1635 0100      AND      P77
0323      1636 1033      TAD      EVAL
0324      1637 3033      DCA      EVAL
0325      1640 5216      JMP      STLOOP
0326      /
0327      1641 0000 STGET, 0
0330      1642 4703      JMS I    GENINI
0331      1643 1024      TAD      VAL
0332      1644 7640      SZA CLA
0333      1645 2241      ISZ      STGET
0334      1646 1122      TAD      M43
0335      1647 1051      TAD      ITEM
0336      1650 7640      SZA CLA
0337      1651 5261      JMP      STNOT
0340      1652 4465      JMS I    PASSEK
0341      1653 4466      JMS I    CURSKA
0342      1654 4532      JMS I    SCLEAR
0343      1655 1705      TAD I    SPT060
0344      1656 3173      DCA      SPCUNT
0345      1657 4530      JMS I    LINENO
0346      1660 5641      JMP I    STGET
0347      1661 1051 STNOT, TAD      ITEM
0350      ASMIFZ TERM44
0351      TAD      STM44
0352      1662 7640      SZA CLA
0353      1663 5641      JMP I    STGET
0354      1664 4702      JMS I    SQERR
0355      1665 4702      JMS I    SQERR
0356      1666 5517      JMP I    MON
0357      /

```

DO  
inclusive OR

/Text pseudo-op

/GET THE CHAR,  
/DO WE WANT A DOLLAR SIGN AT THE END,  
/YEP, ADD IT IN,

|      |      |      |                 |            |
|------|------|------|-----------------|------------|
| 0360 | 1667 | 1033 | STDUMP, TAD     | EVAL       |
| 0361 | 1670 | 4704 | JMS I           | ONEREI     |
| 0362 | 1671 | 4465 | JMS I           | PASSEK     |
| 0363 | 1672 | 4500 | JMS I           | TYCARI     |
| 0364 | 1673 | 1051 | STNODUMP, TAD   | ITEM       |
| 0365 | 1674 | 1122 | TAD             | M43        |
| 0366 | 1675 | 7650 | SNA CLA         |            |
| 0367 | 1676 | 5701 | JMP I           | SQFDRY     |
| 0370 | 1677 | 4703 | JMS I           | GENINI     |
| 0371 | 1700 | 5274 | JMP             | STNODUMP+1 |
| 0372 |      |      | /               |            |
| 0373 |      |      | /               |            |
| 0374 |      |      | /               |            |
| 0375 |      |      | ASMIFZ          | TERMC-44   |
| 0376 |      |      | STM44, -44      |            |
| 0377 | 1701 | 2714 | SQFDRY, FORT    |            |
| 0400 | 1702 | 3201 | SQERR, ERROR    |            |
| 0401 | 1703 | 3670 | GENINI, GETIN   |            |
| 0402 | 1704 | 1436 | ONEREI, ONEREG  |            |
| 0403 | 1705 | 1576 | SPT060, M60     |            |
| 0404 |      |      | /               |            |
| 0405 | 1706 | 0000 | JBTEST, 0       |            |
| 0406 | 1707 | 2012 | JBMSR, ISZ      | CURLIN     |
| 0407 | 1710 | 7300 | CLA CLL         |            |
| 0410 | 1711 | 7240 | CLA CMA         |            |
| 0411 | 1712 | 3003 | DCA             | LISTWD     |
| 0412 | 1713 | 1332 | TAD             | JBSMF      |
| 0413 | 1714 | 3307 | DCA             | JBMSR      |
| 0414 | 1715 | 1012 | TAD             | CURLIN     |
| 0415 | 1716 | 7421 | 7421            |            |
| 0416 | 1717 | 7501 | 7501            |            |
| 0417 | 1720 | 7161 | CLL CML CMA IAC |            |
| 0420 | 1721 | 1333 | TAD             | JSTART     |
| 0421 | 1722 | 7660 | SNL SZA CLA     |            |
| 0422 | 1723 | 5706 | JMP I           | JBTEST     |
| 0423 | 1724 | 1334 | TAD             | JEND       |
| 0424 | 1725 | 7161 | CLL CML CMA IAC |            |
| 0425 | 1726 | 1012 | TAD             | CURLIN     |
| 0426 | 1727 | 7670 | SZL SNA CLA     |            |
| 0427 | 1730 | 3003 | JNOP, DCA       | LISTWD     |
| 0430 | 1731 | 5706 | JMP I           | JBTEST     |
| 0431 |      |      | /               |            |
| 0432 |      |      | /               |            |
| 0433 | 1732 | 2012 | JBSMF, ISZ      | CURLIN     |
| 0434 | 1733 | 0000 | JSTART, 0       |            |
| 0435 | 1734 | 7777 | JEND, 7777      |            |
| 0436 | 1735 | 7777 | M1A, -1         |            |
| 0437 |      |      | /               |            |
| 0440 |      |      | /               |            |
| 0441 |      |      | /               |            |
| 0442 |      |      | /               |            |
| 0443 |      |      | /               |            |
| 0444 |      |      | /               |            |
| 0445 |      |      | /               |            |
| 0446 |      |      | /               |            |
| 0447 | 1736 | 0000 | SAVE, 0         |            |
| 0450 | 1737 | 1026 | TAD SCURR       |            |
| 0451 | 1740 | 3347 | DCA RESTOR-2    |            |
| 0452 | 1741 | 1014 | TAD TYPEA       |            |
| 0453 | 1742 | 3346 | DCA RESTOR-3    |            |
| 0454 | 1743 | 1021 | TAD TYPE        |            |
| 0455 | 1744 | 3350 | DCA RESTOR-1    |            |
| 0456 | 1745 | 5736 | JMP I SAVE      |            |

/DO WE WANT THE LITTERAL IN???

/YEP. ASSEMBLE IT IN.

/THIS IS MADE A NOP FOR THE SEMI-COLON FUDGE

/RESET THE FUDGE SWITCH

/BY REPLACING THE ISZ

/THIS CODE DISPLAYS THE LINE NUMBER BEING

/ASSEMBLED IN THE MQ.

*so we can look back 1 symbol*

|      |      |      |              |
|------|------|------|--------------|
| 0457 | 1746 | 0000 | 0            |
| 0460 | 1747 | 0000 | 0            |
| 0461 | 1750 | 0000 | 0            |
| 0462 | 1751 | 0000 | RESTOR, 0    |
| 0463 | 1752 | 1347 | TAD RESTOR-2 |
| 0464 | 1753 | 4512 | JMS I SDECO1 |
| 0465 | 1754 | 1346 | TAD RESTOR-3 |
| 0466 | 1755 | 3014 | DCA TYPEA    |
| 0467 | 1756 | 1350 | TAD RESTOR-1 |
| 0470 | 1757 | 3021 | DCA TYPE     |
| 0471 | 1760 | 5751 | JMP I RESTOR |
| 0472 |      | /    |              |
| 0473 |      | /    |              |
| 0474 |      | /    |              |
| 0475 |      | /    |              |
| 0476 |      |      | EJECT        |

*restore pointers saved by same*

JMS ENTS  
TYPE  
return

/enter a symbol in table

```
0477 /POP-12 ASSEMBLER PAGE 8
0500 *2000
0501 2000 0000 ENTS, 0
0502 2001 1027 TAD SEND
0503 2002 4512 JMS I SOECO1
0504 2003 6211 6211
0505 2004 1024 TAD VAL
0506 2005 3422 DCA I VADR
0507 2006 1161 TAD TEM1
0510 2007 3423 DCA I SADR
0511 2010 2023 ISZ SADR
0512 2011 1162 TAD TEM2
0513 2012 3423 DCA I SADR
0514 2013 2023 ISZ SADR
0515 2014 1163 TAD TEM3
0516 2015 3423 DCA I SADR
0517 2016 6201 6201
0520 2017 7344 MTWO
0521 2020 1023 TAD SADR
0522 2021 7450 SNA
0523 2022 5633 JMP I SADROV
0524 2023 3023 DCA SADR
0525 2024 1600 TAD I ENTS
0526 2025 3227 DCA ,+2
0527 2026 4511 JMS I TYPAD1
0530 2027 0000 0
0531 2030 2200 ISZ ENTS
0532 2031 2027 ISZ SEND
0533 2032 5600 JMP I ENTS
0534 /
0535 2033 1175 SADROV, STEXD
0536 2034 7721 M57, -57
0537 2035 3670 GETS, GETIN
0540 2036 3701 COMS, COMMEN
0541 /
0542 2037 0000 IDX, 0
0543 2040 4635 JMS I GETS
0544 2041 1234 TAD M57
0545 2042 7650 SNA CLA
0546 2043 5636 JMP I COMS
0547 2044 1051 TAD ITEM
0550 ASMIFZ TERM-44
0551 TAD M44
0552 2045 7650 SNA CLA
0553 2046 5773 JMP I FINEND
0554 2047 4464 BIT, JMS I BIT6A
0555 2050 3044 DCA CHARX
0556 2051 1044 TAD CHARX
0557 2052 3045 DCA CHARX2
0560 2053 5637 JMP I IDX
0561 2054 0000 EMPTY, 0
0562 2055 1003 TAD LISTWD
0563 2056 7040 CMA
0564 2057 0035 AND PASS
0565 2060 7650 SNA CLA
0566 2061 5654 JMP I EMPTY
0567 2062 1057 TAD CHARAC
0570 2063 7450 SNA
0571 2064 5330 JMP EMLEVE
0572 2065 7041 CIA
0573 2066 3057 DCA CHARAC
0574 2067 3175 DCA TBCONT
0575 2070 1174 TAD SY
```

/pointer to 1st free loc.

/RESET TO FIELD 0.  
/BACK UP SADR TO FIRST WORD OF NAME

/IS IT ZERO?  
/YES, TO MANY SYMBOLS, ABORT WITH ERROR  
/OK, RESTORE SADR NOW.

get type

/POINTER TO TOO MANY SYMBOLS ERROR,

/IS END OF FILE CHAR A 44?  
/YEP, ASSEMBLE IT IN NOW,

/GET THE LISTING WORD  
/NEGATE IT,  
/AND WITH THE PASS,  
/LISTING AND PASS2?  
/NOPE, DONT PRINT ANYTHING

|      |      |      |                      |                       |
|------|------|------|----------------------|-----------------------|
| 0576 | 2071 | 3172 | DCA EMPTRA           |                       |
| 0577 | 2072 | 7410 | SKP                  |                       |
| 0600 | 2073 | 2172 | TWOCHR, ISZ EMPTRA   |                       |
| 0601 | 2074 | 1572 | TAD I EMPTRA         |                       |
| 0602 | 2075 | 7012 | RTR                  |                       |
| 0603 | 2076 | 7012 | RTR                  |                       |
| 0604 | 2077 | 7012 | RTR                  |                       |
| 0605 | 2100 | 0160 | AND P77              |                       |
| 0606 | 2101 | 3007 | DCA TEMP             |                       |
| 0607 | 2102 | 7040 | CMA                  |                       |
| 0610 | 2103 | 3170 | DCA SWITCH           |                       |
| 0611 | 2104 | 1007 | TAD TEMP             |                       |
| 0612 | 2105 | 1133 | TAD M47              |                       |
| 0613 | 2106 | 7650 | SNA CLA              |                       |
| 0614 | 2107 | 5356 | NOTAB1, JMP OUTTAB   |                       |
| 0615 | 2110 | 4347 | JMS OUTTP            |                       |
| 0616 | 2111 | 2057 | DRHT, ISZ CHARAC     |                       |
| 0617 | 2112 | 5314 | JMP ,+2              |                       |
| 0620 | 2113 | 5330 | JMP EMLEVE           |                       |
| 0621 | 2114 | 4746 | JMS I LNCKA          |                       |
| 0622 | 2115 | 1572 | TAD I EMPTRA         |                       |
| 0623 | 2116 | 0160 | AND P77              |                       |
| 0624 | 2117 | 3007 | DCA TEMP             |                       |
| 0625 | 2120 | 3170 | DCA SWITCH           |                       |
| 0626 | 2121 | 1007 | TAD TEMP             |                       |
| 0627 | 2122 | 1133 | TAD M47              |                       |
| 0630 | 2123 | 7650 | SNA CLA              |                       |
| 0631 | 2124 | 5356 | NOTAB2, JMP OUTTAB   | /REPLACED BY A TAD M7 |
| 0632 | 2125 | 4347 | JMS OUTTP            |                       |
| 0633 | 2126 | 2057 | DLFT, ISZ CHARAC     |                       |
| 0634 | 2127 | 5344 | JMP TWOCH            |                       |
| 0635 | 2130 | 4500 | EMLEVE, JMS I TYCARI |                       |
| 0636 | 2131 | 5654 | JMP I EMPTY          |                       |
| 0637 | 2132 | 0000 | CLEARR, 0            |                       |
| 0640 | 2133 | 3057 | DCA CHARAC           |                       |
| 0641 | 2134 | 3772 | DCA I SWOTA          |                       |
| 0642 | 2135 | 1167 | TAD ERRCNT           |                       |
| 0643 | 2136 | 1002 | TAD ERTOT            |                       |
| 0644 | 2137 | 3002 | DCA ERTOT            |                       |
| 0645 | 2140 | 3167 | DCA ERRCNT           |                       |
| 0646 | 2141 | 1174 | TAD SY               |                       |
| 0647 | 2142 | 3514 | DCA I SYB            |                       |
| 0650 | 2143 | 5732 | JMP I CLEARR         |                       |
| 0651 | 2144 | 4746 | TWOCH, JMS I LNCKA   |                       |
| 0652 | 2145 | 5273 | JMP TWOCHR           |                       |
| 0653 | 2146 | 3737 | LNCKA, LNCK          |                       |
| 0654 | 2147 | 0000 | OUTTP, 0             |                       |
| 0655 | 2150 | 1007 | TAD TEMP             |                       |
| 0656 | 2151 | 4755 | JMS I ONVRT          |                       |
| 0657 | 2152 | 4401 | JMS I DECONT         |                       |
| 0660 | 2153 | 2175 | ISZ TBCONT           |                       |
| 0661 | 2154 | 5747 | JMP I OUTTP          |                       |
| 0662 | 2155 | 3600 | ONVRT, CONVRT        |                       |
| 0663 | 2156 | 1175 | OUTTAB, TAD TBCONT   |                       |
| 0664 | 2157 | 1371 | TAD M10              |                       |
| 0665 | 2160 | 7500 | SMA                  |                       |
| 0666 | 2161 | 5357 | JMP ,+2              |                       |
| 0667 | 2162 | 4770 | JMS I KLUG2          |                       |
| 0670 | 2163 | 4475 | JMS I SKIP           |                       |
| 0671 | 2164 | 3175 | DCA TBCONT           |                       |
| 0672 | 2165 | 2170 | ISZ SWITCH           |                       |
| 0673 | 2166 | 5326 | JMP DLFT             |                       |
| 0674 | 2167 | 5311 | JMP DRHT             |                       |

|      |      |      |         |       |
|------|------|------|---------|-------|
| 0675 | 2170 | 3744 | KLUG2,  | SPACK |
| 0676 | 2171 | 7770 | M10,    | -10   |
| 0677 | 2172 | 1556 | SWOTA,  | SWOT  |
| 0700 | 2173 | 3717 | FINEND, | ENDMS |
| 0701 | 2174 | 7771 | M7TWO,  | -7    |
| 0702 |      |      |         | EJECT |
| -    |      |      |         |       |

*include precision integers?*

```

0703      /PDP-12 ASSEMBLER PAGE 9
0704      +2200
0705      2200 0000 OPCV, 0
0706      2201 2020      ISZ ANY
0707      2202 3052      DCA HIC      /CLEAR HIGH AND LOW PARTS
0710      2203 3053      DCA LWC
0711      2204 1045      TAD CHARX2      / GET THE NEXT CHARACTER
0712      2205 1126      TAD M67
0713      2206 7450      SNA
0714      2207 5212      JMP DPCS
0715      2210 1242      TAD      OPCVM2
0716      2211 7650      SNA CLA
0717      2212 4504      OPCS, JMS I IDX1      /INDEX CHARACTER POINTER
0720      2213 1045      TAD CHARX2
0721      2214 1125      TAD M45
0722      2215 7500      SMA
0723      2216 5600      JMP I DPCV
0724      2217 1241      TAD P12      /IF LESS OR EQUAL TO 9 CHECK FOR GR
0725      2220 7510      SPA      /EQ, ZERO
0726      2221 5600      JMP I DPCV      /LESS THAN ZERO, EXIT NOT DIGIT
0727      2222 3273      OCA MTDG
0730      2223 4243      JMS MT10
0731      2224 5212      JMP DPCS
0732      2225 4231      DECIM, JMS DECIMS
0733      2226 5503      JMP I B21
0734      2227 4235      OCT, JMS OCTS
0735      2230 5503      JMP I B21
0736      2231 0000      DECIMS, 0
0737      2232 1255      TAD MTSW+4
0740      2233 3251      DCA MTSW
0741      2234 5631      JMP I DECIMS
0742      2235 0000      OCTS, 0
0743      2236 1121      TAD M1000
0744      2237 3251      DCA MTSW
0745      2240 5635      JMP I OCTS
0746      2241 0012      P12, 12
0747      2242 7776      OPCVM2, -2
0750      /MULTIPLY LWC BY 10,
0751      2243 0000      MT10, 0
0752      2244 1053      TAD LWC
0753      2245 0156      AND P777
0754      2246 3274      DCA TIC
0755      2247 4257      JMS MTRL
0756      2250 4257      JMS MTRL
0757      2251 4265      MTSW, JMS MTAD      /NOP FOR OCTAL, JMS MTAD FOR DECIMAL
0760      2252 4257      JMS MTRL
0761      2253 1273      TAD MTDG
0762      2254 3274      DCA TIC
0763      2255 4265      JMS MTAD
0764      2256 5643      JMP I MT10
0765      /ROTATE LWC LEFT 1
0766      2257 0000      MTRL, 0
0767      2260 7300      CLL CLA
0770      2261 1053      TAD LWC
0771      2262 7004      RAL
0772      2263 3053      DCA LWC
0773      2264 5657      JMP I MTRL
0774      /ADD LWC TO TIC
0775      2265 0000      MTAD, 0
0776      2266 7300      CLL CLA
0777      2267 1053      TAD LWC
1000      2270 1274      TAD TIC
1001      2271 3053      DCA LWC

```

|      |      |      |                    |
|------|------|------|--------------------|
| 1002 | 2272 | 5665 | JMP I MTAD         |
| 1003 |      |      | /SOME DATA STORAGE |
| 1004 | 2273 | 0000 | MTDG, 0            |
| 1005 | 2274 | 0000 | TIC, 0             |
| 1006 | 2275 | 0000 | ERR2, 0            |
| 1007 | 2276 | 1021 | TAD TYPE           |
| 1010 | 2277 | 1136 | TAD P6000          |
| 1011 | 2300 | 7650 | SNA CLA            |
| 1012 | 2301 | 5675 | JMP I ERR2         |
| 1013 | 2302 | 1014 | TAD TYPEA          |
| 1014 | 2303 | 7106 | CLL RTL            |
| 1015 | 2304 | 0136 | AND P6000          |
| 1016 | 2305 | 6211 | 6211               |
| 1017 | 2306 | 1422 | TAD I VADR         |
| 1020 | 2307 | 6201 | 6201               |
| 1021 | 2310 | 7041 | CIA                |
| 1022 | 2311 | 1030 | TAD AADR           |
| 1023 | 2312 | 7650 | SNA CLA            |
| 1024 | 2313 | 5675 | JMP I ERR2         |
| 1025 | 2314 | 7326 | CLA CLL CML RTL    |
| 1026 | 2315 | 4476 | JMS I ERR1         |
| 1027 | 2316 | 5675 | JMP I ERR2         |
| 1030 | 2317 | 0000 | UNDERR, 0          |
| 1031 | 2320 | 2035 | ISZ PASS           |
| 1032 | 2321 | 5332 | JMP EQCHK          |
| 1033 | 2322 | 3007 | DCA TEMP           |
| 1034 | 2323 | 7040 | CMA                |
| 1035 | 2324 | 3035 | DCA PASS           |
| 1036 | 2325 | 1007 | TAD TEMP           |
| 1037 | 2326 | 4476 | JMS I ERR1         |
| 1040 | 2327 | 7240 | CLA CMA            |
| 1041 | 2330 | 3035 | DCA PASS           |
| 1042 | 2331 | 5717 | JMP I UNDERR       |
| 1043 | 2332 | 1135 | EQCHK, TAD M4      |
| 1044 | 2333 | 7650 | SNA CLA            |
| 1045 | 2334 | 4740 | JMS I EQCHKI       |
| 1046 | 2335 | 7240 | CLA CMA            |
| 1047 | 2336 | 3006 | DCA EQRET1         |
| 1050 | 2337 | 5330 | JMP EQCHK-2        |
| 1051 | 2340 | 3366 | EQCHKI, EQUCHK     |
| 1052 | 2341 | 0000 | PUSH2, 0           |
| 1053 | 2342 | 3275 | DCA ERR2           |
| 1054 | 2343 | 1142 | TAD P5000          |
| 1055 | 2344 | 1047 | TAD POINT          |
| 1056 | 2345 | 7700 | SNA CLA            |
| 1057 | 2346 | 5364 | JMP ERR12          |
| 1060 | 2347 | 1275 | TAD ERR2           |
| 1061 | 2350 | 3447 | DCA I POINT        |
| 1062 | 2351 | 2047 | ISZ POINT          |
| 1063 | 2352 | 5741 | JMP I PUSH2        |
| 1064 | 2353 | 0000 | CNTRLP, 0          |
| 1065 | 2354 | 6031 | KSF                |
| 1066 | 2355 | 5753 | JMP I CNTRLP       |
| 1067 | 2356 | 6036 | KRB                |
| 1070 | 2357 | 1151 | TAD M215           |
| 1071 | 2360 | 7650 | SNA CLA            |
| 1072 | 2361 | 5763 | JMP I CNTHON       |
| 1073 | 2362 | 5753 | JMP I CNTRLP       |
| 1074 |      |      | /                  |
| 1075 | 2363 | 4075 | CNTHON, KILBUF     |
| 1076 |      |      | /                  |
| 1077 | 2364 | 1015 | ERR12, TAD P11     |
| 1100 | 2365 | 4476 | JMS I ERR1         |

*115 symbol defined?*

*100 NO yes - does old def = new def?*

*1 get old val.*

*1 get new val.*

*1-2 ID error*

*20 if pushdown overflow*

*1 check for car. ret to terminate assembly*

/RETURN TO KILL BUFFER MONITOR RESTART

*20 = spec. char.  
01 = undefined  
10 = defined  
11 = pseudo-op*

|      |      |      |       |            |
|------|------|------|-------|------------|
| 1101 | 2366 | 5517 |       | JMP I MON  |
| 1102 | 2367 | 0000 | ANYY, | 0          |
| 1103 | 2370 | 1021 |       | TAD TYPE   |
| 1104 | 2371 | 1136 |       | TAD P6000  |
| 1105 | 2372 | 7450 |       | SNA        |
| 1106 | 2373 | 2020 |       | ISZ ANY    |
| 1107 | 2374 | 1136 |       | TAD P6000  |
| 1110 | 2375 | 7650 |       | SNA CLA    |
| 1111 | 2376 | 2020 |       | ISZ ANY    |
| 1112 | 2377 | 5767 |       | JMP I ANYY |
| 1113 |      |      |       | EJECT      |

*1. Anything assembled on current line,  
"ANYY" is indexed*

```

1114      /PDP-12 ASSEMBLER PAGE 10
1115      *2400
1116      2400 0000 ALPHA, 0
1117      2401 4500 JMS I TYCARI
1118      2402 3307 SYMBLP, DCA FOUND
1119      2403 1032 TAD PERMA
1120      2404 3026 DCA SCURR
1121      2405 7350 CLA CMA CLL RAR
1122      2406 3040 DCA TEM1A
1123      2407 7350 CLA CLL CMA RAR
1124      2410 3041 DCA TEM1A+1
1125      2411 7350 CLA CMA CLL RAR
1126      2412 3042 DCA TEM1A+2
1127      2413 1026 NXCAND, TAD SCURR
1128      2414 7041 CIA
1129      2415 1027 TAD SEND
1130      2416 7650 SNA CLA
1131      2417 5274 JMP PRSYMB
1132      2420 1026 TAD SCURR
1133      2421 4512 JMS I SDEC01
1134      2422 6211 6211
1135      2423 1423 TAD I SADR
1136      2424 7104 CLL RAL
1137      2425 7020 CML
1138      2426 7530 SZL SPA
1139      2427 5270 JMP REJECT
1140      2430 7070 RAR CMA CML
1141      2431 1040 TAD TEM1A
1142      2432 7040 CMA
1143      2433 7640 SZA CLA
1144      2434 5252 JMP JUDGE
1145      2435 2023 ISZ SADR
1146      2436 7350 CLA CLL CMA RAR
1147      2437 0423 AND I SADR
1148      2440 7040 CMA
1149      2441 1041 TAD TEM1A+1
1150      2442 7040 CMA
1151      2443 7640 SZA CLA
1152      2444 5252 JMP JUDGE
1153      2445 2023 ISZ SADR
1154      2446 7350 CLA CLL CMA RAR
1155      2447 0423 AND I SADR
1156      2450 7160 CMA CLL CML
1157      2451 1042 TAD TEM1A+2
1158      2452 7630 JUDGE, SZL CLA
1159      2453 5270 JMP REJECT
1160      2454 6201 CDF 0
1161      2455 1026 TAD SCURR
1162      2456 4512 JMS I SDEC01
1163      2457 1023 TAD SADR
1164      2460 4474 JMS I MOVE1
1165      2461 0040 TEM1A
1166      2462 6211 6211
1167      2463 6211 6211
1168      2464 1416 TAD I 16
1169      2465 3024 DCA VAL
1170      2466 1023 TAD SADR
1171      2467 3307 DCA FOUND
1172      2470 7200 REJECT, CLA
1173      2471 6201 CDF 0
1174      2472 2026 ISZ SCURR
1175      2473 5213 JMP NXCAND
1176
1177
1178
1179
1180
1181
1182

```

*output symbols in alphabetical order*

```

1213 2474 1307 PRSYMB, TAD FOUND
1214 2475 7450 SNA
1215 2476 5600 JMP I ALPHA
1216 2477 3023 DCA SADR
1217 2500 7240 CLA CMA
1220 2501 6211 6211
1221 2502 3707 DCA I FOUND
1222 2503 6201 CDF 0
1223 2504 4706 JMS I POSANDV
1224 2505 5202 JMP SYMBLP
1225 /
1226 /
1227 /
1230 2506 2600 POSANDV, OS ANDV
1231 2507 0000 FOUND, 0
1232 /
1233 /
1234 /
1235 /
1236 2510 0000 TYCAR, 0
1237 2511 1154 TAD P215
1240 2512 4401 JMS I DECOU
1241 2513 1153 TAD P212
1242 2514 4401 JMS I DECOU
1243 2515 4470 JMS I CHEKER
1244 2516 4404 JMS I CNTLPI
1245 2517 5710 JMP I TYCAR
1246 2520 0000 CURSKP, 0
1247 2521 1012 TAD CURLIN
1250 2522 4462 JMS I SOPS
1251 2523 1335 TAD M22
1252 2524 4475 JMS I SKIP
1253 2525 4461 JMS I EMPTY1
1254 2526 5720 JMP I CURSKP
1255 2527 0000 TYP0, 0
1256 2530 6041 TSF
1257 2531 5330 JMP , -1
1260 2532 6046 TLS
1261 2533 7200 CLA
1262 2534 5727 JMP I TYP0
1263 2535 7756 M22, -22
1264 TEMP A=TYCAR
1265 /
1266 /
1267 /
1270 /
1271 /
1272 /
1273 /
1274 2536 4747 XCHAIN, JMS I OXWRIT
1275 2537 2555 XPSEU
1276 2540 4746 JMS I OXREAD
1277 2541 2551 XINIT
1300 2542 4750 JMS I XPP
1301 2543 4746 JMS I OXREAD
1302 2544 2555 XPSEU
1303 2545 5501 JMP I MAIN1
1304 /
1305 /
1306 /
1307 /
1310 2546 7774 OXREAD, 7774
1311 2547 7775 OXWRIT, 7775

```

/TEST FOR CARRAGE RETURN

|      |      |      |                 |          |                               |
|------|------|------|-----------------|----------|-------------------------------|
| 1312 | 2550 | 5000 | XPP,            | XXXX     |                               |
| 1313 | 2551 | 0100 | XINIT,          | DIALUNIT |                               |
| 1314 | 2552 | 0012 |                 | 12       |                               |
| 1315 | 2553 | 0024 |                 | BCHAIN   |                               |
| 1316 | 2554 | 0001 |                 | 1        |                               |
| 1317 |      |      | /               |          |                               |
| 1320 |      |      | /               |          |                               |
| 1321 | 2555 | 0100 | XPSEU,          | DIALUNIT |                               |
| 1322 | 2556 | 0012 |                 | 12       |                               |
| 1323 | 2557 | 0025 |                 | SCRATCH  |                               |
| 1324 | 2560 | 0001 |                 | 1        |                               |
| 1325 |      |      | /               |          |                               |
| 1326 |      |      | /               |          |                               |
| 1327 |      |      | /               |          |                               |
| 1330 |      |      | /               |          |                               |
| 1331 |      |      | /               |          |                               |
| 1332 |      |      | /               |          |                               |
| 1333 |      |      | /               |          |                               |
| 1334 |      |      | /               |          |                               |
| 1335 |      |      | /               |          |                               |
| 1336 |      |      | /               |          |                               |
| 1337 |      |      | /               |          |                               |
| 1340 | 2561 | 0000 | SEMITX, 0       |          |                               |
| 1341 | 2562 | 1375 | TAD             | XM13     |                               |
| 1342 | 2563 | 7450 | SNA             |          |                               |
| 1343 | 2564 | 5367 | JMP             | ,+3      |                               |
| 1344 | 2565 | 1376 | TAD             | XP46     |                               |
| 1345 | 2566 | 5761 | JMP I           | SEMITX   |                               |
| 1346 | 2567 | 1373 | TAD             | SEMPDG   |                               |
| 1347 | 2570 | 3774 | OCA I           | SEMP TF  | /REPLACE LINE BOPPER BY A NOP |
| 1350 | 2571 | 1376 | TAD             | XP46     |                               |
| 1351 | 2572 | 5761 | JMP I           | SEMITX   |                               |
| 1352 |      |      | /               |          |                               |
| 1353 |      |      | /               |          |                               |
| 1354 |      |      | /               |          |                               |
| 1355 | 2573 | 7000 | SEMPDG, NOP     |          |                               |
| 1356 | 2574 | 1707 | SEMP TF, JBMSTR |          | /POINTER TO THE ISZ,          |
| 1357 | 2575 | 7765 | XM13,           | -13      |                               |
| 1360 | 2576 | 0046 | XP46,           | 46       |                               |
| 1361 |      |      | /               |          |                               |
| 1362 |      |      | /               |          |                               |
| 1363 |      |      | /               |          |                               |
| 1364 |      |      | /               |          |                               |
| 1365 |      |      | /               |          |                               |
| 1366 |      |      | /               |          |                               |
| 1367 |      |      |                 | EJECT    |                               |

```

1370      /PDP-12 ASSEMBLER PAGE 11
1371      /OUTPUT A SYMBOL AND ITS VALUE
1372      *2600
1373      OSANDV, 0
1374      2600 0000      JMS OSYM      /OUTPUT SYMBOL
1375      2601 4213      TAD TBCONT
1376      2602 1175      TAD M7
1377      2603 1302      JMS I SKIP      /OUTPUT TAB
1378      2604 4475      JMS I LINK8
1379      2605 4624      TAD VAL
1380      2606 1024      JMS OPS      /OUTPUT NUMBER
1381      2607 4261      JMS I TYCARI
1382      2610 4500      JMS I CNTLPI
1383      2611 4404      JMP I OSANDV
1384      2612 5600      OSYM, 0
1385      2613 0000      DCA TBCONT
1386      2614 3175      TAD TEM1A
1387      2615 1040      JMS OX
1388      2616 4225      TAD TEM1A+1
1389      2617 1041      JMS OX
1390      2620 4225      TAD TEM1A+2
1391      2621 1042      JMS OX
1392      2622 4225      JMP I OSYM
1393      2623 5613      LINK8, LINK8A
1394      2624 3551      OX, 0
1395      2625 0000      DCA LWC
1396      2626 3053      DCA HIC
1397      2627 3052      TAD LWC
1398      2630 1053      SNA
1399      2631 7450      JMP I OSYM
1400      2632 5613      TAD M45
1401      2633 1125      SPA
1402      2634 7510      JMP DV3
1403      2635 5240      ISZ HIC
1404      2636 2052      JMP , -4
1405      2637 5233      TAD P45
1406      2640 1301      DV3, DCA LWC
1407      2641 3053      TAD HIC
1408      2642 1052      JMS UNTRAN
1409      2643 4247      TAD LWC
1410      2644 1053      JMS UNTRAN
1411      2645 4247      JMP I OX
1412      2646 5625      UNTRAN, 0
1413      2647 0000      SNA
1414      2650 7450      JMP I OSYM
1415      2651 5613      TAD M33
1416      2652 1123      SPA
1417      2653 7510      TAD P53
1418      2654 1303      TAD P260
1419      2655 1304      JMS I DECOU
1420      2656 4401      ISZ TBCONT
1421      2657 2175      JMP I UNTRAN
1422      2660 5647      /OCTAL PRINT SUBROUTINE
1423      2661 0000      OPS, 0
1424      2662 3213      DCA OSYM
1425      2663 1135      TAD M4
1426      2664 3175      DCA TBCONT
1427      2665 1213      TAD OSYM
1428      2666 7006      RTL
1429      2667 7004      RAL
1430      2670 3213      DCA OSYM
1431      2671 1213      TAD OSYM
1432      2672 7004      RAL

```

*Derode a word of symbol by  $\div 45$*

*convert from 6 bit internal to ASCII*

|      |      |      |                    |
|------|------|------|--------------------|
| 1467 | 2673 | 0157 | AND P7             |
| 1470 | 2674 | 1304 | TAD P260           |
| 1471 | 2675 | 4401 | JMS I DECOUT       |
| 1472 | 2676 | 2175 | ISZ TBCONT         |
| 1473 | 2677 | 5265 | JMP OPS+4          |
| 1474 | 2700 | 5661 | JMP I OPS          |
| 1475 | 2701 | 0045 | P45, 45            |
| 1476 | 2702 | 7771 | M7, -7             |
| 1477 | 2703 | 0053 | P53, 53            |
| 1500 | 2704 | 0260 | P260, 260          |
| 1501 | 2705 | 4477 | FLDUP, JMS I EXPR1 |
| 1502 | 2706 | 1037 | TAD MODE           |
| 1503 | 2707 | 7700 | SMA CLA            |
| 1504 | 2710 | 5314 | JMP FORT           |
| 1505 | 2711 | 1020 | TAD ANY            |
| 1506 | 2712 | 7640 | SZA CLA            |
| 1507 | 2713 | 5317 | JMP FIELDM         |
| 1510 | 2714 | 4465 | FDRT, JMS I PASSEK |
| 1511 | 2715 | 4466 | JMS I CURSKA       |
| 1512 | 2716 | 5501 | JMP I MAIN1        |
| 1513 | 2717 | 1033 | FIELDM, TAD EVAL   |
| 1514 | 2720 | 7710 | SPA CLA            |
| 1515 | 2721 | 5314 | JMP FORT           |
| 1516 | 2722 | 7344 | MTWO               |
| 1517 | 2723 | 1033 | TAD EVAL           |
| 1520 | 2724 | 7700 | SMA CLA            |
| 1521 | 2725 | 5314 | JMP FORT           |
| 1522 | 2726 | 1033 | TAD EVAL           |
| 1523 | 2727 | 3036 | DCA FLDWD          |
| 1524 | 2730 | 5314 | JMP FORT           |
| 1525 |      |      | LISTA=,            |
| 1526 |      |      | EJECT              |

```

1527      /PDP-12 ASSEMBLER PAGE 12
1530      *3000
1531      3000 2661      OPS
1532      3001 0000      PUNONE, 0
1533      3002 5601      JMP I PUNONE      /NOP FOR PASS 2
1534      3003 1030      TAD AADR
1535      3004 0155      AND P7400
1536      3005 7106      CLL RTL
1537      3006 7006      RTL
1540      3007 7004      RAL
1541      3010 1176      TAD BINTAB
1542      3011 3010      DCA BINPTR
1543      3012 1036      TAD FLOWD
1544      3013 7650      SNA CLA
1545      3014 5220      JMP BINCK
1546      3015 1010      TAD BINPTR
1547      3016 1300      TAD P20
1550      3017 3010      DCA BINPTR
1551      3020 1010      BINCK, TAD BINPTR
1552      3021 7041      CIA
1553      3022 1011      TAD BINBLK
1554      3023 7640      SZA CLA
1555      3024 4676      JMS I NOTBL1
1556      3025 4577      JMS I SETINU      /SET BITS NOWWIN HEADER BLOCK
1557      3026 1030      TAD AADR
1560      3027 0277      AND P377
1561      3030 1142      TAD P5000
1562      3031 3140      DCA LOWTMP
1563      3032 1033      TAD EVAL
1564      3033 3540      DCA I LOWTMP
1565      3034 1003      TAD LISTWO
1566      3035 7710      SPA CLA
1567      3036 5601      JMP I PUNONE
1570      3037 1167      TAD ERRCNT
1571      3040 7640      SZA CLA
1572      3041 5246      JMP NONER
1573      3042 1012      ABNOP1, TAD CURLIN
1574      3043 4600      ABNOP2, JMS I PUNONE-1
1575      3044 1301      ABNOP3, TAD M6
1576      3045 4475      ABNOP4, JMS I SKIP
1577      3046 1037      NONER, TAD MODE
1600      3047 7700      SHA CLA
1601      3050 5262      JMP LNKA
1602      3051 1030      TAD AADR
1603      3052 4600      AADRL, JMS I PUNONE-1
1604      3053 7344      MTWO
1605      3054 4475      JMS I SKIP
1606      3055 1033      TAD EVAL
1607      3056 4600      JMS I PUNONE-1
1610      3057 7344      MTWO
1611      3060 4475      JMS I SKIP
1612      3061 5601      JMP I PUNONE
1613      3062 1030      LNKA, TAD AADR
1614      3063 0136      AND P6000
1615      3064 7041      CIA
1616      3065 1046      TAD ABANK
1617      3066 7650      SNA CLA
1620      3067 5273      JMP SAME5
1621      3070 1030      TAD AADR
1622      3071 0131      AND P3777
1623      3072 5252      JMP AADRL
1624      3073 1030      SAME5, TAD AADR
1625      3074 0144      AND P1777

```

*outputs contents of EVAL  
(the generated binary)*

```
1626      3075 5252      JMP AADRL
1627      3076 3616      NOTBL1, NOTBLK
1630      3077 0377      P377, 377
1631      3100 0020      P20, 20
1632      3101 7772      H6, -6
1633      SYMBUF=,
1634      /
1635      /
1636      /
1637      /
1640      /
1641      /
1642      /
1643      /
1644      /
1645      /
1646      /
1647      EJECT
-
```

1650 /PDP-12 ASSEMBLER PAGE 13  
1651 \*3200

1652 /  
1653 /  
1654 3200 3271 TABLE

1655 /  
1656 3201 0000 ERROR, 0  
1657 3202 1200 TAD , -2  
1658 3203 3237 OCA UNREF-1  
1659 3204 4465 JMS I PASSEK  
1660 3205 1167 TAD ERRCNT  
1661 3206 7640 SZA CLA  
1662 3207 4500 JMS I TYCARI  
1663 3210 1012 TAD CURLIN  
1664 3211 4462 JMS I SOPS  
1665 3212 7344 MTWO  
1666 3213 4475 JMS I SKIP  
1667 3214 1637 TAD I UNREF-1  
1668 3215 0160 AND P77  
1669 3216 1143 TAD P300  
1670 3217 4401 JMS I DECOUT  
1671 3220 1637 TAD I UNREF-1  
1672 3221 7012 RTR  
1673 3222 7012 RTR  
1674 3223 7012 RTR  
1675 3224 0160 AND P77  
1676 3225 1143 TAD P300  
1677 3226 4401 JMS I DECOUT  
1678 3227 7344 MTWO  
1679 3230 4475 JMS I SKIP  
1680 3231 2167 ISZ ERRCNT  
1681 3232 7000 NOP  
1682 3233 4465 JMS I PASSEK  
1683 3234 7410 SKP  
1684 3235 4500 JMS I TYCARI  
1685 3236 5601 JMP I ERROR

/CHECK FOR LISTING AND PASS2.

1714 /  
1715 3237 0000 0  
1716 3240 2317 UNREF, UNDERR  
1717 3241 1332 AAS  
1718 3242 0000 GETASY, 0  
1719 3243 7200 CLA  
1720 3244 3050 OCA IN  
1721 3245 4641 JMS I GETASY-1  
1722 3246 1050 TAD IN  
1723 3247 7650 SNA CLA  
1724 3250 5642 JMP I GETASY  
1725 3251 4670 JMS I SEAR1  
1726 3252 7610 SKP CLA  
1727 3253 5262 JMP CKUN  
1728 3254 3024 OCA VAL  
1729 3255 4667 JMS I ENTS1  
1730 3256 2000 USYMB  
1731 3257 7307 CLA CLL IAC RTL /4  
1732 3260 4640 JMS I GETASY-2  
1733 3261 5642 JMP I GETASY  
1734 3262 1021 CKUN, TAD TYPE  
1735 3263 1136 TAD P6000  
1736 3264 7640 SZA CLA  
1737 3265 5642 JMP I GETASY  
1738 3266 5257 JMP CKUN-3  
1739 3267 2000 ENTS1, ENTS  
1740 3270 0266 SEAR1, SEARCH

/FOUND

/US

|      |      |      |                          |                                                 |
|------|------|------|--------------------------|-------------------------------------------------|
| 1747 |      |      | /TABLE OF ERROR MESSAGES |                                                 |
| 1750 | 3271 | 0511 | TABLE, 0511              | /IE ILLEGAL EQUALS                              |
| 1751 | 3272 | 2211 |                          | /IR ILLEGAL REFERENCE                           |
| 1752 | 3273 | 0411 |                          | /ID ILLEGAL REDEFINITION                        |
| 1753 | 3274 | 0311 |                          | /IC ILLEGAL CHARACTER                           |
| 1754 | 3275 | 2325 |                          | /US REFERENCE TO UNDEFINED SYMBOL               |
| 1755 | 3276 | 1503 |                          | /CM COMMA USED INCORRECTLY                      |
| 1756 | 3277 | 1003 |                          | /CH CHAINING ERROR,                             |
| 1757 | 3300 | 0523 |                          | /SE SYSDOL TABLE EXCEEDED                       |
| 1760 | 3301 | 0127 |                          | /WA WORKING AREA EXCEEDED                       |
| 1761 | 3302 | 2320 |                          | /PS PUSH DOWN LIST EXCEEDED                     |
| 1762 | 3303 | 7610 | LNKMOD, CLA SKP          |                                                 |
| 1763 | 3304 | 7240 | PDP0MD, CLA CMA          |                                                 |
| 1764 | 3305 | 3037 |                          | DCA MODE                                        |
| 1765 | 3306 | 1037 |                          | TAD MODE                                        |
| 1766 | 3307 | 1316 |                          | TAD P6321                                       |
| 1767 | 3310 | 3375 |                          | DCA INSYM                                       |
| 1770 | 3311 | 4772 |                          | JMS I SYREAD                                    |
| 1771 | 3312 | 3373 |                          | SYCRAP                                          |
| 1772 | 3313 | 4463 |                          | JMS I RESET1                                    |
| 1773 | 3314 | 4502 |                          | JMS I PAGE1                                     |
| 1774 | 3315 | 5503 |                          | JMP I B21                                       |
| 1775 | 3316 | 0042 | P6321, D6+42-300         | /SWAP THE CORRECT SYMBOL TABLE IN NOW           |
| 1776 | 3317 | 0000 | SYSIN, 0                 | /CLEAN UP THE SYSTEM, WE RE DONE WITH THIS ONE. |
| 1777 | 3320 | 7300 |                          | CLA CLL                                         |
| 2000 | 3321 | 2351 |                          | ISZ WDHAF                                       |
| 2001 | 3322 | 5340 |                          | JMP LEFTHF                                      |
| 2002 | 3323 | 1752 |                          | TAD I PTBUFP                                    |
| 2003 | 3324 | 0160 |                          | AND P77                                         |
| 2004 | 3325 | 3007 |                          | DCA TEMP                                        |
| 2005 | 3326 | 2352 |                          | ISZ PTBUFP                                      |
| 2006 | 3327 | 2354 |                          | ISZ CHRC                                        |
| 2007 | 3330 | 5336 |                          | JMP RTHFT                                       |
| 2010 | 3331 | 1155 |                          | TAD P7400                                       |
| 2011 | 3332 | 3354 |                          | DCA CHRC                                        |
| 2012 | 3333 | 1353 |                          | TAD PTBUFP                                      |
| 2013 | 3334 | 3352 |                          | DCA PTBUFP                                      |
| 2014 | 3335 | 4750 |                          | JMS I BUF1                                      |
| 2015 | 3336 | 1007 | RTHFT, TAD TEMP          |                                                 |
| 2016 | 3337 | 5717 |                          | JMP I SYSIN                                     |
| 2017 | 3340 | 7040 | LEFTHF, CMA              |                                                 |
| 2020 | 3341 | 3351 |                          | DCA WDHAF                                       |
| 2021 | 3342 | 1752 |                          | TAD I PTBUFP                                    |
| 2022 | 3343 | 7012 |                          | RTR                                             |
| 2023 | 3344 | 7012 |                          | RTR                                             |
| 2024 | 3345 | 7012 |                          | RTR                                             |
| 2025 | 3346 | 0160 |                          | AND P77                                         |
| 2026 | 3347 | 5717 |                          | JMP I SYSIN                                     |
| 2027 | 3350 | 4006 | BUF1, BUF1N              |                                                 |
| 2030 | 3351 | 0000 | WDHALF, 0                |                                                 |
| 2031 | 3352 | 0000 | PTBUFP, 0                |                                                 |
| 2032 | 3353 | 4400 | PTBUFP, 4400             |                                                 |
| 2033 | 3354 | 0000 | CHRC, 0                  |                                                 |
| 2034 | 3355 | 0000 | PASSER, 0                |                                                 |
| 2035 | 3356 | 1003 |                          | TAD LISTWD                                      |
| 2036 | 3357 | 7710 |                          | SPA CLA                                         |
| 2037 | 3360 | 5364 |                          | JMP ,+4                                         |
| 2040 | 3361 | 1035 |                          | TAD PASS                                        |
| 2041 | 3362 | 7710 |                          | SPA CLA                                         |
| 2042 | 3363 | 5755 |                          | JMP I PASSER                                    |
| 2043 | 3364 | 2355 |                          | ISZ PASSER                                      |
| 2044 | 3365 | 5755 |                          | JMP I PASSER                                    |
| 2045 | 3366 | 0000 | EQUCHK, 0                |                                                 |

|      |      |      |                  |
|------|------|------|------------------|
| 2046 | 3367 | 2006 | ISZ EQRET1       |
| 2047 | 3370 | 4476 | JMS I ERR1       |
| 2050 | 3371 | 5766 | JMP I EQUCHK     |
| 2051 | 3372 | 7774 | SYREAD, READ     |
| 2052 |      |      | /                |
| 2053 |      |      | /                |
| 2054 |      |      | /                |
| 2055 |      |      | /                |
| 2056 |      |      | /                |
| 2057 |      |      | /                |
| 2060 | 3373 | 0100 | SYCRAP, DIALUNIT |
| 2061 | 3374 | 0036 | 36               |
| 2062 | 3375 | 0000 | INSYM, 0         |
| 2063 | 3376 | 0001 | 1                |
| 2064 |      |      | /                |
| 2065 |      |      | /                |
| 2066 |      |      | /                |
| 2067 |      |      | /                |
| 2070 |      |      | /                |
| 2071 |      |      | /                |
| 2072 |      |      | /                |
| 2073 |      |      | /                |
| 2074 |      |      | /                |
| 2075 |      |      | EJECT            |

```

2076 /PDP-12 ASSEMBLER PAGE 14
2077 /
2100 /
2101 *3400
2102 3400 0000 BIT6, 0
2103 3401 1051 TAD ITEM
2104 3402 1013 TAD M40
2105 3403 7510 SPA
2106 3404 5225 JMP P1T037
2107 3405 1254 TAD M20
2110 3406 7510 SPA
2111 3407 5212 JMP P40T57
2112 3410 4640 JMS I SEMITZ
2113 3411 5600 JMP I BIT6
2114 3412 1015 P40T57, TAD P11
2115 3413 7450 SNA
2116 3414 5241 JMP TABA
2117 3415 1256 TAD P2
2120 3416 7450 SNA
2121 3417 5250 JMP P367A
2122 3420 1256 TAD P2
2123 3421 7450 SNA
2124 3422 5243 JMP CARET
2125 3423 1257 TAD P57
2126 3424 5600 JMP I BIT6
2127 3425 7200 P1T037, CLA
2130 3426 1051 TAD ITEM
2131 3427 1123 TAD M33
2132 3430 7510 SPA
2133 3431 5245 JMP ALPHAB
2134 3432 1262 TAD BIT6M3
2135 3433 7700 SNA CLA
2136 3434 5250 JMP P367A
2137 3435 1051 TAD ITEM
2140 3436 1260 TAD P41
2141 3437 5600 JMP I BIT6
2142 /
2143 /
2144 3440 2561 SEMITZ, SEMITX
2145 /
2146 /
2147 3441 1255 TABA, TAD P54
2150 3442 5600 JMP I BIT6
2151 3443 1261 CARET, TAD P46
2152 3444 5600 JMP I BIT6
2153 3445 7200 ALPHAB, CLA
2154 3446 1051 TAD ITEM
2155 3447 5600 JMP I BIT6
2156 3450 7325 P367A, PTHREE
2157 3451 4476 JMS I ERR1
2160 3452 5653 JMP I ,+1
2161 3453 2040 IDX+1
2162 3454 7760 M20, -20
2163 3455 0054 P54, 54
2164 3456 0002 P2, 2
2165 3457 0057 P57, 57
2166 3460 0041 P41, 41
2167 3461 0046 P46, 46
2170 3462 7775 BIT6M3, -3
2171 3463 0000 DISRET, 0
2172 3464 4500 JMS I TYCARI
2173 3465 1002 TAD ERTOT
2174 3466 7450 SNA

```

*converts ASCII to 6-bit internal*

*47 = TMB? (DIM)*

*Percent?*

*143 car net?*

/GIVE A CARRIAGE RETURN BEFORE THE SYMBOL TABLE  
 /GET THE TOTAL NUMBER OF ERRORS  
 /ZERO?

|      |      |      |         |           |                                                           |
|------|------|------|---------|-----------|-----------------------------------------------------------|
| 2175 | 3467 | 5277 | JMP     | LWD       | /YEP, PRINT OUT MESSAGE WITH "NO" PREFIX                  |
| 2176 | 3470 | 7110 | CLL     | RAR       | /JUST 1 ERROR?                                            |
| 2177 | 3471 | 7450 | SNA     |           |                                                           |
| 2200 | 3472 | 3320 | DCA     | MESSS     | /REMOVE THE FINAL S NOW                                   |
| 2201 | 3473 | 7004 | RAL     |           |                                                           |
| 2202 | 3474 | 4462 | JMS I   | SOPS      | /OUTPUT THE OCTAL NUMBER OF ERRORS                        |
| 2203 | 3475 | 2277 | ISZ     | LWD       | /PUSH PAST THE "NO" PART OF THE MESSAGE                   |
| 2204 | 3476 | 2277 | ISZ     | LWD       |                                                           |
| 2205 |      |      | /       |           |                                                           |
| 2206 | 3477 | 1310 | LWD,    | TAD       | MESS                                                      |
| 2207 | 3500 | 7450 | SNA     |           | /GET A WORD OF THE MESSAGE                                |
| 2210 | 3501 | 5304 | JMP     | LWDE      | /ZERO??                                                   |
| 2211 | 3502 | 4401 | JMS I   | DECOU     | /YEP, THATS ALL                                           |
| 2212 | 3503 | 5276 | JMP     | LWD-1     | /OUTPUT THE LETTER                                        |
| 2213 |      |      | /       |           | /GET THE NEXT CHARACTER NOW                               |
| 2214 | 3504 | 1322 | LWDE,   | TAD       | PCONT                                                     |
| 2215 | 3505 | 3322 | DCA     | PCONT     | /OVERLAYED INSTRUCTION,                                   |
| 2216 | 3506 | 5663 | JMP I   | DISRET    | /AND RETURN TO CALLER                                     |
| 2217 |      |      |         |           | /SYSTEM WILL GIVE THE EXTRA CRLF AFTER                    |
| 2220 |      |      |         |           | /DECIDING IF THERES ANY BINARY,                           |
| 2221 |      |      | /       |           |                                                           |
| 2222 |      |      | /       |           |                                                           |
| 2223 |      |      | /       |           |                                                           |
| 2224 | 3507 | 7677 | LWDEFD, | 7677      | /MAGIC NUMBER FOR FORM FEED PROBLEM AT SYMBOL TABLE TIME, |
| 2225 |      |      | /       |           |                                                           |
| 2226 |      |      | /       |           |                                                           |
| 2227 |      |      | /       |           |                                                           |
| 2230 |      |      | /       |           |                                                           |
| 2231 |      |      | /       |           |                                                           |
| 2232 | 3510 | 0316 | MESS,   | 316       | /"NO"                                                     |
| 2233 | 3511 | 0317 |         | 317       |                                                           |
| 2234 | 3512 | 0240 |         | 240       | /" ERRORS"                                                |
| 2235 | 3513 | 0305 |         | 305       |                                                           |
| 2236 | 3514 | 0322 |         | 322       |                                                           |
| 2237 | 3515 | 0322 |         | 322       |                                                           |
| 2240 | 3516 | 0317 |         | 317       |                                                           |
| 2241 | 3517 | 0322 |         | 322       |                                                           |
| 2242 | 3520 | 0323 | MESSS,  | 323       | /FINAL OPTIONAL S                                         |
| 2243 | 3521 | 0000 |         | 0000      | /TERMINATING ZERO WORD                                    |
| 2244 |      |      | /       |           |                                                           |
| 2245 |      |      | /       |           |                                                           |
| 2246 |      |      | /       |           |                                                           |
| 2247 |      |      | /       |           |                                                           |
| 2250 |      |      | /       |           |                                                           |
| 2251 | 3522 | 7777 | PCONT,  | 7777      |                                                           |
| 2252 | 3523 | 0043 | P43,    | 43        |                                                           |
| 2253 | 3524 | 0000 | PAGSET, | 0         |                                                           |
| 2254 | 3525 | 2322 | ISZ     | PCONT     |                                                           |
| 2255 | 3526 | 5341 | JMP     | COSA      |                                                           |
| 2256 | 3527 | 1135 | TAD     | M4        |                                                           |
| 2257 | 3530 | 3016 | DCA     | 16        |                                                           |
| 2260 | 3531 | 1323 | TAD     | P43       |                                                           |
| 2261 | 3532 | 1153 | TATA,   | TAD P212  |                                                           |
| 2262 | 3533 | 4401 | JMS I   | DECOU     |                                                           |
| 2263 | 3534 | 1154 | TAD     | P215      |                                                           |
| 2264 | 3535 | 4401 | JMS I   | DECOU     |                                                           |
| 2265 | 3536 | 2016 | ISZ     | 16        |                                                           |
| 2266 | 3537 | 5332 | JMP     | TATA      |                                                           |
| 2267 | 3540 | 5724 | JMP I   | PAGSET    |                                                           |
| 2270 | 3541 | 1322 | COSA,   | TAD PCONT |                                                           |
| 2271 | 3542 | 7041 | CIA     |           |                                                           |
| 2272 | 3543 | 1127 | TAD     | P76       |                                                           |
| 2273 | 3544 | 7640 | SZA     | CLA       |                                                           |

|      |      |      |               |
|------|------|------|---------------|
| 2274 | 3545 | 5724 | JMP I PAGSET  |
| 2275 | 3546 | 7240 | CLA CMA       |
| 2276 | 3547 | 3322 | DCA PCONT     |
| 2277 | 3550 | 5724 | JMP I PAGSET  |
| 2300 | 3551 | 0000 | LINK8A, 0     |
| 2301 | 3552 | 6211 | 6211          |
| 2302 | 3553 | 4767 | JMS I GYP     |
| 2303 | 3554 | 6201 | 6201          |
| 2304 | 3555 | 1014 | TAD TYPEA     |
| 2305 | 3556 | 0144 | AND P1777     |
| 2306 | 3557 | 7650 | SNA CLA       |
| 2307 | 3560 | 5751 | JMP I LINK8A  |
| 2310 | 3561 | 1014 | TAD TYPEA     |
| 2311 | 3562 | 7106 | RTL CLL       |
| 2312 | 3563 | 0136 | AND P6000     |
| 2313 | 3564 | 1024 | TAD VAL       |
| 2314 | 3565 | 3024 | DCA VAL       |
| 2315 | 3566 | 5751 | JMP I LINK8A  |
| 2316 | 3567 | 3623 | GYP, GETYPE   |
| 2317 | 3570 | 7300 | THOU, CLA CLL |
| 2320 | 3571 | 1053 | TAD LWC       |
| 2321 | 3572 | 7012 | RTR           |
| 2322 | 3573 | 7012 | RTR           |
| 2323 | 3574 | 3164 | DCA CON       |
| 2324 | 3575 | 5776 | JMP I C1A     |
| 2325 | 3576 | 1055 | C1A, C        |
| 2326 |      |      | /             |
| 2327 |      |      | /             |
| 2330 |      |      | /             |
| 2331 |      |      | /             |
| 2332 |      |      | EJECT         |

*11 line mode, add on the two wind bits  
for printout*

```

2333      /PDP-12 ASSEMBLER PAGE 15
2334      /
2335      /
2336      *3600
2337      /
2340      3600 0000  CONVRT, 0
2341      3601 0160      AND P77
2342      3602 1013      TAD M40
2343      3603 7510      SPA
2344      3604 1351      TAD P100
2345      3605 1134      TAD P240
2346      3606 5600      JMP I CONVRT
2347      /
2350      /
2351      /
2352      /
2353      3607 0000  UPLN, 0
2354      3610 4500      JMS I TYCARI
2355      3611 1261      TAD M26
2356      3612 4475      JMS I SKIP
2357      3613 1262      TAD M60A
2360      3614 3173      DCA SPCUNT
2361      3615 5607      JMP I UPLN
2362      3616 0000  NOTBLK, 0
2363      3617 1011      TAD BINBLK
2364      3620 4513      JMS I SRITIC
2365      3621 4253      JMS SETUSE
2366      3622 5616      JMP I NOTBLK
2367      3623 0000  GETYPE, 0
2370      3624 1423      TAD I SADR
2371      3625 0136      AND P6000
2372      3626 3014      DCA TYPEA
2373      3627 2023      ISZ SADR
2374      3630 1423      TAD I SADR
2375      3631 0150      AND P4000
2376      3632 7112      CLL RTR
2377      3633 1014      TAD TYPEA
2400      3634 3014      DCA TYPEA
2401      3635 2023      ISZ SADR
2402      3636 1423      TAD I SADR
2403      3637 7012      RTR
2404      3640 7010      RAR
2405      3641 0124      AND P400
2406      3642 1014      TAD TYPEA
2407      3643 3014      DCA TYPEA
2410      3644 1014      TAD TYPEA
2411      3645 0136      AND P6000
2412      3646 3021      DCA TYPE
2413      3647 7344      MTHO
2414      3650 1023      TAD SADR
2415      3651 3023      DCA SADR
2416      3652 5623      JMP I GETYPE
2417      3653 0000  SETUSE, 0
2420      3654 1010      TAD BINPTR
2421      3655 4515      JMS I SRD1
2422      3656 1010      TAD BINPTR
2423      3657 3011      DCA BINBLK
2424      3660 5653      JMP I SETUSE
2425      3661 7752      M26, -26
2426      3662 7716      M60A, -62
2427      3663 0000  RESETL, 0
2430      3664 1030      TAD AADR
2431      3665 0136      AND P6000

```

*get 1st word, current symbol*  
*1st type of symbol & place in "TYPE"*  
*get 2nd word, current symbol*  
*get 3rd word*  
*reset to 1st word*

|      |      |      |         |         |
|------|------|------|---------|---------|
| 2432 | 3666 | 3046 | DCA     | ABANK   |
| 2433 | 3667 | 5663 | JMP I   | RESETL  |
| 2434 | 3670 | 0000 | GETIN,  | 0       |
| 2435 | 3671 | 4467 | JMS I   | GETCHR  |
| 2436 | 3672 | 3051 | DCA     | ITEM    |
| 2437 | 3673 | 1035 | TAD     | PASS    |
| 2440 | 3674 | 7710 | CONSKP, | SPA CLA |
| 2441 | 3675 | 4471 | JMS I   | SYMLST  |
| 2442 | 3676 | 1051 | TAD     | ITEM    |
| 2443 | 3677 | 5670 | JMP I   | GETIN   |
| 2444 | 3700 | 7710 | CONYES, | SPA CLA |
| 2445 | 3701 | 1300 | COMMEN, | TAD     |
| 2446 | 3702 | 3274 | DCA     | CONYES  |
| 2447 | 3703 | 4270 | JMS     | CONSKP  |
| 2450 | 3704 | 1122 | TAD     | GETIN   |
| 2451 | 3705 | 7640 | SZA     | M43     |
| 2452 | 3706 | 5301 | JMP     | COMMEN  |
| 2453 | 3707 | 1325 | TAD     | CONREG  |
| 2454 | 3710 | 3274 | DCA     | CONSKP  |
| 2455 | 3711 | 5712 | JMP I   | BITT    |
| 2456 | 3712 | 2047 | BITT,   | BIT     |
| 2457 |      |      | /       |         |
| 2460 |      |      | /       |         |
| 2461 | 3713 | 0000 | RAND,   | 0       |
| 2462 | 3714 | 0164 | AND     | CON     |
| 2463 | 3715 | 3164 | DCA     | CON     |
| 2464 | 3716 | 5713 | JMP I   | RAND    |
| 2465 |      |      | /       |         |
| 2466 |      |      | /       |         |
| 2467 |      |      | /       |         |
| 2470 |      |      | /       |         |
| 2471 | 3717 | 1035 | ENDMS,  | TAD     |
| 2472 | 3720 | 7710 | SPA     | PASS    |
| 2473 | 3721 | 5516 | JMP I   | CLA     |
| 2474 | 3722 | 5472 | JMP I   | RET1    |
| 2475 |      |      | /       |         |
| 2476 |      |      | /       |         |
| 2477 |      |      | /       |         |
| 2500 | 3723 | 0000 | PGJS,   | 0       |
| 2501 | 3724 | 1003 | TAD     | LISTWD  |
| 2502 | 3725 | 7710 | CONREG, | SPA CLA |
| 2503 | 3726 | 5723 | JMP I   | PGJS    |
| 2504 | 3727 | 1736 | TAD I   | PCOT    |
| 2505 | 3730 | 1356 | TAD     | M77     |
| 2506 | 3731 | 3017 | DCA     | 17      |
| 2507 | 3732 | 4500 | JMS I   | TYCARI  |
| 2510 | 3733 | 2017 | ISZ     | 17      |
| 2511 | 3734 | 5332 | JMP     | , -2    |
| 2512 | 3735 | 5723 | JMP I   | PGJS    |
| 2513 | 3736 | 3522 | PCOT,   | PCONT   |
| 2514 |      |      | /       |         |
| 2515 |      |      | /       |         |
| 2516 |      |      | /       |         |
| 2517 | 3737 | 0000 | LNCK,   | 0       |
| 2520 | 3740 | 2173 | ISZ     | SPCUNT  |
| 2521 | 3741 | 5343 | JMP ,+2 |         |
| 2522 | 3742 | 4207 | JMS     | UPLN    |
| 2523 | 3743 | 5737 | JMP I   | LNCK    |
| 2524 | 3744 | 0000 | SPACK,  | 0       |
| 2525 | 3745 | 3007 | DCA     | TEMP    |
| 2526 | 3746 | 1007 | TAD     | TEMP    |
| 2527 | 3747 | 7040 | CMA     |         |
| 2530 | 3750 | 1173 | TAD     | SPCUNT  |

/GET A CHARACTER

*from input*

/SPA FOR NORMAL, SKP FOR ABBREVIATED!!

*Do "AND"*

/FINAL UPPER

/-(TEMP+1)

|      |      |      |           |        |          |                      |
|------|------|------|-----------|--------|----------|----------------------|
| 2531 | 3751 | 7500 | P100,     | SMA    |          | /TOO FAR???          |
| 2532 | 3752 | 7240 |           | CLA    | CMA      |                      |
| 2533 | 3753 | 3173 |           | DCA    | SPCUNT   |                      |
| 2534 | 3754 | 1007 |           | TAD    | TEMP     |                      |
| 2535 | 3755 | 5744 |           | JMP    | I SPACK  |                      |
| 2536 |      |      | /         |        |          |                      |
| 2537 | 3756 | 7701 | M77,      | -77    |          |                      |
| 2540 |      |      | /         |        |          |                      |
| 2541 |      |      | /         |        |          |                      |
| 2542 |      |      | /         |        |          |                      |
| 2543 |      |      | /         |        |          |                      |
| 2544 |      |      | /         |        |          |                      |
| 2545 | 3757 | 0000 | STOCHK, 0 |        |          |                      |
| 2546 | 3760 | 1171 |           | TAD    | SNUM     |                      |
| 2547 | 3761 | 7040 |           | CMA    |          |                      |
| 2550 | 3762 | 1032 |           | TAD    | PERMA    |                      |
| 2551 | 3763 | 7710 |           | SPA    | CLA      |                      |
| 2552 | 3764 | 5757 |           | JMP    | I STOCHK |                      |
| 2553 | 3765 | 1370 |           | TAD    | P5       |                      |
| 2554 |      |      | /         |        |          |                      |
| 2555 | 3766 | 4476 | EQUERR,   | JMS    | I ERR1   | /GIVE ERROR MESSAGE, |
| 2556 | 3767 | 5771 |           | JMP    | I EDEFIN |                      |
| 2557 |      |      | /         |        |          |                      |
| 2560 | 3770 | 0005 | P5,       | 5      |          |                      |
| 2561 | 3771 | 1416 | EDEFIN,   | MAINSF |          |                      |
| 2562 |      |      | /         |        |          |                      |
| 2563 |      |      | /         |        |          |                      |
| 2564 |      |      | /         |        |          |                      |
| 2565 |      |      | /         |        |          |                      |
| 2566 |      |      |           | EJECT  |          |                      |

```

2567      /
2570      /
2571      /
2572      /
2573      /
2574
2575      /ASSEMBLER PAGE16      !!!!
2576      *4001      /4000 IS THE JMP RET STORE IN LMODE.
2577      4001 3463      DISRE, DISRET
2578      *4002
2600      4002 1370      FSBLK, 1000+D7
2601      4003 0000      FSUNIT, 0
2602      4004 2400      AALPHA, ALPHA
2603      4005 7774      LTREAD, READ
2604      4006 0000      BUFIN, 0      /GETS NEXT BUFFER FROM INPUT UNIT.
2605      4007 4605      JMS I LTREAD
2606      4010 4175      BBLOCK
2607      4011 2377      ISZ BUFINZ
2610      4012 5606      JMP I BUFIN
2611      4013 0000      INIT, 0      /INITIALIZATION ROUTINE: CALLED AT BEGINNING OF EACH PASS.
2612      4014 4500      JMS I TYCARI
2613      4015 1202      TAD FSBLK
2614      4016 0374      AND L777
2615      4017 3377      DCA BUFINZ
2616      4020 1203      TAD FSUNIT
2617      4021 3375      DCA BBLOCK
2620      4022 1241      TAD P2270
2621      4023 3010      DCA BINPTR
2622      4024 4643      JMS I TOUSE      /SET UP POINTER, SETUSE WILL SET BINBLK
2623      4025 1155      TAD P7400      /CALL SET USE NOW.
2624      4026 3645      DCA I CHO
2625      4027 3642      DCA I WDH
2626      4030 1246      TAD P4400
2627      4031 3647      DCA I PTB
2630      4032 4206      JMS BUFIN
2631      4033 3037      DCA MODE
2632      4034 4605      JMS I LTREAD      /RESET TO LMODE AND READ IN SYMBOL TABLE.
2633      4035 4170      LTAB
2634      4036 7240      CLA CMA
2635      4037 3012      DCA CURLIN
2636      4040 5613      JMP I INIT
2637      4041 2400      P2270, 2000+D7+10
2640      4042 3351      WDH, WDHAF
2641      4043 3653      TOUSE, SETUSE
2642      4044 4337      USES, USEDNO
2643      4045 3354      CHO, CHRC
2644      4046 4400      P4400, 4400
2645      4047 3352      PTB, PTBUFP
2646      4050 5737      OCHECK, CHECKO
2647      4051 1011      RETURN, TAD BINBLK
2650      4052 4513      JMS I SRITIC
2651      4053 1244      TAD USES
2652      4054 3016      DCA 16
2653      4055 1013      TAD M40
2654      4056 3017      DCA 17
2655      4057 3644      DCA I USES
2656      4060 1416      TAD I 16
2657      4061 7710      SPA CLA
2660      4062 2644      ISZ I USES
2661      4063 2017      ISZ 17
2662      4064 5260      JMP ,+4
2663      4065 4763      JMS I RWRITE
2664      4066 4164      BINHDR
2665      4067 4601      JMS I DISRE

```

|      |      |      |         |          |                                           |
|------|------|------|---------|----------|-------------------------------------------|
| 2666 | 4070 | 4650 | JMS I   | OCHECK   | /CHECK FOR NO BINARY, ALSO GIVE CARRIAGE  |
| 2667 |      |      |         |          | /RETURN AFTER ERROS MESSAGE, MUST GO      |
| 2670 |      |      |         |          | /AFTER CALL TO DISRET                     |
| 2671 | 4071 | 4604 | JMS I   | AALPHA   |                                           |
| 2672 | 4072 | 4674 | JMS I   | PSTS     |                                           |
| 2673 | 4073 | 5517 | JMP I   | MON      |                                           |
| 2674 | 4074 | 3723 | PGJS    |          |                                           |
| 2675 |      |      | PSTS,   |          |                                           |
| 2676 |      |      | /       |          |                                           |
| 2677 |      |      | /       |          |                                           |
| 2700 |      |      | /       |          |                                           |
| 2701 | 4075 | 3733 | KILBUF, | DCA I    | PCOUNT                                    |
| 2702 | 4076 | 1731 | TAD I   | PINP     |                                           |
| 2703 | 4077 | 3732 | DCA I   | POUTP    |                                           |
| 2704 | 4100 | 3314 | DCA     | KLOOP    |                                           |
| 2705 |      |      | /       |          |                                           |
| 2706 |      |      | /       |          |                                           |
| 2707 |      |      | /       |          |                                           |
| 2710 |      |      | /       |          |                                           |
| 2711 |      |      | /       |          |                                           |
| 2712 | 4101 | 4736 | MONIT,  | JMS I    | MMMOVE                                    |
| 2713 | 4102 | 6201 | COF     | 0        |                                           |
| 2714 | 4103 | 7000 | 7000    |          |                                           |
| 2715 | 4104 | 6211 | COF     | 10       |                                           |
| 2716 | 4105 | 7000 | 7000    |          |                                           |
| 2717 | 4106 | 1000 | 1000    |          |                                           |
| 2720 | 4107 | 1334 | TAD     | PL215    | /GIVE AN EXTRA C.R.L.F FOR GOOD MEASURE   |
| 2721 | 4110 | 4401 | JMS I   | DECOU    |                                           |
| 2722 | 4111 | 1335 | TAD     | PL212    |                                           |
| 2723 | 4112 | 4401 | JMS I   | DECOU    |                                           |
| 2724 | 4113 | 4401 | JMS I   | DECOU    |                                           |
| 2725 | 4114 | 4404 | KLOOP,  | JMS I    | CNTLPI                                    |
| 2726 | 4115 | 4541 | JMS I   | PCHECK   | /CHECK FOR A C.R. TO TERMINATE THE OUTPUT |
| 2727 | 4116 | 1733 | TAD I   | PCOUNT   |                                           |
| 2730 | 4117 | 7640 | SZA CLA |          |                                           |
| 2731 | 4120 | 5314 | JMP     | KLOOP    | /WAIT UNTIL THE BUFFER IS EMPTY,          |
| 2732 | 4121 | 1330 | TAD     | PL17     |                                           |
| 2733 | 4122 | 6664 | LPR     |          |                                           |
| 2734 | 4123 | 0000 | 0       |          |                                           |
| 2735 | 4124 | 2000 | ISZ     | 0        |                                           |
| 2736 | 4125 | 4323 | JMS     | ,-2      | /WAIT FOR LPR TO TAPE EFFECT              |
| 2737 | 4126 | 6212 | CIF     | 10       |                                           |
| 2740 | 4127 | 5730 | JMP I   | ,-+1     |                                           |
| 2741 | 4130 | 7777 | PL17,   | 7777     |                                           |
| 2742 |      |      | /       |          |                                           |
| 2743 |      |      | /       |          |                                           |
| 2744 | 4131 | 0722 | PINP,   | INP      |                                           |
| 2745 | 4132 | 0705 | POUTP,  | OUTP     |                                           |
| 2746 | 4133 | 0760 | PCOUNT, | COUNT    |                                           |
| 2747 | 4134 | 0215 | PL215,  | 215      |                                           |
| 2750 | 4135 | 0212 | PL212,  | 212      |                                           |
| 2751 | 4136 | 7200 | MMMOVE, | 7200     |                                           |
| 2752 |      |      | /       |          |                                           |
| 2753 |      |      | /       |          |                                           |
| 2754 |      |      | /       |          |                                           |
| 2755 | 4137 | 1030 | NOS,    | TAD AADR |                                           |
| 2756 | 4140 | 0131 | AND     | P3777    |                                           |
| 2757 | 4141 | 5347 | JMP     | PDLK     |                                           |
| 2760 | 4142 | 2020 | PERIOD, | ISZ ANY  |                                           |
| 2761 | 4143 | 1037 | TAD     | MODE     |                                           |
| 2762 | 4144 | 7700 | SMA     | CLA      |                                           |
| 2763 | 4145 | 5352 | JMP     | LINKDT   |                                           |
| 2764 | 4146 | 1030 | TAD     | AAADR    |                                           |

|      |      |      |         |            |
|------|------|------|---------|------------|
| 2765 | 4147 | 4434 | PDLK,   | JMS I SGN  |
| 2766 | 4150 | 4504 |         | JMS I IDX1 |
| 2767 | 4151 | 5503 |         | JMP I B21  |
| 2770 | 4152 | 1030 | LINKDT, | TAD AADR   |
| 2771 | 4153 | 0136 |         | AND P6000  |
| 2772 | 4154 | 7041 |         | CIA        |
| 2773 | 4155 | 1046 |         | TAD ABANK  |
| 2774 | 4156 | 7640 |         | SZA CLA    |
| 2775 | 4157 | 5337 |         | JMP NOS    |
| 2776 | 4160 | 1030 |         | TAD AADR   |
| 2777 | 4161 | 0144 |         | AND P1777  |
| 3000 | 4162 | 5347 |         | JMP PDLK   |
| 3001 |      |      | /       |            |
| 3002 | 4163 | 7775 | RWRITE, | WRITE      |
| 3003 |      |      | /       |            |
| 3004 | 4164 | 0111 | BINH0R, | DIALBINARY |
| 3005 | 4165 | 0010 |         | 10         |
| 3006 | 4166 | 0057 |         | D7+57-370  |
| 3007 | 4167 | 0001 |         | 1          |
| 3010 |      |      | /       |            |
| 3011 | 4170 | 0100 | LTAB,   | DIALUNIT   |
| 3012 | 4171 | 0036 |         | 36         |
| 3013 | 4172 | 0042 |         | D6+42-300  |
| 3014 | 4173 | 0002 |         | 2          |
| 3015 |      |      | /       |            |
| 3016 |      |      | /       |            |
| 3017 | 4174 | 0777 | L777,   | 777        |
| 3020 |      |      | /       |            |
| 3021 | 4175 | 0000 | BBLOCK, | 0          |
| 3022 | 4176 | 0011 |         | 11         |
| 3023 | 4177 | 0000 | BUFINZ, | 0          |
| 3024 | 4200 | 0001 |         | 1          |
| 3025 |      |      | /       |            |
| 3026 |      |      | /       |            |
| 3027 |      |      |         | EJECT      |

```

3030 /ASSEMBLER PAGE 17
3031 /
3032 /
3033 /
3034 /
3035 ASMIFM , -4200
3036 *4200
3037 /
3038 /
3039 /
3040 /
3041 4201 0000 SWRC, 0
3042 4202 1305 TAD SM370
3043 4203 0304 AND SL777
3044 4204 3317 DCA SWOUT
3045 4205 4610 JMS I WWRITE
3046 4206 4315 SIT
3047 4207 5601 JMP I SWRC
3050 /
3051 4210 7775 WWRITE, WRITE
3052 4211 0000 RSUB, 0
3053 4212 3227 DCA SOMEW
3054 4213 1164 TAD CON
3055 4214 3232 DCA SUBCON
3056 4215 1037 TAD MODE
3057 4216 7700 SMA CLA
3060 4217 5225 JMP LNKSUB
3061 4220 1227 TAD SOMEW
3062 4221 7041 CIA
3063 4222 1164 TAD CON
3064 4223 3164 RSUBRT, DCA CON
3065 4224 5611 JMP I RSUB
3066 4225 6141 LNKSUB, LINC
3067 LMODE
3070 0226 1020 LDA I
3071 0227 0000 SOMEW, 0
3072 0230 0017 COM
3073 0231 1120 ADA I
3074 0232 0000 SUBCON, 0
3075 0233 0002 MSC 2
3076 PMODE
3077 4234 5223 JMP RSUBRT
3100 4235 0000 RADD, 0
3101 4236 3252 DCA WHERE
3102 4237 1164 TAD CON
3103 4240 3254 DCA ADDCON
3104 4241 1037 TAD MODE
3105 4242 7700 SMA CLA
3106 4243 5250 JMP LNKADD
3107 4244 1252 TAD WHERE
3110 4245 1164 TAD CON
3111 4246 3164 RADDRT, DCA CON
3112 4247 5635 JMP I RADD
3113 4250 6141 LNKADD, LINC
3114 LMODE
3115 0251 1020 LDA I
3116 0252 0000 WHERE, 0
3117 0253 1120 ADA I
3120 0254 0000 ADDCON, 0
3121 0255 0002 MSC 2
3122 PMODE
3123 4256 5246 JMP RADDRT
3124 /
3125 /
3126 /

```

$c(CON) - c(AC)$

$c(CON) + c(AC)$

PSEUDO OP SETTER AND PAGER

```

3127      /
3130      /
3131      /ROUTINE PAGES IN PSEUDO OP PROCESSOR WITH NUMBER IN AC, PROCESSOR DOES THE REST.
3132      /FOR 8K, PSNOPS ARE MADE INTO NOPS, AND PPSEUDO IS CHANGED,
3133      /
3134      4257 2273 PSUD07, ISZ      OP1
3135      4260 2273 PSUD06, ISZ      OP1
3136      4261 2273 PSUD05, ISZ      OP1
3137      4262 2273 PSUD04, ISZ      OP1
3140      4263 2273 PSUD03, ISZ      OP1
3141      4264 2273 PSUD02, ISZ      OP1
3142      4265 7300 PSUD01, CLA CLL
3143      4266 1273      TAD      OP1
3144      4267 4674      JMS I      PPSEUDO
3145      4270 3201 PAFTER, DCA      PSRET
3146      4271 3273      DCA      OP1
3147      4272 5601      JMP I      PSRET
3150      4273 0000 OP1,      0
3151      PSRET=SWRC
3152      4274 5410 PPSEUDO, PSEUDO      /PSEUDO + CORRECT ORIGIN
3153      4275 0000 SREAD1, 0
3154      4276 1305      TAD      SM370      /SUBTRACT OFF THE EXTRANEIOUS CRAP
3155      4277 0304      AND      SL777
3156      4300 3317      DCA      SWOUT
3157      4301 4706      JMS I      WREAD
3160      4302 4315      SIT
3161      4303 5675      JMP I      SREAD1
3162      /
3163      4304 0777 SL777, 777
3164      4305 7410 SM370, -370
3165      /
3166      4306 7774 WREAD, READ
3167      4307 0000 POPUP1, 0      /PUT LAST ENTRY IN PUSH
3170      4310 7240      CLA CMA
3171      4311 1047      TAD POINT
3172      4312 3047      DCA POINT
3173      4313 1447      TAD I POINT
3174      4314 5707      JMP I POPUP1
3175      /
3176      4315 0111 SIT,      DIALBINARY
3177      4316 0012      12
3200      4317 0000 SWOUT, 0
3201      4320 0001      1
3202      /
3203      /
3204      /
3205      /
3206      /
3207      /
3210      /
3211      4321 0000 UBITS, 0
3212      4322 1011      TAD      BINBLK
3213      4323 1330      TAD      FACTOR
3214      4324 3007      DCA      TEMP
3215      4325 7240      CLA CMA
3216      4326 3407      DCA I      TEMP
3217      4327 5721      JMP I      UBITS
3220      /
3221      /
3222      4330 1750 FACTOR, BLKUSE=2000-D7      /CORE HEADER BLOCK MAPPER,
3223      /
3224      /
3225      /

```

|      |      |      |                 |
|------|------|------|-----------------|
| 3226 |      | /    |                 |
| 3227 |      | /    |                 |
| 3230 |      | /    |                 |
| 3231 |      | /    |                 |
| 3232 |      | /    |                 |
| 3233 |      | /    |                 |
| 3234 |      | /    |                 |
| 3235 |      | /    |                 |
| 3236 |      | /    |                 |
| 3237 |      | /    |                 |
| 3240 |      | /    |                 |
| 3241 |      | /    |                 |
| 3242 |      | /    |                 |
| 3243 | 4331 | 0100 | VSAVE, DIALUNIT |
| 3244 | 4332 | 0015 | 15              |
| 3245 | 4333 | 0025 | SCRATCH         |
| 3246 | 4334 | 0001 | 1               |
| 3247 |      | /    |                 |
| 3250 |      | /    |                 |
| 3251 |      | /    |                 |
| 3252 |      | /    |                 |
| 3253 |      | /    |                 |
| 3254 |      | /    |                 |
| 3255 |      | /    |                 |
| 3256 |      | /    |                 |
| 3257 |      | /    |                 |
| 3260 |      | /    |                 |
| 3261 |      | /    |                 |
| 3262 |      | /    |                 |
| 3263 |      | /    |                 |
| 3264 |      | /    |                 |
| 3265 |      | /    |                 |
| 3266 |      | /    |                 |
| 3267 |      | /    |                 |
| 3270 |      | /    |                 |
| 3271 |      | /    |                 |
| 3272 |      | /    |                 |
| 3273 |      | /    |                 |
| 3274 |      | /    |                 |
| 3275 |      | /    |                 |
| 3276 |      |      | *4337           |
| 3277 | 4337 | 0000 | USEDNO, 0       |
| 3300 |      |      | *4340           |
| 3301 |      |      | BL KUSE =,      |
| 3302 | 4340 | 0000 | 0               |
| 3303 | 4341 | 0000 | 0               |
| 3304 | 4342 | 0000 | 0               |
| 3305 | 4343 | 0000 | 0               |
| 3306 | 4344 | 0000 | 0               |
| 3307 | 4345 | 0000 | 0               |
| 3310 | 4346 | 0000 | 0               |
| 3311 | 4347 | 0000 | 0               |
| 3312 | 4350 | 0000 | 0               |
| 3313 | 4351 | 0000 | 0               |
| 3314 | 4352 | 0000 | 0               |
| 3315 | 4353 | 0000 | 0               |
| 3316 | 4354 | 0000 | 0               |
| 3317 | 4355 | 0000 | 0               |
| 3320 | 4356 | 0000 | 0               |
| 3321 | 4357 | 0000 | 0               |
| 3322 | 4360 | 0000 | 0               |
| 3323 | 4361 | 0000 | 0               |
| 3324 | 4362 | 0000 | 0               |

/THE ACTUAL DIAL AREA  
 /LOCATIONS 6400-6777  
 /A SCRATCH AREA ON THE TAPE  
 /1 BLOCK IN OR OUT,

|      |      |      |   |
|------|------|------|---|
| 3325 | 4363 | 0000 | 0 |
| 3326 | 4364 | 0000 | 0 |
| 3327 | 4365 | 0000 | 0 |
| 3330 | 4366 | 0000 | 0 |
| 3331 | 4367 | 0000 | 0 |
| 3332 | 4370 | 0000 | 0 |
| 3333 | 4371 | 0000 | 0 |
| 3334 | 4372 | 0000 | 0 |
| 3335 | 4373 | 0000 | 0 |
| 3336 | 4374 | 0000 | 0 |
| 3337 | 4375 | 0000 | 0 |
| 3340 | 4376 | 0000 | 0 |
| 3341 | 4377 | 0000 | 0 |
| 3342 |      |      | / |
| 3343 |      |      | / |
| 3344 |      |      | / |
| 3345 |      |      | / |
| 3346 |      |      | / |
| 3347 |      |      | / |
| 3350 |      |      | / |
| 3351 |      |      | / |
| 3352 |      |      | / |
| 3353 |      |      | / |
| 3354 |      |      | / |
| 3355 |      |      |   |

EJECT

```
3356 /
3357 /
3360 /
3361 /
3362 /
3363 /
3364 /
3365 /
3366 /
3367 /
3370 /
3371 /
3372 /
3373 /
3374 /
3375 /
3376 /
3377 /
3400 /
3401 /
3402 /      END OF ASSEMTWO, FETCH ASSEM3.
3403 /
3404 /
3405 /
3406 /
3407 /
3410 /
3411 /
3412 /
3413 /
3414 /
3415 /
3416 /      CHAIN  "ASSEM3"
```

```
0000      *20
0001      PMODE
0002      /
0003      /
0004      /
0005      /
0006      /
0007      /
0010      /
0011      BEGINNING OF ASSETHREE
0012      /
0013      /
0014      /
0015      /
0016      /
0017      /
0020      /
0021      /
0022      /
0023      /
0024      /
0025      /
0026      /
0027      /
0030      /
0031      /
0032      EJECT
-
```

```

0033      /      M A I N   I N I T I A L I Z A T I O N   O F   T H E   A S S E M B L E R
0034      /
0035      /
0036      /
0037      /
0040      /
0041      /
0042      *5000
0043      5000 7200 BEGNLI, CLA      /ENTRY IF LIST IS DESIRED.
0044      5001 7410 SKP
0045      5002 7240 BEGNAS, CLA CMA /ENTRY IF JUST ASSEMBLY.
0046      5003 3003 DCA LISTWD /SAVE LIST STATUS CONDITION,
0047      5004 1621 TAD I P4775 /4775 CONTAINS STARTING LINE NUM. OF LISTING
0050      5005 3620 DCA I PJSTART
0051      5006 1622 TAD I P4776 /4776 CONTAINS ENDING LINE NUM, STASH IT AWAY PERMANENTLY.
0052      5007 3617 DCA I PJEND
0053      5010 1003 TAD LISTWD
0054      5011 7650 SNA CLA
0055      5012 5216 JMP ,+4
0056      5013 1214 TAD ,+1
0057      5014 7000 JBNOP, NOP
0060      5015 3637 DCA I PJNOP /IF NO LIST THEN FIX JBTEST SO IT CAN NEVER LIST OUT.
0061      5016 5240 JMP PJNOP+1
0062      5017 1734 PJEND, JEND
0063      5020 1733 PJSTART, JSTART
0064      5021 4775 P4775, 4775
0065      5022 4776 P4776, 4776
0066      5023 4774 P4774, 4774
0067      5024 3042 BANOP1, ABNOP1
0070      5025 3043 BANOP2, ABNOP2
0071      5026 3044 BANOP3, ABNOP3
0072      5027 3045 BANOP4, ABNOP4
0073      5030 3700 BANOP5, CONYES
0074      5031 7610 BAND0, SKP CLA
0075      5032 1374 BLTAD, 1200+M7TWO-ENTS
0076      5033 2107 BLTAD1, NOTAB1
0077      5034 2124 BLTAD2, NOTAB2
0100      5035 7770 BLM10, -10
0101      5036 2535 BLPM22, M22
0102      5037 1730 PJNOP, JNOP
0103      5040 2623 ISZ I P4774 /4774=7777 IF QUICK LIST,
0104      5041 5262 JMP LISREG /NO QUICK LIST,
0105      5042 1214 TAD JBNOP /CHANGE LIST ROUTINE FOR QL BY ALL FOLLOWING CRAP.
0106      5043 3624 DCA I BANOP1
0107      5044 1214 TAD JBNOP
0110      5045 3625 DCA I BANOP2
0111      5046 1214 TAD JBNOP
0112      5047 3626 DCA I BANOP3
0113      5050 1214 TAD JBNOP
0114      5051 3627 DCA I BANOP4
0115      5052 1231 TAD BAND0
0116      5053 3630 DCA I BANOP5
0117      5054 1232 TAD BLTAD
0120      5055 3633 DCA I BLTAD1
0121      5056 1232 TAD BLTAD
0122      5057 3634 DCA I BLTAD2
0123      5060 1235 TAD BLM10
0124      5061 3636 DCA I BLPM22
0125      5062 1473 LISREG, TAD I INAS
0126      5063 3472 DCA I MAINEX /INITILIZE TO PASS1,
0127      5064 6141 LINC
0130      LMODE
0131      1065 0641 LDF 1 /SET UP PRINTER CHECK

```

|      |      |      |                                         |                                                               |
|------|------|------|-----------------------------------------|---------------------------------------------------------------|
| 0132 | 1066 | 1020 | LDA I                                   | /NOW CHANGING TTY ROUTINE TO CALL LINE PRINTER CHECK ROUTINE. |
| 0133 | 1067 | 5731 | TYPO+3202                               | /TYPE=2571+3202=5773(JMP I TYPO+2)                            |
| 0134 | 1070 | 1040 | STA                                     |                                                               |
| 0135 | 1071 | 2530 | TYPO+1                                  |                                                               |
| 0136 | 1072 | 1020 | LDA I                                   |                                                               |
| 0137 | 1073 | 6244 | LPTEST                                  |                                                               |
| 0140 | 1074 | 1040 | STA                                     |                                                               |
| 0141 | 1075 | 2531 | TYPO+2                                  |                                                               |
| 0142 | 1076 | 0643 | LDF 3                                   |                                                               |
| 0143 | 1077 | 1000 | LDA                                     |                                                               |
| 0144 | 1100 | 0777 | 777                                     |                                                               |
| 0145 | 1101 | 4003 | STC FSUNIT&1777                         |                                                               |
| 0146 | 1102 | 1000 | LDA                                     |                                                               |
| 0147 | 1103 | 0002 | FSBLK&1777                              |                                                               |
| 0150 | 1104 | 1560 | BCL I                                   |                                                               |
| 0151 | 1105 | 7000 | -777                                    |                                                               |
| 0152 | 1106 | 4002 | STC FSBLK&1777                          |                                                               |
| 0153 | 1107 | 0002 | PDP                                     |                                                               |
| 0154 |      |      | PMODE                                   |                                                               |
| 0155 | 5110 | 6212 | CIF 10                                  | /PREPARE TO MOVE DOWN THEREAD ROUTINES.                       |
| 0156 | 5111 | 4730 | JMS I IMMOVE                            |                                                               |
| 0157 | 5112 | 6211 | CDF 10                                  |                                                               |
| 0160 | 5113 | 7000 | 7000                                    |                                                               |
| 0161 | 5114 | 6201 | CDF 0                                   |                                                               |
| 0162 | 5115 | 7000 | 7000                                    |                                                               |
| 0163 | 5116 | 1000 | 1000                                    |                                                               |
| 0164 | 5117 | 4731 | JMS I IMREAD                            |                                                               |
| 0165 | 5120 | 5133 | ITABIN                                  |                                                               |
| 0166 | 5121 | 1072 | TAD MAINEX                              | /FAKE START OF ASSEMBLY BY PLACING                            |
| 0167 | 5122 | 3460 | DCA I INIT1                             | /STARTING ADDRESS OF ASSEMBLER IN IT                          |
| 0170 | 5123 | 1326 | TAD IM215                               | /GET A C.R. TO INITIALIZE THE SYSTEM                          |
| 0171 | 5124 | 4727 | JMS I IMTYPO                            | /AND SEND IT DIRECTLY TO THE OUTPUTER                         |
| 0172 | 5125 | 5732 | JMP I IMINIT                            | /AND BY JUMPING TO THE SECOND LOC OF INIT                     |
| 0173 |      |      | /START THE ASSEMBLER GOING RIGHT NOW... |                                                               |
| 0174 |      |      | /                                       |                                                               |
| 0175 |      |      | /                                       |                                                               |
| 0176 |      |      | /                                       |                                                               |
| 0177 | 5126 | 0215 | IM215, 215                              | /C.R.                                                         |
| 0200 | 5127 | 2527 | IMTYPO, TYPO                            |                                                               |
| 0201 | 5130 | 7200 | IMMOVE, 7200                            |                                                               |
| 0202 | 5131 | 7774 | IMREAD, READ                            |                                                               |
| 0203 | 5132 | 4014 | IMINIT, INIT+1                          |                                                               |
| 0204 |      |      | /                                       |                                                               |
| 0205 |      |      | /                                       |                                                               |
| 0206 |      |      | /                                       |                                                               |
| 0207 | 5133 | 0100 | ITABIN, DIALUNIT                        |                                                               |
| 0210 | 5134 | 0013 | 13                                      |                                                               |
| 0211 | 5135 | 0026 | FUDGE1                                  |                                                               |
| 0212 | 5136 | 0002 | 2                                       |                                                               |
| 0213 |      |      | /                                       |                                                               |
| 0214 |      |      | /                                       |                                                               |
| 0215 |      |      | /                                       |                                                               |
| 0216 |      |      | /                                       |                                                               |
| 0217 |      |      | /                                       |                                                               |
| 0220 |      |      | EJECT                                   |                                                               |

```

0221      /          PSEUDO OP PROCESSOR, IT REALLY IS ORIGINATED AT
0222      /          5000 FOR 4K, BUT IT WOULD THEN OVERLAY THE INITIALIZATION ROUTINE. (AT ASSEMBLY).
0223      /THUS ITS ASSEMBLED AT 7000 BUT PFUDGE MAKES THE INTERPAGE REFERENCE LOOK LIKE IT S AT 5000.
0224      / THE 8K VERSION IS REALLY ORIGINATED AT 7000. I GENERALLY CHANGE THE INTERPAGE
0225      /REFERENCES BY HAND(THERE S ABOUT 6 ON THIS PAGE AND 3 ON THE NEXT PAGE.)
0226      /ALL INTERBANK REFERENCE HAVE A
0227      /TO THE DATA ITEMS.
0230      /REMEMBER, THIS IS FINE AS IS FOR "FUDGE4". FOR FUDGE5 ADD 2000 TO INTERPAGE REFERENCES.
0231      /
0232      /
0233      /
0234      /
0235      /
0236      /
0237      /
0240      *5400          /NEW LOCATION OF PSEUDO PROCESSOR
0241      /
0242      /
0243      /
0244      /          MAIN PSEUDO OP PROCESSOR, PAGED IN!
0245      /
0246      /
0247      /
0250      5400 1202 ACCOUNT, 1202          /"JB" ON THE TAPE; IN CORE IT S A TEMPORARY
0251      5401 4377 GETVAL, 4360+VERSION    /VERSION NUMBER ON TAPE, IN CORE IT S
0252      /A SUBROUTINE ENTRY,
0253      /NO.7
0254      / (CHECKS UNIT 1 TO SEE
0255      /IF PRESENT AND IF IT IS USES UNIT
0256      /1 TO READ IN THIS PAGE
0257      /
0260      5402 4477      JMS I  EXPR1
0261      5403 1020      TAD  ANY
0262      5404 7650      SNA CLA
0263      5405 5303      JMP  ASSNO
0264      5406 1033      TAD  EVAL
0265      5407 5601      JMP I  GETVAL
0266      /
0267      /GETVAL EVALUATES THE EXPRESION TO THE RIGHT OF A PSEUDO OP,
0270      /IF NO EXPRESSION THEN IT EXITS.
0271      /BACK TO THE MAIN ASSEMBLER WITHOUT DOING ANYTHING.
0272      /
0273      /
0274      /
0275      /
0276      /
0277      5410 4270 PSEUDO, PAFTER          /RETURN IF OVERLAYED BY ITSELF.
0280      5411 1214      TAD  PJMP
0281      5412 3213      DCA  ,+1
0282      5413 5214      JMP  ,+1          /OVERLAYED BY PSEUDO NO,
0283      5414 5215      PJMP, JMP  ,+1    /MAIN TABLE POINTER
0284      5415 5224      JMP  NASSIF      /EVIL IF SPELLED RIGHT
0285      5416 5225      JMP  ZASSIF
0286      5417 5226      JMP  MASSIF
0287      5420 5353      JMP  SYML0D
0288      5421 5316      JMP  AEJECT
0289      5422 5333      JMP  SYMSAV
0290      5423 5235      JMP  SKIPAS
0291      /
0292      /
0293      /END OF JUMP TABLE. PSEUDO WAS CALLED WITH PSEUDO OP NUMBER IN AC.
0294      /
0295      /

```

```

0320      /
0321      /
0322      /
0323      5424 1305 MASSIF, TAD AMM10
0324      5425 1013 MASSIF, TAD M40
0325      5426 1320 MASSIF, TAD L7710
0326      5427 3231 DCA ,+2 /STORE CORRECT SKP AFTER EVALUATING EXPR.
0327      5430 4201 JMS GETVAL
0330      5431 0000 0
0331      5432 5303 JMP ASSNO /WRONG SKIP CONDITION. ASSEMBLE NEXT.
0332      5433 7240 CLA CMA /DO AN ASMSKP 1.
0333      5434 5241 JMP ASSYIN
0334      5435 4201 SKIPAS, JMS GETVAL
0335      5436 7041 CIA
0336      5437 7450 SNA /NUMBER TO BE SKKIPED. IF ZERO EXIT,
0337      5440 5303 JMP ASSNO
0340      5441 3200 ASSYIN, DCA ACCOUNT
0341      5442 1051 TAD ITEM
0342      5443 1122 TAD M43 /WAS DOLLAR TERMINATOR,
0343      5444 7640 SZA CLA
0344      5445 7240 CLA CMA /NO, ADD 1 TO SKIP BECAUSE OF END OF THIS STATEMENT.
0345      5446 1200 TAD ACCOUNT
0346      5447 3200 DCA ACCOUNT
0347      5450 1312 TAD PACHECK
0350      5451 3711 DCA I PSWITCH
0351      5452 1306 TAD ASEND
0352      5453 3714 DCA I PDOLLAR
0353      5454 1307 TAD AMAINPS
0354      5455 3310 DCA PMAINSA
0355      5456 5657 JMP I ,+1 /AFTER SETTING SWITHCES EXIT TO PICK UP NEXT STATEMENT.
0356      5457 2714 AFDR, FORT
0357      5460 7300 ACHECK, CLA CLL
0360      5461 1051 TAD ITEM
0361      5462 1122 TAD M43
0362      5463 7650 SNA CLA
0363      5464 5267 JMP ASPAST
0364      5465 4504 JMS I IDX1
0365      5466 5261 JMP , -5 /WAIT FOR END OF THIS LINE.
0366      5467 2200 ASPAST, ISZ ACCOUNT
0367      5470 5710 JMP I PMAINSA /NOT DONE YET, PICK UP NEXT STATEMENT TO SKIP.
0370      5471 1313 ASSRST, TAD PDOCAL
0371      5472 3711 DCA I PSWITCH
0372      5473 1315 TAD PENOMS
0373      5474 3714 DCA I PDOLLAR
0374      5475 1310 TAD PMAINSA
0375      5476 5610 JMP I PSEUDO /RESET SWITCHES AND EXIT.
0376      5477 7300 APAEND, CLA CLL
0377      5500 1315 TAD PENOMS
0400      5501 3310 DCA PMAINSA
0401      5502 5271 JMP ASSRST /END OF MANUSCRIPT
0402      /
0403      /
0404      /
0405      /
0406      /
0407      /
0410      /
0411      5503 1257 ASSNO, TAD AFDR
0412      5504 5610 JMP I PSEUDO /GENERAL EXIT.
0413      /
0414      /
0415      /
0416      /

```

```

0417 5505 7770 AMM10, -10
0420 5506 5477 ASEND, APAEND
0421 5507 1504 AMAINPS, MAINSA
0422 5510 1504 PMAINSA, MAINSA
0423 5511 1575 PSWITCH, PSINTER
0424 5512 5460 PACHECK, ACHECK
0425 5513 1426 PDOCAL, DOCAL
0426 5514 2173 PDOLLAR, FINEND
0427 5515 3717 PENDMS, ENDMS
0430 /
0431 /
0432 /
0433 /
0434 5516 4477 AEJECT, JMS I EXPRI /JUST TO SCAN TILL END OF LINE,
0435 5517 4465 JMS I PASSEK /GET THE PASS NO.
0436 5520 7710 L7710, SPA CLA /ALWAYS SKIPS BECAUSE AC=0
0437 5521 5303 JMP ASSNO /NOT LIST OR NOT PASS2
0440 5522 1537 TAD I PM60A /IF LP THEN M60=-140
0441 5523 1331 TAD AEP62 /SEE IF GREATER THAN 62
0442 5524 7700 SMA CLA /IS PRINTER THERE???
0443 5525 5303 JMP ASSNO /NOPE. LINE PRINTER ISN T THERE
0444 5526 7240 CLA CMA /YES. SET PAGE POINTER TO EJECT,
0445 5527 3732 DCA I AEPONT
0446 5530 5303 JMP ASSNO
0447 /
0450 /
0451 5531 0062 AEP62, 62
0452 5532 3522 AEPONT, PCONT
0453 /
0454 /
0455 /
0456 /
0457 /
0460 /
0461 5533 4201 SYMSAV, JMS GETVAL
0462 5534 1035 TAD PASS
0463 5535 7010 RAR
0464 5536 7620 SNL CLA
0465 5537 5303 JMP ASSNO /INCORRECT PASS NO,!!
0466 5540 4772 JMS I VOUT /SAVE 6400 FOR A SECOND,
0467 5541 1027 TAD SEND /NUMBER OF DEFINED SYMBOLS,
0470 5542 3751 DCA I VL6777
0471 5543 4747 JMS I YWRITE
0472 5544 5566 YP1
0473 5545 4773 JMS I VIN /RESTORE 6400 NOW,
0474 5546 5752 JMP I PV8MOVE
0475 /
0476 5547 7775 YWRITE, WRITE
0477 5550 7774 YREAD, READ
0500 5551 6777 VL6777, 6777
0501 5552 5600 PV8MOVE, V8MOVE
0502 5553 1035 SYMLOD, TAD PASS
0503 5554 7640 SZA CLA
0504 5555 5224 JMP NASSIF /DO NOT LOAD SYMBOLS IF PASS2
0505 5556 4772 JMS I VOUT /PRESERVE 6400 NOW
0506 5557 4750 JMS I YREAD
0507 5560 5566 YP1
0510 5561 1751 TAD I VL6777
0511 5562 3027 DCA SEND /GET NUMBER OF SAVED SYMBOLS
0512 5563 4773 JMS I VIN /NOW RESTORE 6400
0513 5564 5765 JMP I VINIT /NOW READ IN THE SYMBOL TABLE
0514 5565 5604 VINIT, VVINIT
0515 /

```

|      |      |      |       |            |
|------|------|------|-------|------------|
| 0516 |      | /    |       |            |
| 0517 |      | /    |       |            |
| 0520 | 5566 | 0111 | YP1,  | DIALBINARY |
| 0521 | 5567 | 0015 |       | 15         |
| 0522 | 5570 | 0077 |       | FUDGE2     |
| 0523 | 5571 | 0001 |       | 1          |
| 0524 |      | /    |       |            |
| 0525 |      | /    |       |            |
| 0526 |      | /    |       |            |
| 0527 |      | /    |       |            |
| 0530 |      | /    |       |            |
| 0531 |      | /    |       |            |
| 0532 |      | /    |       |            |
| 0533 |      | /    |       |            |
| 0534 |      | /    |       |            |
| 0535 |      | /    |       |            |
| 0536 | 5572 | 0773 | VOUT, | VPRSRV     |
| 0537 | 5573 | 0572 | VIN,  | VRSTR      |
| 0540 |      | /    |       |            |
| 0541 |      | /    |       |            |
| 0542 |      | /    |       |            |
| 0543 |      | /    |       |            |
| 0544 |      | /    |       |            |
| 0545 |      | /    |       |            |
| 0546 |      | /    |       |            |
| 0547 |      | /    |       |            |
| 0550 |      | /    |       |            |
| 0551 |      | /    |       |            |
| 0552 |      | /    |       |            |
| 0553 |      | /    |       |            |
| 0554 |      |      |       | EJECT      |

/POINTS TO THE PRESERVE ROUTINE  
/POINTS TO THE RESTORE ROUTINE

|      |      |      |                  |        |                                           |
|------|------|------|------------------|--------|-------------------------------------------|
| 0555 |      |      |                  | *5600  |                                           |
| 0556 |      |      | /                |        |                                           |
| 0557 | 5600 | 4607 | V8MOVE, JMS I    | VWRITE |                                           |
| 0560 | 5601 | 5611 | VTAB             |        |                                           |
| 0561 |      |      | /                |        |                                           |
| 0562 | 5602 | 5603 | VCLEAN, JMP I    | VASSNO |                                           |
| 0563 |      |      | /                |        |                                           |
| 0564 | 5603 | 5503 | VASSNO, ASSNO    |        |                                           |
| 0565 |      |      | /                |        |                                           |
| 0566 |      |      | /                |        |                                           |
| 0567 | 5604 | 4610 | VVINIT, JMS I    | VREAD  |                                           |
| 0570 | 5605 | 5611 | VTAB             |        |                                           |
| 0571 | 5606 | 5202 | JMP              | VCLEAN |                                           |
| 0572 |      |      | /                |        |                                           |
| 0573 |      |      | /                |        |                                           |
| 0574 |      |      | /                |        |                                           |
| 0575 | 5607 | 7775 | VWRITE, WRITE    |        |                                           |
| 0576 | 5610 | 7774 | VREAD, READ      |        |                                           |
| 0577 |      |      | /                |        |                                           |
| 0600 |      |      | /                |        |                                           |
| 0601 |      |      | /                |        |                                           |
| 0602 | 5611 | 0111 | VTAB, DIALBINARY |        |                                           |
| 0603 | 5612 | 0020 | 20               |        |                                           |
| 0604 | 5613 | 0060 | BOOUT            |        |                                           |
| 0605 | 5614 | 0016 | 16               |        |                                           |
| 0606 |      |      | /                |        |                                           |
| 0607 |      |      | /                |        |                                           |
| 0610 |      |      | /                |        |                                           |
| 0611 |      |      | /                |        |                                           |
| 0612 |      |      | /                |        |                                           |
| 0613 |      |      | /                |        |                                           |
| 0614 |      |      | /                |        |                                           |
| 0615 |      |      | /                |        |                                           |
| 0616 |      |      | /                |        |                                           |
| 0617 |      |      | /                |        |                                           |
| 0620 | 5615 | 4724 | OUTPUT, JMS I    | OGTVAL | /THIS ROUTINE ALLOWS OPTIONAL TAPE OUTPUT |
| 0621 | 5616 | 3254 | DCA              | OPUT   | /SAVE FOR A SECOND                        |
| 0622 | 5617 | 1035 | TAD              | PASS   | /GET THE PASS NUMBER                      |
| 0623 | 5620 | 7640 | SZA CLA          |        | /IS IT 1                                  |
| 0624 | 5621 | 5725 | JMP I            | OASSNO | /NOPE, IGNORE                             |
| 0625 | 5622 | 1254 | TAD              | OPUT   | /REGET THE USERS VALUE,                   |
| 0626 | 5623 | 7510 | SPA              |        | /IF POSITIVE THEN OUTPUT                  |
| 0627 | 5624 | 5306 | JMP              | OBSET  | /IF NEGATIVE THEN READ IN BIT TABLE       |
| 0630 | 5625 | 0326 | AND              | OL17   | /JUST GET THE UNIT NUMBER                 |
| 0631 | 5626 | 3333 | DCA              | OUNIT  |                                           |
| 0632 | 5627 | 3335 | DCA              | OBLOCK | /CLEAR THE BLOCK TO 0                     |
| 0633 | 5630 | 1331 | TAD              | OL6400 | /RESET THE LOCATION COUNTER TO 6400       |
| 0634 | 5631 | 3330 | DCA              | OLOC   | /WHICH IS THE BUFFER AREA                 |
| 0635 | 5632 | 7240 | CLA CMA          |        | /SET UP THE TEST TO FOLLOW                |
| 0636 | 5633 | 6141 | LINC             |        |                                           |
| 0637 |      |      | LMODE            |        |                                           |
| 0640 | 1634 | 0460 | SNS I            | 0      | /TEST FOR BOTH                            |
| 0641 | 1635 | 0441 | SNS              | 1      | /SWITCHES 0 AND 1 UP                      |
| 0642 | 1636 | 0011 | CLR              |        |                                           |
| 0643 | 1637 | 0462 | SNS I            | 2      | /AND SO ON DOWN THE LINE                  |
| 0644 | 1640 | 0443 | SNS              | 3      |                                           |
| 0645 | 1641 | 0011 | CLR              |        |                                           |
| 0646 | 1642 | 0464 | SNS I            | 4      |                                           |
| 0647 | 1643 | 0445 | SNS              | 5      |                                           |
| 0650 | 1644 | 0011 | CLR              |        |                                           |
| 0651 | 1645 | 0002 | PDP              |        |                                           |
| 0652 |      |      | PHODE            |        |                                           |
| 0653 | 5646 | 7040 | CMA              |        |                                           |

|      |      |      |         |        |                                  |                                                            |
|------|------|------|---------|--------|----------------------------------|------------------------------------------------------------|
| 0654 | 5647 | 7640 | SZA CLA |        |                                  |                                                            |
| 0655 | 5650 | 5725 | JMP I   | OASSNO | /SWITCHES NOT UP, IGNORE REQUEST |                                                            |
| 0656 | 5651 | 1327 | TAD     | OP0INT | /RESET DECOUT TO POINT TO HERE   |                                                            |
| 0657 | 5652 | 3001 | DCA     | DECOUT |                                  |                                                            |
| 0660 | 5653 | 5725 | JMP I   | OASSNO | /NOW RETURN TO THE USER,         |                                                            |
| 0661 |      |      | /       |        |                                  |                                                            |
| 0662 |      |      | /       |        |                                  |                                                            |
| 0663 |      |      | /       |        |                                  |                                                            |
| 0664 |      |      | /       |        |                                  |                                                            |
| 0665 | 5654 | 0000 | OPUT,   | 0      |                                  |                                                            |
| 0666 | 5655 | 7450 | SNA     |        |                                  |                                                            |
| 0667 | 5656 | 5261 | JMP     | OEND   |                                  |                                                            |
| 0670 | 5657 | 4267 | JMS     | OOPUT  |                                  |                                                            |
| 0671 | 5660 | 5654 | JMP I   | OPUT   |                                  |                                                            |
| 0672 |      |      | /       |        |                                  |                                                            |
| 0673 | 5661 | 4267 | OEND,   | JMS    | OOPUT                            | /PUT IN A ZERO WORD                                        |
| 0674 | 5662 | 1330 | TAD     | OL0C   |                                  |                                                            |
| 0675 | 5663 | 0332 | AND     | OL377  |                                  |                                                            |
| 0676 | 5664 | 7650 | SNA CLA |        |                                  | /CHOP OFF THE HIGH BITS NOW, JUST LEAVE THE POSITION BITS. |
| 0677 | 5665 | 5654 | JMP I   | OPUT   |                                  |                                                            |
| 0700 | 5666 | 5261 | JMP     | OEND   |                                  |                                                            |
| 0701 |      |      | /       |        |                                  |                                                            |
| 0702 |      |      | /       |        |                                  |                                                            |
| 0703 |      |      | /       |        |                                  |                                                            |
| 0704 |      |      | /       |        |                                  |                                                            |
| 0705 | 5667 | 0000 | OOPUT,  | 0      |                                  |                                                            |
| 0706 | 5670 | 3730 | DCA I   | OL0C   |                                  |                                                            |
| 0707 | 5671 | 1330 | TAD     | OL0C   |                                  |                                                            |
| 0710 | 5672 | 7001 | IAC     |        |                                  |                                                            |
| 0711 | 5673 | 0332 | AND     | OL377  |                                  |                                                            |
| 0712 | 5674 | 1331 | TAD     | OL6400 |                                  |                                                            |
| 0713 | 5675 | 3330 | DCA     | OL0C   |                                  |                                                            |
| 0714 | 5676 | 1330 | TAD     | OL0C   |                                  |                                                            |
| 0715 | 5677 | 0332 | AND     | OL377  |                                  | /CHOP OFF THE BAD BITS,                                    |
| 0716 | 5700 | 7640 | SZA CLA |        |                                  |                                                            |
| 0717 | 5701 | 5667 | JMP I   | OOPUT  |                                  |                                                            |
| 0720 | 5702 | 4607 | JMS I   | VWRITE |                                  |                                                            |
| 0721 | 5703 | 5733 | OUNIT   |        |                                  |                                                            |
| 0722 | 5704 | 2335 | ISZ     | OBLOCK |                                  |                                                            |
| 0723 | 5705 | 5667 | JMP I   | OOPUT  |                                  |                                                            |
| 0724 |      |      | /       |        |                                  |                                                            |
| 0725 |      |      | /       |        |                                  |                                                            |
| 0726 |      |      | /       |        |                                  |                                                            |
| 0727 |      |      | /       |        |                                  |                                                            |
| 0730 |      |      | /       |        |                                  |                                                            |
| 0731 |      |      | /       |        |                                  |                                                            |
| 0732 |      |      | /       |        |                                  |                                                            |
| 0733 | 5706 | 7200 | OBSET,  | CLA    |                                  | /ALSO POINTS TO MOVE ROUTINE                               |
| 0734 | 5707 | 4610 | JMS I   | VREAD  |                                  | /READ IN THE OLD HEADER BLOCK                              |
| 0735 | 5710 | 5720 | OBIN    |        |                                  |                                                            |
| 0736 | 5711 | 4706 | JMS I   | OBSET  |                                  | /MOVE DOWN THE HEADER BLOCK NOW                            |
| 0737 | 5712 | 6201 | COF     | 0      |                                  |                                                            |
| 0740 | 5713 | 5337 | 5337    |        |                                  |                                                            |
| 0741 | 5714 | 6201 | COF     | 0      |                                  |                                                            |
| 0742 | 5715 | 4337 | 4337    |        |                                  |                                                            |
| 0743 | 5716 | 0041 | 41      |        |                                  |                                                            |
| 0744 | 5717 | 5725 | JMP I   | OASSNO |                                  | /AND RETURN TO THE USER                                    |
| 0745 |      |      | /       |        |                                  |                                                            |
| 0746 |      |      | /       |        |                                  |                                                            |
| 0747 |      |      | /       |        |                                  |                                                            |
| 0750 |      |      | /       |        |                                  |                                                            |
| 0751 |      |      | /       |        |                                  |                                                            |
| 0752 |      |      | /       |        |                                  |                                                            |

```

0753      /
0754      /
0755      /
0756      5720 0011 0BIN, 111
0757      5721 0012 12
0760      5722 0057 447-370
0761      5723 0001 1
0762      /
0763      /
0764      /
0765      /
0766      /
0767      /
0770      /
0771      /
0772      /
0773      /
0774      /
0775      /
0776      /
0777      /
1000      /

```

```

1001      5724 5401 OCTVAL, GETVAL
1002      5725 5503 OASSNO, ASSNO
1003      5726 0017 OL17, 17
1004      5727 5654 OPOINT, OPUT
1005      5730 6400 OLOC, 6400
1006      5731 6400 OL6400, 6400
1007      5732 0377 OL377, 377
1010      /

```

```

1011      5733 0000 OUNIT, 0
1012      5734 0015 15
1013      5735 0000 OBLOCK, 0
1014      5736 0001 1

```

/15=6400 FOR THE BUFFER AREA

```

1015      /
1016      /
1017      /
1020      /
1021      /
1022      /
1023      /
1024      /
1025      /
1026      /
1027      /
1030      /
1031      /

```

```

1032      5737 0000 CHECKO, 0
1033      5740 1776 TAD I OCUSES
1034      5741 7640 SEA CLA
1035      5742 5351 JHP OCCRLF
1036      /

```

/THIS ROUTINE CHECKS FOR NO BINARY OUTPUT.  
 /GET THE NUMBER OF BLOCKS OF OUTPUT.  
 /AND THERE?  
 /YEP, JUST GIVE A C.R.L.F AND RETURN TO THE CALLER

```

1037      5743 1353 OCDOIT, TAD OCPNT
1040      5744 7450 SNA
1041      5745 5351 JHP OCCRLF
1042      5746 4401 JMS I DECOIT
1043      5747 2343 ISZ OCDOIT
1044      5750 5343 JHP OCDOIT

```

/GET A CHARACTER FROM THE MESSEAGER,  
 /ZERO = THATS ALL FOLKS  
 /GIVE THE LINEFEED NOW  
 /OUT GOES THE CHARACTER NOW  
 /BOP TO THE NEXT CHARACTER,  
 /GO GET THE NEXT CHARACTER

```

1045      /
1046      5751 4500 OCCRLF, JMS I TYCARI
1047      5752 5737 JHP I CHECKO

```

/GIVE THE CARRIGE RETURN NOW  
 /RETURN TO THE CALLER NOW

```

1050      /
1051      /

```

|      |      |      |                |                                              |
|------|------|------|----------------|----------------------------------------------|
| 1052 |      | /    |                |                                              |
| 1053 |      | /    |                |                                              |
| 1054 | 5753 | 0254 | 0CPNT,         | 254                                          |
| 1055 | 5754 | 0240 |                | 240                                          |
| 1056 | 5755 | 0316 |                | 316                                          |
| 1057 | 5756 | 0317 |                | 317                                          |
| 1060 | 5757 | 0240 |                | 240                                          |
| 1061 | 5760 | 0302 |                | 302                                          |
| 1062 | 5761 | 0311 |                | 311                                          |
| 1063 | 5762 | 0316 |                | 316                                          |
| 1064 | 5763 | 0301 |                | 301                                          |
| 1065 | 5764 | 0322 |                | 322                                          |
| 1066 | 5765 | 0331 |                | 331                                          |
| 1067 | 5766 | 0240 |                | 240                                          |
| 1070 | 5767 | 0317 |                | 317                                          |
| 1071 | 5770 | 0325 |                | 325                                          |
| 1072 | 5771 | 0324 |                | 324                                          |
| 1073 | 5772 | 0320 |                | 320                                          |
| 1074 | 5773 | 0325 |                | 325                                          |
| 1075 | 5774 | 0324 |                | 324                                          |
| 1076 | 5775 | 0000 |                | 0                                            |
| 1077 |      | /    |                |                                              |
| 1100 |      | /    |                |                                              |
| 1101 |      | /    |                |                                              |
| 1102 | 5776 | 4337 | OCUSES, USEDNO | /POINTS TO THE NUMBER OF BINARY BLOCKS USED. |
| 1103 |      | /    |                |                                              |
| 1104 |      | /    |                |                                              |
| 1105 |      | /    |                |                                              |
| 1106 |      | /    |                |                                              |
| 1107 |      | /    |                |                                              |
| 1110 |      | /    |                |                                              |
| 1111 |      | /    |                |                                              |
| 1112 |      | /    |                |                                              |
| 1113 |      | /    |                |                                              |
| 1114 |      | /    |                |                                              |
| 1115 |      | /    |                |                                              |
| 1116 |      | /    |                |                                              |
| 1117 |      | /    |                |                                              |
| 1120 |      | /    |                |                                              |
| 1121 |      | /    |                |                                              |
| 1122 |      | /    |                |                                              |
| 1123 |      | /    |                |                                              |
| 1124 |      | /    |                |                                              |
| 1125 |      | /    |                |                                              |
| 1126 |      | /    |                |                                              |
| 1127 |      | /    |                |                                              |
| 1130 |      | /    |                |                                              |
| 1131 |      |      | EJECT          |                                              |

```

1132          / LINE PRINTER ROUTINE
1133          /
1134          /
1135          /
1136          /
1137          *6000
1140      6000 0000 PRINT, 0
1141      6001 1151 TAD M215
1142      6002 7450 SNA
1143      6003 5210 JMP CRFOUND
1144      6004 1154 TAD P215
1145      6005 6661 LSD
1146      6006 5241 JMP LPBACK
1147      6007 5640 JMP I PCLP
1150          /
1151      6010 1235 CRFOUND, TAD SW1
1152      6011 3201 DCA PRINT+1
1153      6012 3306 DCA NUM
1154      6013 5600 JMP I PRINT
1155      6014 3234 LFTEST, DCA TEMP1
1156      6015 1234 TAD TEMP1
1157      6016 1305 TAD M212
1160      6017 7650 SNA CLA
1161      6020 5232 JMP LFF
1162      6021 1234 TAD TEMP1
1163      6022 1151 TAD M215
1164      6023 7650 SNA CLA
1165      6024 5233 JMP LFF2
1166      6025 1236 TAD SW2
1167      6026 3201 DCA PRINT+1
1170      6027 7240 CLA CMA
1171      6030 1306 TAD NUM
1172      6031 5637 JMP I PCFIG
1173          /
1174      6032 2306 LFF, ISZ NUM
1175      6033 5600 LFF2, JMP I PRINT
1176          /
1177          /
1200          /
1201      6034 0000 TEMP1, 0
1202      6035 5214 SW1, JMP LFTEST
1203      6036 1151 SW2, TAD M215
1204          /
1205          /
1206          /
1207          /
1210          /
1211          /
1212      6037 6057 PCFIG, ANLXE
1213      6040 6054 PCLP, ANLXC
1214          /
1215          /
1216          /
1217          /
1220      6041 2700 LPBACK, ISZ I LCOUNT
1221      6042 7240 CLA CMA
1222      6043 1701 TAD I LOUTP
1223      6044 3701 DCA I LOUTP
1224      6045 1302 TAD LFIRST
1225      6046 1701 TAD I LOUTP
1226      6047 7650 SNA CLA
1227      6050 1303 TAD LBAD
1230      6051 1701 TAD I LOUTP

```

```

/THIS ROUTINE LOOKS
/SORT OF LINE A TTY, TAD CHAR; JMS PRINT,
/IT HANDLES CR AND LF,
/IS IT CARRIAGE RETURN,
/NO!
/WAIT FOR CLEAR SIGNAL,
/LINE PRINTER IS BUSY, BACK SPACE 1 CHAR

/SET TOCHECK FOR LF

/IS IT A LINEFFED.

/NOT AN LF OR CR, NEW LINE, OUTPUT LAST,

/LF, BOP UP LF COUNT,

/INCREMENT THE COUNTER BY 1
/-1 TO DECREMENT THE POINTER

/COMPARE AGAINST LOWER LIMIT

/ITS TOO FAR, COMPENSTATE TO TOP OF BUFFER

```

```

1231      6052 3701      DCA I  LOUTP
1232      6053 5600      JMP I  PRINT
1233      /
1234      /
1235      /
1236      /
1237      /
1240      6054 6656  ANLXC, LCF:LLB
1241      6055 7200      CLA
1242      6056 5600      JMP I  PRINT
1243      /
1244      6057 0157  ANLXE, AND  P7
1245      6060 7640  M140, SZA CLA
1246      6061 1157      TAD  P7
1247      6062 1304      TAD  L10
1250      6063 6652      LCF
1251      6064 6664      LPR
1252      6065 7200  LPRENT, CLA
1253      6066 3306      DCA  NUM
1254      6067 1234      TAD  TEMP1
1255      6070 5201      JMP  PRINT+1
1256      /
1257      /
1260      6071 6666  LP00C, LPR:LCB
1261      6072 5255      JMP  ANLXC+1
1262      /
1263      6073 7640  LP00E, SZA CLA
1264      6074 7305      CLA CLL IAC RAL
1265      6075 1153      TAD  P212
1266      6076 6666      LPR:LCB
1267      6077 5265      JMP  LPRENT
1270      /
1271      /
1272      6100 0760  LCOUNT, COUNT
1273      6101 0705  LOUTP,  OUTP
1274      6102 1671  LFIRST, -FIRST+1
1275      6103 0270  LBAD,  BAD-FIRST+1-1
1276      /
1277      /
1300      6104 0010  L10,  10
1301      6105 7566  M212,  -212
1302      /
1303      /
1304      /
1305      /
1306      /
1307      /
1310      /
1311      /
1312      /
1313      6106 0000  NUM,  0
1314      6107 0000  PRINIT, 0
1315      /
1316      /
1317      /
1320      /
1321      /
1322      /
1323      /
1324      /
1325      FIRST=,      /DEFINE THE START OF THE BUFFER
1326      BAD=6400      /AND THE END+1
1327      /

```

```

1330 /
1331 /
1332 /
1333 /
1334 / PAGE
1335 /
1336 /
1337 /
1340 /
1341 /
1342 /
1343 6200 1305 LOK, TAD L17 /LP IS THERE, EJECT TO START AS,
1344 6201 6652 LCF
1345 6202 6664 LPR
1346 6203 7200 CLA
1347 6204 1311 TAD M154 /SET LINE COUNTER FOR 154 COLUMNS WIDE.
1350 6205 3706 DCA I PC1
1351 6206 1706 TAD I PC1
1352 6207 3707 DCA I PC2
1353 /
1354 6210 1335 LOK0, TAD SYMFIX
1355 6211 3736 DCA I SYMOVR
1356 /
1357 6212 1233 LOK1, TAD R1 /RESET TYPO TO LP OR TTY ROUTINE.
1360 6213 3626 DCA I PR1
1361 6214 1235 TAD R2
1362 6215 3627 DCA I PR2
1363 6216 1237 TAD R3
1364 6217 3630 DCA I PR3
1365 6220 1241 TAD R4
1366 6221 3631 DCA I PR4
1367 6222 1243 TAD R5
1370 6223 3632 DCA I PR5
1371 6224 1154 TAD P215
1372 6225 5626 JMP I ,+1
1373 6226 2530 PR1, TYPO+1
1374 6227 2531 PR2, TYPO+2
1375 6230 2532 PR3, TYPO+3
1376 6231 2533 PR4, TYPO+4
1377 6232 2534 PR5, TYPO+5
1400 6233 4732 R1, JMS I LOK+TYPO-TYBASE+3
1401 6234 6041 TSF
1402 6235 7410 R2, SKP
1403 6236 5330 JMP LOK+TYPO-TYBASE+1
1404 6237 6000 R3, PRINT
1405 6240 6046 TLS
1406 6241 5727 R4, JMP I LOK+TYPO-TYBASE
1407 6242 7200 CLA
1410 6243 5727 R5, JMP I LOK+TYPO-TYBASE
1411 /
1412 /
1413 6244 7300 LPTEST, CLA CLL
1414 6245 1135 TAD M4 /WAIT FOR PRINTER TEST
1415 6246 3313 DCA PT2
1416 6247 3312 DCA PT1
1417 6250 6652 LCF
1420 6251 6662 LCB /CLEAR BUFFER, IF LP THERE FLAG RAISES IN 300 MILLSEC.
1421 6252 6661 TEST, LSD
1422 6253 7410 SKP
1423 6254 5200 JMP LOK /FLAG SI UP, LP THERE, WHAT A LUCKY BASTARD.
1424 6255 2312 ISZ PT1
1425 6256 5252 JMP TEST
1426 6257 2313 ISZ PT2

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1427      6260 5252      /      JMP      TEST
1430
1431      6261 1135      /      TAD      M4
1432      6262 3313      /      DCA      PT2
1433      6263 3312      /      DCA      PT1
1434      6264 1134      /      TAD      P240
1435      6265 6666      /      LCB!LPR
1436      6266 7200      /      CLA
1437      6267 6661      /      TESTC2, LSD
1440      6270 7410      /      SKP
1441      6271 5314      /      JMP      LOKA2
1442      6272 2312      /      ISZ      PT1
1443      6273 5267      /      JMP      TESTC2
1444      6274 2313      /      ISZ      PT2
1445      6275 5267      /      JMP      TESTC2
1446
1447      6276 2212      /      ISZ      LOK1
1450      6277 2214      /      ISZ      LOK1+2
1451      6300 2216      /      ISZ      LOK1+4
1452      6301 2220      /      ISZ      LOK1+6
1453      6302 3704      /      DCA I   CCNOP
1454      6303 5212      /      JMP      LOK1
1455
1456      6304 0747      /      CCNOP, CNOP
1457      /
1460      /
1461      /
1462      /
1463      /
1464      /
1465      /
1466      6305 0017      /      L17, 17
1467      6306 1576      /      PC1, M60
1470      6307 3662      /      PC2, M60A
1471      6310 7706      /      M72, -72
1472      6311 7624      /      M154, -154
1473      6312 0000      /      PT1, 0
1474      6313 0000      /      PT2, 0
1475      /
1476      /
1477      /
1500      /
1501      /
1502      /
1503      /
1504      6314 7305      /      LOKA2, CLL CLA IAC RAL
1505      6315 1153      /      TAD      P212
1506      6316 6676      /      LCF!LPR
1507      6317 7200      /      CLA
1510      6320 1310      /      TAD      M72
1511      6321 3706      /      DCA I   PC1
1512      6322 1310      /      TAD      M72
1513      6323 3707      /      DCA I   PC2
1514      6324 1331      /      TAD      PLP08C
1515      6325 3733      /      DCA I   PXL1
1516      6326 1332      /      TAD      PLP08E
1517      6327 3734      /      DCA I   PXL2
1520      6330 5210      /      JMP      LOK0
1521      /
1522      /
1523      6331 6071      /      PLP08C, LP08C
1524      6332 6073      /      PLP08E, LP08E
1525      6333 6040      /      PXL1, PCLP

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/TIME HAS RUN OUT, CHANGE TO SET UT TTY MODE.

/ZAP OUT THE LPTST ON THE BUFFERING ROUTINE

|      |      |      |              |                                         |
|------|------|------|--------------|-----------------------------------------|
| 1526 | 6334 | 6037 | PXL2, PCFIG  |                                         |
| 1527 | 6335 | 1307 | SYMFIX, TAD  | LWDEFD&177*LOK                          |
| 1530 | 6336 | 3504 | SYMOVR, LWDE |                                         |
| 1531 |      |      | /            |                                         |
| 1532 |      |      | /            |                                         |
| 1533 |      |      | /            |                                         |
| 1534 |      |      | /            |                                         |
| 1535 |      |      | /            |                                         |
| 1536 |      |      | /            |                                         |
| 1537 |      |      | /            |                                         |
| 1540 |      |      | /            |                                         |
| 1541 |      |      | /            |                                         |
| 1542 |      |      | /            |                                         |
| 1543 |      |      | /            |                                         |
| 1544 |      |      | /            | SOME BASIC DEFINITIONS FOR THE ROUTINES |
| 1545 |      |      | /            |                                         |
| 1546 |      |      | TYBASE=2400  | /MAIN BASE FOR REFS.                    |
| 1547 |      |      | LLB=6654     |                                         |
| 1550 |      |      | LCF=6652     |                                         |
| 1551 |      |      | LSD=6661     |                                         |
| 1552 |      |      | LCB=6662     |                                         |
| 1553 |      |      | LPR=6664     |                                         |
| 1554 |      |      | /            |                                         |
| 1555 |      |      | /            |                                         |
| 1556 |      |      | /            |                                         |
| 1557 |      |      | /            |                                         |
| 1560 |      |      | /            |                                         |
| 1561 |      |      | /            |                                         |
| 1562 |      |      | /            |                                         |
| 1563 |      |      | /            |                                         |
| 1564 |      |      | /            |                                         |
| 1565 |      |      | /            |                                         |
| 1566 |      |      | /            |                                         |
| 1567 |      |      | /            |                                         |
| 1570 |      |      | /            |                                         |
| 1571 |      |      |              | EJECT                                   |

```

1572      /
1573      /
1574      /
1575      /  S Y M B O L   T A B L E   ! ! ! !
1576      /
1577      /
1600      /
1601      /
1602      /
1603      /
1604      /
1605      /
1606      /
1607      6400  7777  7777      *6400
1610      6401  7777  7777
1611      6402  7777  7777
1612      6403  7777  7777
1613      6404  7777  7777
1614      6405  7777  7777
1615      6406  7777  7777
1616      6407  7777  7777
1617      6410  7777  7777
1620      6411  7777  7777
1621      6412  7777  7777
1622      6413  7777  7777
1623      6414  5143  5143      /PSF
1624      6415  0336  0336
1625      6416  0000  0000
1626      6417  6021  6021
1627      6420  5123  5123      /PCF
1630      6421  0336  0336
1631      6422  0000  0000
1632      6423  6022  6022
1633      6424  5140  5140      /PPC
1634      6425  0157  0157
1635      6426  0000  0000
1636      6427  6024  6024
1637      6430  5134  5134      /PLS
1640      6431  1277  1277
1641      6432  0000  0000
1642      6433  6026  6026
1643      6434  5255  5255      /RCF
1644      6435  0336  0336
1645      6436  0000  0000
1646      6437  6011  6011
1647      6440  5254  5254      /RRB
1650      6441  0112  0112
1651      6442  0000  0000
1652      6443  6012  6012
1653      6444  5240  5240      /RFC
1654      6445  0157  0157
1655      6446  0000  0000
1656      6447  6014  6014
1657      6450  4170  4170      /CIF
1660      6451  0336  0336
1661      6452  0000  0000
1662      6453  6202  6202
1663      6454  4163  4163      /CDF
1664      6455  0336  0336
1665      6456  0000  0000
1666      6457  6201  6201
1667      6460  5243  5243      /RIB
1670      6461  0112  0112

```

|      |      |      |           |       |
|------|------|------|-----------|-------|
| 1671 | 6462 | 0000 | 0000      |       |
| 1672 | 6463 | 6234 | 6234      |       |
| 1673 | 6464 | 5247 | 5247      | /RMF  |
| 1674 | 6465 | 0336 | 0336      |       |
| 1675 | 6466 | 0000 | 0000      |       |
| 1676 | 6467 | 6244 | 6244      |       |
| 1677 | 6470 | 5243 | 5243      | /RIF  |
| 1700 | 6471 | 0336 | 0336      |       |
| 1701 | 6472 | 0000 | 0000      |       |
| 1702 | 6473 | 6224 | 6224      |       |
| 1703 | 6474 | 5236 | 5236      | /RDF  |
| 1704 | 6475 | 0336 | 0336      |       |
| 1705 | 6476 | 0000 | 0000      |       |
| 1706 | 6477 | 6214 | 6214      |       |
| 1707 | 6500 | 5360 | 5360      | /TLS  |
| 1710 | 6501 | 1277 | 1277      |       |
| 1711 | 6502 | 0000 | 0000      |       |
| 1712 | 6503 | 6046 | 6046      |       |
| 1713 | 6504 | 5364 | 5364      | /TPC  |
| 1714 | 6505 | 0157 | 0157      |       |
| 1715 | 6506 | 0000 | 0000      |       |
| 1716 | 6507 | 6044 | 6044      |       |
| 1717 | 6510 | 5347 | 5347      | /TCF  |
| 1720 | 6511 | 0336 | 0336      |       |
| 1721 | 6512 | 0000 | 0000      |       |
| 1722 | 6513 | 6042 | 6042      |       |
| 1723 | 6514 | 5367 | 5367      | /TSF  |
| 1724 | 6515 | 0336 | 0336      |       |
| 1725 | 6516 | 0000 | 0000      |       |
| 1726 | 6517 | 6041 | 6041      |       |
| 1727 | 6520 | 4651 | 4651      | /KRB  |
| 1730 | 6521 | 0112 | 0112      |       |
| 1731 | 6522 | 0000 | 0000      |       |
| 1732 | 6523 | 6036 | 6036      |       |
| 1733 | 6524 | 4651 | 4651      | /KRS  |
| 1734 | 6525 | 1277 | 1277      |       |
| 1735 | 6526 | 0000 | 0000      |       |
| 1736 | 6527 | 6034 | 6034      |       |
| 1737 | 6530 | 4632 | 4632      | /KCC  |
| 1740 | 6531 | 0157 | 0157      |       |
| 1741 | 6532 | 0000 | 0000      |       |
| 1742 | 6533 | 6032 | 6032      |       |
| 1743 | 6534 | 4652 | 4652      | /KSF  |
| 1744 | 6535 | 0336 | 0336      |       |
| 1745 | 6536 | 0000 | 0000      |       |
| 1746 | 6537 | 6031 | 6031      |       |
| 1747 | 6540 | 4534 | 4534      | /IOF  |
| 1750 | 6541 | 0336 | 0336      |       |
| 1751 | 6542 | 0000 | 0000      |       |
| 1752 | 6543 | 6002 | 6002      |       |
| 1753 | 6544 | 4534 | 4534      | /ION  |
| 1754 | 6545 | 1006 | 1006      |       |
| 1755 | 6546 | 0000 | 0000      |       |
| 1756 | 6547 | 6001 | 6001      |       |
| 1757 | 6550 | 4705 | 4705      | /LINC |
| 1760 | 6551 | 1011 | 1011      |       |
| 1761 | 6552 | 0000 | 0000      |       |
| 1762 | 6553 | 6141 | 6141      |       |
| 1763 | 6554 | 4173 | 4173/CLSK |       |
| 1764 | 6555 | 1312 | 1312      |       |
| 1765 | 6556 | 0000 | 0         |       |
| 1766 | 6557 | 6131 | 6131      |       |
| 1767 | 6560 | 4173 | 4173/CLLR |       |

|      |      |      |      |       |
|------|------|------|------|-------|
| 1770 | 6561 | 0716 | 716  |       |
| 1771 | 6562 | 0000 | 0    |       |
| 1772 | 6563 | 6132 | 6132 |       |
| 1773 | 6564 | 4173 | 4173 |       |
| 1774 | 6565 | 0047 | 47   |       |
| 1775 | 6566 | 0000 | 0    |       |
| 1776 | 6567 | 6133 | 6133 |       |
| 1777 | 6570 | 4173 | 4173 |       |
| 2000 | 6571 | 0307 | 307  |       |
| 2001 | 6572 | 0000 | 0    |       |
| 2002 | 6573 | 6134 | 6134 |       |
| 2003 | 6574 | 4173 | 4173 | /CLSA |
| 2004 | 6575 | 1300 | 1300 |       |
| 2005 | 6576 | 0000 | 0    |       |
| 2006 | 6577 | 6135 | 6135 |       |
| 2007 | 6600 | 4173 | 4173 | /CLCA |
| 2010 | 6601 | 0160 | 160  |       |
| 2011 | 6602 | 0000 | 0    |       |
| 2012 | 6603 | 6137 | 6137 |       |
| 2013 | 6604 | 4173 | 4173 | /CLBA |
| 2014 | 6605 | 0113 | 113  |       |
| 2015 | 6606 | 0000 | 0    |       |
| 2016 | 6607 | 6136 | 6136 |       |
| 2017 | 6610 | 4417 | 4417 | /GLK  |
| 2020 | 6611 | 0627 | 0627 |       |
| 2021 | 6612 | 0000 | 0000 |       |
| 2022 | 6613 | 7204 | 7204 |       |
| 2023 | 6614 | 5323 | 5323 | /STL  |
| 2024 | 6615 | 0674 | 0674 |       |
| 2025 | 6616 | 0000 | 0000 |       |
| 2026 | 6617 | 7120 | 7120 |       |
| 2027 | 6620 | 5323 | 5323 | /STA  |
| 2030 | 6621 | 0045 | 0045 |       |
| 2031 | 6622 | 0000 | 0000 |       |
| 2032 | 6623 | 7240 | 7240 |       |
| 2033 | 6624 | 4675 | 4675 | /LAS  |
| 2034 | 6625 | 1277 | 1277 |       |
| 2035 | 6626 | 0000 | 0000 |       |
| 2036 | 6627 | 7604 | 7604 |       |
| 2037 | 6630 | 4170 | 4170 | /CIA  |
| 2040 | 6631 | 0045 | 0045 |       |
| 2041 | 6632 | 0000 | 0000 |       |
| 2042 | 6633 | 7041 | 7041 |       |
| 2043 | 6634 | 4464 | 4464 | /HLT  |
| 2044 | 6635 | 1344 | 1344 |       |
| 2045 | 6636 | 0000 | 0000 |       |
| 2046 | 6637 | 7402 | 7402 |       |
| 2047 | 6640 | 5076 | 5076 | /OSR  |
| 2050 | 6641 | 1232 | 1232 |       |
| 2051 | 6642 | 0000 | 0000 |       |
| 2052 | 6643 | 7404 | 7404 |       |
| 2053 | 6644 | 5312 | 5312 | /SKP  |
| 2054 | 6645 | 1120 | 1120 |       |
| 2055 | 6646 | 0000 | 0000 |       |
| 2056 | 6647 | 7410 | 7410 |       |
| 2057 | 6650 | 5331 | 5331 | /SZL  |
| 2060 | 6651 | 0674 | 0674 |       |
| 2061 | 6652 | 0000 | 0000 |       |
| 2062 | 6653 | 7430 | 7430 |       |
| 2063 | 6654 | 5315 | 5315 | /SNL  |
| 2064 | 6655 | 0674 | 0674 |       |
| 2065 | 6656 | 0000 | 0000 |       |
| 2066 | 6657 | 7420 | 7420 |       |

|      |      |      |      |      |
|------|------|------|------|------|
| 2067 | 6660 | 5315 | 5315 | /SNA |
| 2070 | 6661 | 0045 | 0045 |      |
| 2071 | 6662 | 0000 | 0000 |      |
| 2072 | 6663 | 7450 | 7450 |      |
| 2073 | 6664 | 5317 | 5317 | /SPA |
| 2074 | 6665 | 0045 | 0045 |      |
| 2075 | 6666 | 0000 | 0000 |      |
| 2076 | 6667 | 7510 | 7510 |      |
| 2077 | 6670 | 5331 | 5331 | /SZA |
| 2100 | 6671 | 0045 | 0045 |      |
| 2101 | 6672 | 0000 | 0000 |      |
| 2102 | 6673 | 7440 | 7440 |      |
| 2103 | 6674 | 5314 | 5314 | /SMA |
| 2104 | 6675 | 0045 | 0045 |      |
| 2105 | 6676 | 0000 | 0000 |      |
| 2106 | 6677 | 7500 | 7500 |      |
| 2107 | 6700 | 4516 | 4516 | /IAC |
| 2110 | 6701 | 0157 | 0157 |      |
| 2111 | 6702 | 0000 | 0000 |      |
| 2112 | 6703 | 7001 | 7001 |      |
| 2113 | 6704 | 5256 | 5256 | /RTL |
| 2114 | 6705 | 0674 | 0674 |      |
| 2115 | 6706 | 0000 | 0000 |      |
| 2116 | 6707 | 7006 | 7006 |      |
| 2117 | 6710 | 5233 | 5233 | /RAL |
| 2120 | 6711 | 0674 | 0674 |      |
| 2121 | 6712 | 0000 | 0000 |      |
| 2122 | 6713 | 7004 | 7004 |      |
| 2123 | 6714 | 5256 | 5256 | /RTR |
| 2124 | 6715 | 1232 | 1232 |      |
| 2125 | 6716 | 0000 | 0000 |      |
| 2126 | 6717 | 7012 | 7012 |      |
| 2127 | 6720 | 5233 | 5233 | /RAR |
| 2130 | 6721 | 1232 | 1232 |      |
| 2131 | 6722 | 0000 | 0000 |      |
| 2132 | 6723 | 7010 | 7010 |      |
| 2133 | 6724 | 4174 | 4174 | /CML |
| 2134 | 6725 | 0674 | 0674 |      |
| 2135 | 6726 | 0000 | 0000 |      |
| 2136 | 6727 | 7020 | 7020 |      |
| 2137 | 6730 | 4174 | 4174 | /CMA |
| 2140 | 6731 | 0045 | 0045 |      |
| 2141 | 6732 | 0000 | 0000 |      |
| 2142 | 6733 | 7040 | 7040 |      |
| 2143 | 6734 | 4173 | 4173 | /CLL |
| 2144 | 6735 | 0674 | 0674 |      |
| 2145 | 6736 | 0000 | 0000 |      |
| 2146 | 6737 | 7100 | 7100 |      |
| 2147 | 6740 | 4173 | 4173 | /CLA |
| 2150 | 6741 | 0045 | 0045 |      |
| 2151 | 6742 | 0000 | 0000 |      |
| 2152 | 6743 | 7200 | 7200 |      |
| 2153 | 6744 | 5025 | 5025 | /NOP |
| 2154 | 6745 | 1120 | 1120 |      |
| 2155 | 6746 | 0000 | 0000 |      |
| 2156 | 6747 | 7000 | 7000 |      |
| 2157 | 6750 | 4577 | 4577 | /JMP |
| 2160 | 6751 | 1120 | 1120 |      |
| 2161 | 6752 | 0000 | 0000 |      |
| 2162 | 6753 | 5000 | 5000 |      |
| 2163 | 6754 | 4577 | 4577 | /JMS |
| 2164 | 6755 | 1277 | 1277 |      |
| 2165 | 6756 | 0000 | 0000 |      |

|      |      |      |      |       |
|------|------|------|------|-------|
| 2166 | 6757 | 4000 | 4000 |       |
| 2167 | 6760 | 4227 | 4227 | /DCA  |
| 2170 | 6761 | 0045 | 0045 |       |
| 2171 | 6762 | 0000 | 0000 |       |
| 2172 | 6763 | 3000 | 3000 |       |
| 2173 | 6764 | 4540 | 4540 | /ISE  |
| 2174 | 6765 | 1702 | 1702 |       |
| 2175 | 6766 | 0000 | 0000 |       |
| 2176 | 6767 | 2000 | 2000 |       |
| 2177 | 6770 | 5345 | 5345 | /TAD  |
| 2200 | 6771 | 0224 | 0224 |       |
| 2201 | 6772 | 0000 | 0000 |       |
| 2202 | 6773 | 1000 | 1000 |       |
| 2203 | 6774 | 4063 | 4063 | /AND  |
| 2204 | 6775 | 0224 | 0224 |       |
| 2205 | 6776 | 0000 | 0000 |       |
| 2206 | 6777 | 0000 | 0    |       |
| 2207 |      |      |      | EJECT |

|      |      |      |                 |      |
|------|------|------|-----------------|------|
| 2210 |      |      | /LINC SYM TABLE |      |
| 2211 |      |      | *7000           |      |
| 2212 |      |      | /               |      |
| 2213 |      |      | /               |      |
| 2214 | 7000 | 5607 | 5607            | /XOA |
| 2215 | 7001 | 0045 | 0045            |      |
| 2216 | 7002 | 0000 | 0000            |      |
| 2217 | 7003 | 0021 | 0021            |      |
| 2220 | 7004 | 5201 | 5201            | /QLZ |
| 2221 | 7005 | 1702 | 1702            |      |
| 2222 | 7006 | 0000 | 0000            |      |
| 2223 | 7007 | 0455 | 0455            |      |
| 2224 | 7010 | 4352 | 4352            | /FLO |
| 2225 | 7011 | 1053 | 1053            |      |
| 2226 | 7012 | 0000 | 0000            |      |
| 2227 | 7013 | 0454 | 0454            |      |
| 2230 | 7014 | 4652 | 4652            | /KST |
| 2231 | 7015 | 1344 | 1344            |      |
| 2232 | 7016 | 0000 | 0000            |      |
| 2233 | 7017 | 0415 | 0415            |      |
| 2234 | 7020 | 5305 | 5305            | /SFA |
| 2235 | 7021 | 0045 | 45              |      |
| 2236 | 7022 | 0000 | 0               |      |
| 2237 | 7023 | 0024 | 24              |      |
| 2240 | 7024 | 4075 | 4075            | /AXO |
| 2241 | 7025 | 1053 | 1053            |      |
| 2242 | 7026 | 0000 | 0000            |      |
| 2243 | 7027 | 0001 | 0001            |      |
| 2244 | 7030 | 4314 | 4314            | /ESF |
| 2245 | 7031 | 0336 | 0336            |      |
| 2246 | 7032 | 0000 | 0000            |      |
| 2247 | 7033 | 0004 | 0004            |      |
| 2250 | 7034 | 4464 | 4464            | /HLT |
| 2251 | 7035 | 1344 | 1344            |      |
| 2252 | 7036 | 0000 | 0000            |      |
| 2253 | 7037 | 0000 | 0000            |      |
| 2254 | 7040 | 5166 | 5166            | /QAC |
| 2255 | 7041 | 0157 | 0157            |      |
| 2256 | 7042 | 0000 | 0000            |      |
| 2257 | 7043 | 0005 | 0005            |      |
| 2260 | 7044 | 4173 | 4173            | /CLR |
| 2261 | 7045 | 1232 | 1232            |      |
| 2262 | 7046 | 0000 | 0000            |      |
| 2263 | 7047 | 0011 | 0011            |      |
| 2264 | 7050 | 4071 | 4071            | /ATR |
| 2265 | 7051 | 1232 | 1232            |      |
| 2266 | 7052 | 0000 | 0000            |      |
| 2267 | 7053 | 0014 | 0014            |      |
| 2270 | 7054 | 5256 | 5256            | /RTA |
| 2271 | 7055 | 0045 | 0045            |      |
| 2272 | 7056 | 0000 | 0000            |      |
| 2273 | 7057 | 0015 | 0015            |      |
| 2274 | 7060 | 5025 | 5025            | /NOP |
| 2275 | 7061 | 1120 | 1120            |      |
| 2276 | 7062 | 0000 | 0000            |      |
| 2277 | 7063 | 0016 | 0016            |      |
| 2300 | 7064 | 4176 | 4176            | /COM |
| 2301 | 7065 | 0741 | 0741            |      |
| 2302 | 7066 | 0000 | 0000            |      |
| 2303 | 7067 | 0017 | 0017            |      |
| 2304 | 7070 | 5304 | 5304            | /SET |
| 2305 | 7071 | 1344 | 1344            |      |
| 2306 | 7072 | 0000 | 0000            |      |

|      |      |      |      |          |
|------|------|------|------|----------|
| 2307 | 7073 | 0040 | 0040 |          |
| 2310 | 7074 | 5300 | 5300 | /SAM     |
| 2311 | 7075 | 0741 | 0741 |          |
| 2312 | 7076 | 0000 | 0000 |          |
| 2313 | 7077 | 0100 | 0100 |          |
| 2314 | 7100 | 4235 | 4235 | /DIS     |
| 2315 | 7101 | 1277 | 1277 |          |
| 2316 | 7102 | 0000 | 0000 |          |
| 2317 | 7103 | 0140 | 0140 |          |
| 2320 | 7104 | 5613 | 5613 | /XSK     |
| 2321 | 7105 | 0627 | 0627 |          |
| 2322 | 7106 | 0000 | 0000 |          |
| 2323 | 7107 | 0200 | 0200 |          |
| 2324 | 7110 | 5251 | 5251 | /ROL     |
| 2325 | 7111 | 0674 | 0674 |          |
| 2326 | 7112 | 0000 | 0000 |          |
| 2327 | 7113 | 0240 | 0240 |          |
| 2330 | 7114 | 5251 | 5251 | /ROR     |
| 2331 | 7115 | 1232 | 1232 |          |
| 2332 | 7116 | 0000 | 0000 |          |
| 2333 | 7117 | 0300 | 0300 |          |
| 2334 | 7120 | 5302 | 5302 | /SCR     |
| 2335 | 7121 | 1232 | 1232 |          |
| 2336 | 7122 | 0000 | 0000 |          |
| 2337 | 7123 | 0340 | 0340 |          |
| 2340 | 7124 | 5327 | 5327 | /SXL     |
| 2341 | 7125 | 0674 | 0674 |          |
| 2342 | 7126 | 0000 | 0000 |          |
| 2343 | 7127 | 0400 | 0400 |          |
| 2344 | 7130 | 5315 | 5315 | /SNS     |
| 2345 | 7131 | 1277 | 1277 |          |
| 2346 | 7132 | 0000 | 0000 |          |
| 2347 | 7133 | 0440 | 0440 |          |
| 2350 | 7134 | 5312 | 5312 | /SKP (L) |
| 2351 | 7135 | 1120 | 1120 |          |
| 2352 | 7136 | 0000 | 0000 |          |
| 2353 | 7137 | 0456 | 0456 |          |
| 2354 | 7140 | 4077 | 4077 | /AZE     |
| 2355 | 7141 | 0271 | 0271 |          |
| 2356 | 7142 | 0000 | 0000 |          |
| 2357 | 7143 | 0450 | 0450 |          |
| 2360 | 7144 | 4065 | 4065 | /APO     |
| 2361 | 7145 | 1053 | 1053 |          |
| 2362 | 7146 | 0000 | 0000 |          |
| 2363 | 7147 | 0451 | 0451 |          |
| 2364 | 7150 | 4726 | 4726 | /LZE     |
| 2365 | 7151 | 0271 | 0271 |          |
| 2366 | 7152 | 0000 | 0000 |          |
| 2367 | 7153 | 0452 | 0452 |          |
| 2370 | 7154 | 4236 | 4236 | /DJR     |
| 2371 | 7155 | 1232 | 1232 |          |
| 2372 | 7156 | 0000 | 0000 |          |
| 2373 | 7157 | 0006 | 0006 |          |
| 2374 | 7160 | 4705 | 4705 | /LIF     |
| 2375 | 7161 | 0336 | 0336 |          |
| 2376 | 7162 | 0000 | 0000 |          |
| 2377 | 7163 | 0600 | 0600 |          |
| 2400 | 7164 | 4700 | 4700 | /LDF     |
| 2401 | 7165 | 0336 | 0336 |          |
| 2402 | 7166 | 0000 | 0000 |          |
| 2403 | 7167 | 0640 | 0640 |          |
| 2404 | 7170 | 5236 | 5236 | /RDC     |
| 2405 | 7171 | 0157 | 0157 |          |

|      |      |      |      |      |
|------|------|------|------|------|
| 2406 | 7172 | 0000 | 0000 |      |
| 2407 | 7173 | 0700 | 0700 |      |
| 2410 | 7174 | 5235 | 5235 | /RCG |
| 2411 | 7175 | 0403 | 0403 |      |
| 2412 | 7176 | 0000 | 0000 |      |
| 2413 | 7177 | 0701 | 0701 |      |
| 2414 | 7200 | 5236 | 5236 | /RDE |
| 2415 | 7201 | 0271 | 0271 |      |
| 2416 | 7202 | 0000 | 0000 |      |
| 2417 | 7203 | 0702 | 0702 |      |
| 2420 | 7204 | 4765 | 4765 | /MTB |
| 2421 | 7205 | 0112 | 0112 |      |
| 2422 | 7206 | 0000 | 0000 |      |
| 2423 | 7207 | 0703 | 0703 |      |
| 2424 | 7210 | 5545 | 5545 | /WRC |
| 2425 | 7211 | 0157 | 0157 |      |
| 2426 | 7212 | 0000 | 0000 |      |
| 2427 | 7213 | 0704 | 0704 |      |
| 2430 | 7214 | 5526 | 5526 | /WCG |
| 2431 | 7215 | 0403 | 0403 |      |
| 2432 | 7216 | 0000 | 0000 |      |
| 2433 | 7217 | 0705 | 0705 |      |
| 2434 | 7220 | 5545 | 5545 | /WRI |
| 2435 | 7221 | 0515 | 0515 |      |
| 2436 | 7222 | 0000 | 0000 |      |
| 2437 | 7223 | 0706 | 0706 |      |
| 2440 | 7224 | 4167 | 4167 | /CHK |
| 2441 | 7225 | 0627 | 0627 |      |
| 2442 | 7226 | 0000 | 0000 |      |
| 2443 | 7227 | 0707 | 0707 |      |
| 2444 | 7230 | 5300 | 5300 | /SAE |
| 2445 | 7231 | 0271 | 0271 |      |
| 2446 | 7232 | 0000 | 0000 |      |
| 2447 | 7233 | 1440 | 1440 |      |
| 2450 | 7234 | 5321 | 5321 | /SRO |
| 2451 | 7235 | 1053 | 1053 |      |
| 2452 | 7236 | 0000 | 0000 |      |
| 2453 | 7237 | 1500 | 1500 |      |
| 2454 | 7240 | 4115 | 4115 | /BCL |
| 2455 | 7241 | 0674 | 0674 |      |
| 2456 | 7242 | 0000 | 0000 |      |
| 2457 | 7243 | 1540 | 1540 |      |
| 2460 | 7244 | 4135 | 4135 | /BSE |
| 2461 | 7245 | 0271 | 0271 |      |
| 2462 | 7246 | 0000 | 0000 |      |
| 2463 | 7247 | 1600 | 1600 |      |
| 2464 | 7250 | 4115 | 4115 | /BCO |
| 2465 | 7251 | 1053 | 1053 |      |
| 2466 | 7252 | 0000 | 0000 |      |
| 2467 | 7253 | 1640 | 1640 |      |
| 2470 | 7254 | 4247 | 4247 | /DSC |
| 2471 | 7255 | 0157 | 0157 |      |
| 2472 | 7256 | 0000 | 0000 |      |
| 2473 | 7257 | 1740 | 1740 |      |
| 2474 | 7260 | 4051 | 4051 | /ADD |
| 2475 | 7261 | 0224 | 0224 |      |
| 2476 | 7262 | 0000 | 0000 |      |
| 2477 | 7263 | 2000 | 2000 |      |
| 2500 | 7264 | 5323 | 5323 | /STC |
| 2501 | 7265 | 0157 | 0157 |      |
| 2502 | 7266 | 0000 | 0000 |      |
| 2503 | 7267 | 4000 | 4000 |      |
| 2504 | 7270 | 4577 | 4577 | /JMP |

|      |      |      |      |      |
|------|------|------|------|------|
| 2505 | 7271 | 1120 | 1120 |      |
| 2506 | 7272 | 0000 | 0000 |      |
| 2507 | 7273 | 6000 | 6000 |      |
| 2510 | 7274 | 4764 | 4764 | /HSC |
| 2511 | 7275 | 0157 | 0157 |      |
| 2512 | 7276 | 0000 | 0000 |      |
| 2513 | 7277 | 0000 | 0000 |      |
| 2514 | 7300 | 4700 | 4700 | /LDA |
| 2515 | 7301 | 0045 | 0045 |      |
| 2516 | 7302 | 0000 | 0000 |      |
| 2517 | 7303 | 1000 | 1000 |      |
| 2520 | 7304 | 5323 | 5323 | /STA |
| 2521 | 7305 | 0045 | 0045 |      |
| 2522 | 7306 | 0000 | 0000 |      |
| 2523 | 7307 | 1040 | 1040 |      |
| 2524 | 7310 | 4051 | 4051 | /ADA |
| 2525 | 7311 | 0045 | 0045 |      |
| 2526 | 7312 | 0000 | 0000 |      |
| 2527 | 7313 | 1100 | 1100 |      |
| 2530 | 7314 | 4051 | 4051 | /ADM |
| 2531 | 7315 | 0741 | 0741 |      |
| 2532 | 7316 | 0000 | 0000 |      |
| 2533 | 7317 | 1140 | 1140 |      |
| 2534 | 7320 | 4675 | 4675 | /LAM |
| 2535 | 7321 | 0741 | 0741 |      |
| 2536 | 7322 | 0000 | 0000 |      |
| 2537 | 7323 | 1200 | 1200 |      |
| 2540 | 7324 | 4766 | 4766 | /MUL |
| 2541 | 7325 | 0674 | 0674 |      |
| 2542 | 7326 | 0000 | 0000 |      |
| 2543 | 7327 | 1240 | 1240 |      |
| 2544 | 7330 | 4700 | 4700 | /LDH |
| 2545 | 7331 | 0450 | 0450 |      |
| 2546 | 7332 | 0000 | 0000 |      |
| 2547 | 7333 | 1300 | 1300 |      |
| 2550 | 7334 | 5323 | 5323 | /STH |
| 2551 | 7335 | 0450 | 0450 |      |
| 2552 | 7336 | 0000 | 0000 |      |
| 2553 | 7337 | 1340 | 1340 |      |
| 2554 | 7340 | 5307 | 5307 | /SHD |
| 2555 | 7341 | 0224 | 0224 |      |
| 2556 | 7342 | 0000 | 0000 |      |
| 2557 | 7343 | 1400 | 1400 |      |
| 2560 | 7344 | 5345 | 5345 | /TAC |
| 2561 | 7345 | 0157 | 0157 |      |
| 2562 | 7346 | 0000 | 0000 |      |
| 2563 | 7347 | 0003 | 0003 |      |
| 2564 | 7350 | 5361 | 5361 | /TMA |
| 2565 | 7351 | 0045 | 0045 |      |
| 2566 | 7352 | 0000 | 0000 |      |
| 2567 | 7353 | 0023 | 0023 |      |
| 2570 | 7354 | 5124 | 5124 | /PDP |
| 2571 | 7355 | 1120 | 1120 |      |
| 2572 | 7356 | 0000 | 0000 |      |
| 2573 | 7357 | 0002 | 0002 |      |
| 2574 | 7360 | 5323 | 5323 | /STD |
| 2575 | 7361 | 0224 | 0224 |      |
| 2576 | 7362 | 0000 | 0000 |      |
| 2577 | 7363 | 0416 | 0416 |      |
| 2600 | 7364 | 4717 | 4717 | /LSW |
| 2601 | 7365 | 1523 | 1523 |      |
| 2602 | 7366 | 0000 | 0000 |      |
| 2603 | 7367 | 0517 | 0517 |      |

|      |      |      |      |       |
|------|------|------|------|-------|
| 2604 | 7370 | 5255 | 5255 | /RSW  |
| 2605 | 7371 | 1523 | 1523 |       |
| 2606 | 7372 | 0000 | 0000 |       |
| 2607 | 7373 | 0516 | 0516 |       |
| 2610 | 7374 | 4534 | 4534 | /IOB  |
| 2611 | 7375 | 0112 | 0112 |       |
| 2612 | 7376 | 0000 | 0000 |       |
| 2613 | 7377 | 0500 | 0500 |       |
| 2614 |      |      |      | EJECT |

-

| /SPECIAL CHARACTERS AND PSEUDO-INSTRUCTIONS |      |      |        |                                                                      |
|---------------------------------------------|------|------|--------|----------------------------------------------------------------------|
| *7400                                       |      |      |        |                                                                      |
| 2615                                        |      |      |        |                                                                      |
| 2616                                        |      |      |        |                                                                      |
| 2617                                        |      | /    |        |                                                                      |
| 2620                                        | 7400 | 6167 | 6167   | /CHAIN                                                               |
| 2621                                        | 7401 | 0056 | 0056   |                                                                      |
| 2622                                        | 7402 | 1006 | 1006   |                                                                      |
| 2623                                        | 7403 | 2536 | XCHAIN |                                                                      |
| 2624                                        | 7404 | 6705 | 6705   | /LISTAPE                                                             |
| 2625                                        | 7405 | 1323 | 1323   |                                                                      |
| 2626                                        | 7406 | 0065 | 0065   |                                                                      |
| 2627                                        | 7407 | 5615 | OUTPUT |                                                                      |
| 2630                                        | 7410 | 6303 | 6303   | /EJECT                                                               |
| 2631                                        | 7411 | 0274 | 0274   |                                                                      |
| 2632                                        | 7412 | 1344 | 1344   |                                                                      |
| 2633                                        | 7413 | 4261 | PSUD05 | /PSEUDO OP 5                                                         |
| 2634                                        | 7414 | 6070 | 6070   | /ASMIFZ (I HAD IT ASSIFZ ORGINALLY BUT THE PRUDES THOUGHT IT WAS TOO |
| 2635                                        | 7415 | 0752 | 0752   | /SUGGESTIVE! ! ! ! ! ! ! ! ! !)                                      |
| 2636                                        | 7416 | 0354 | 0354   |                                                                      |
| 2637                                        | 7417 | 4265 | PSUD01 | /PSEUDO OP 1                                                         |
| 2640                                        | 7420 | 6070 | 6070   | /ASMIFN                                                              |
| 2641                                        | 7421 | 0752 | 0752   |                                                                      |
| 2642                                        | 7422 | 0370 | 0370   |                                                                      |
| 2643                                        | 7423 | 4264 | PSUD02 | /PSEUDO OP 2                                                         |
| 2644                                        | 7424 | 6070 | 6070   | /ASMIFM                                                              |
| 2645                                        | 7425 | 0752 | 0752   |                                                                      |
| 2646                                        | 7426 | 0353 | 0353   |                                                                      |
| 2647                                        | 7427 | 4263 | PSUD03 | /PSEUDO OP 3                                                         |
| 2650                                        | 7430 | 6070 | 6070   | /ASMSKP                                                              |
| 2651                                        | 7431 | 0764 | 0764   |                                                                      |
| 2652                                        | 7432 | 0647 | 0647   |                                                                      |
| 2653                                        | 7433 | 4257 | PSUD07 | /PSEUDO OP 7                                                         |
| 2654                                        | 7434 | 7300 | 7300   | /SAVSYM                                                              |
| 2655                                        | 7435 | 1501 | 1501   |                                                                      |
| 2656                                        | 7436 | 1652 | 1652   |                                                                      |
| 2657                                        | 7437 | 4260 | PSUD06 | /PSEUDO OP 6                                                         |
| 2660                                        | 7440 | 6347 | 6347   | /FIELD                                                               |
| 2661                                        | 7441 | 0305 | 0305   |                                                                      |
| 2662                                        | 7442 | 0224 | 0224   |                                                                      |
| 2663                                        | 7443 | 2705 | FLDUP  |                                                                      |
| 2664                                        | 7444 | 7135 | 7135   | /PMODE                                                               |
| 2665                                        | 7445 | 1057 | 1057   |                                                                      |
| 2666                                        | 7446 | 0271 | 0271   |                                                                      |
| 2667                                        | 7447 | 3304 | PDP8MD |                                                                      |
| 2670                                        | 7450 | 6711 | 6711   | /LMODE                                                               |
| 2671                                        | 7451 | 1057 | 1057   |                                                                      |
| 2672                                        | 7452 | 0271 | 0271   |                                                                      |
| 2673                                        | 7453 | 3303 | LNKMOD |                                                                      |
| 2674                                        | 7454 | 7304 | 7304   | /SEGMNT                                                              |
| 2675                                        | 7455 | 0420 | 0420   |                                                                      |
| 2676                                        | 7456 | 1032 | 1032   |                                                                      |
| 2677                                        | 7457 | 0200 | BANKCH |                                                                      |
| 2700                                        | 7460 | 6713 | 6713   | /LODSYM                                                              |
| 2701                                        | 7461 | 0247 | 0247   |                                                                      |
| 2702                                        | 7462 | 1652 | 1652   |                                                                      |
| 2703                                        | 7463 | 4262 | PSUD04 | /PSEUDO OP 4                                                         |
| 2704                                        | 7464 | 0000 | 0000   | /AND                                                                 |
| 2705                                        | 7465 | 0000 | 0000   |                                                                      |
| 2706                                        | 7466 | 0062 | 0062   |                                                                      |
| 2707                                        | 7467 | 3400 | 3400   |                                                                      |
| 2710                                        | 7470 | 7056 | 7056   | /OCTAL                                                               |
| 2711                                        | 7471 | 1345 | 1345   |                                                                      |
| 2712                                        | 7472 | 0674 | 0674   |                                                                      |
| 2713                                        | 7473 | 2227 | OCT    |                                                                      |

|      |      |      |        |                    |
|------|------|------|--------|--------------------|
| 2714 | 7474 | 6231 | 6231   | /DECIMAL           |
| 2715 | 7475 | 0170 | 0170   |                    |
| 2716 | 7476 | 0742 | 0742   |                    |
| 2717 | 7477 | 2225 | DECIM  |                    |
| 2720 | 7500 | 7351 | 7351   | /TEXT              |
| 2721 | 7501 | 1614 | 1614   |                    |
| 2722 | 7502 | 0000 | 0      |                    |
| 2723 | 7503 | 1610 | STRING |                    |
| 2724 | 7504 | 7025 | 7025   | /NOLIST            |
| 2725 | 7505 | 0705 | 0705   |                    |
| 2726 | 7506 | 1323 | 1323   |                    |
| 2727 | 7507 | 0555 | PLIST  |                    |
| 2730 | 7510 | 6705 | 6705   | /LIST              |
| 2731 | 7511 | 1323 | 1323   |                    |
| 2732 | 7512 | 0000 | 0000   |                    |
| 2733 | 7513 | 0554 | DOLIST |                    |
| 2734 | 7514 | 7121 | 7121   | /PAGE              |
| 2735 | 7515 | 0410 | 0410   |                    |
| 2736 | 7516 | 0000 | 0      |                    |
| 2737 | 7517 | 0477 | PAGEC  |                    |
| 2740 | 7520 | 0000 | 0000   | /SEMICOLON         |
| 2741 | 7521 | 0000 | 0000   |                    |
| 2742 | 7522 | 0046 | 0046   |                    |
| 2743 | 7523 | 0200 | 0200   |                    |
| 2744 | 7524 | 0000 | 0000   | /EXCLAMATION POINT |
| 2745 | 7525 | 0000 | 0000   |                    |
| 2746 | 7526 | 0055 | 0055   |                    |
| 2747 | 7527 | 2200 | 2200   |                    |
| 2750 | 7530 | 6000 | 6000   | /STAR              |
| 2751 | 7531 | 0000 | 0000   |                    |
| 2752 | 7532 | 0066 | 0066   |                    |
| 2753 | 7533 | 1345 | ORGIC  |                    |
| 2754 | 7534 | 6000 | 6000   | /POINT             |
| 2755 | 7535 | 0000 | 0      |                    |
| 2756 | 7536 | 0072 | 0072   |                    |
| 2757 | 7537 | 4142 | PERIOD |                    |
| 2760 | 7540 | 6000 | 6000   | //                 |
| 2761 | 7541 | 0000 | 0000   |                    |
| 2762 | 7542 | 0075 | 0075   |                    |
| 2763 | 7543 | 3570 | THOU   |                    |
| 2764 | 7544 | 7411 | 7411   | /U                 |
| 2765 | 7545 | 0000 | 0000   |                    |
| 2766 | 7546 | 0000 | 0000   |                    |
| 2767 | 7547 | 1105 | UN     |                    |
| 2770 | 7550 | 0000 | 0000   | /COMMA             |
| 2771 | 7551 | 0000 | 0000   |                    |
| 2772 | 7552 | 0070 | 0070   |                    |
| 2773 | 7553 | 0600 | 0600   |                    |
| 2774 | 7554 | 0000 | 0000   | /=                 |
| 2775 | 7555 | 0000 | 0000   |                    |
| 2776 | 7556 | 0050 | 0050   |                    |
| 2777 | 7557 | 1400 | 1400   |                    |
| 3000 | 7560 | 7702 | 7702   | /Z                 |
| 3001 | 7561 | 0000 | 0      |                    |
| 3002 | 7562 | 0000 | 0      |                    |
| 3003 | 7563 | 1025 | A+1    |                    |
| 3004 | 7564 | 6515 | 6515   | /I                 |
| 3005 | 7565 | 0000 | 0      |                    |
| 3006 | 7566 | 0000 | 0      |                    |
| 3007 | 7567 | 1066 | H      |                    |
| 3010 | 7570 | 0000 | 0000   | /-                 |
| 3011 | 7571 | 0000 | 0000   |                    |
| 3012 | 7572 | 0071 | 0071   |                    |

|      |      |      |      |        |
|------|------|------|------|--------|
| 3013 | 7573 | 5200 | 5200 |        |
| 3014 | 7574 | 0000 | 0000 | /+     |
| 3015 | 7575 | 0000 | 0000 |        |
| 3016 | 7576 | 0067 | 0067 |        |
| 3017 | 7577 | 4600 | 4600 |        |
| 3020 | 7600 | 0000 | 0000 | /SPACE |
| 3021 | 7601 | 0000 | 0000 |        |
| 3022 | 7602 | 0054 | 0054 |        |
| 3023 | 7603 | 5600 | 5600 |        |
| 3024 |      |      |      | /      |
| 3025 |      |      |      | /      |
| 3026 |      |      |      | /      |
| 3027 |      |      |      | /      |
| 3030 |      |      |      | /      |
| 3031 |      |      |      | /      |
| 3032 |      |      |      | /      |
| 3033 |      |      |      | /      |
| 3034 |      |      |      | /      |
| 3035 |      |      |      | /      |
| 3036 |      |      |      | /      |
| 3037 |      |      |      | EJECT  |

```

3040      /
3041      /
3042      /
3043      /
3044      /
3045      /
3046      FIELD 1          /THIS IS REALY IN 5000 OF FIELD 0.
3047      /
3050      /
3051      /
3052      /
3053      /
3054      /
3055      *5000
3056      /
3057      /
3060      /
3061      /
3062      /
3063      /
3064      /
3065      /
3066      5000 0000 XXXX, 0
3067      5001 1754 TAD I XBLOCK
3070      5002 0353 AND XL77
3071      5003 3366 DCA XTHISBLOCK
3072      5004 4312 JMS XRESET
3073      5005 7240 CLA CMA
3074      5006 4264 JMS XPUT
3075      5007 5205 JMP -2
3076      5010 4312 JMS XRESET
3077      5011 4322 XL00P1, JMS XGET
3100      5012 5334 JMP XCER
3101      5013 5225 JMP XOK1
3102      5014 1051 TAD ITEM
3103      5015 1350 TAD XM47
3104      5016 7650 SNA CLA
3105      5017 5211 JMP XL00P1
3106      5020 1051 TAD ITEM
3107      5021 1346 TAD XM40
3110      5022 7650 SNA CLA
3111      5023 5211 JMP XL00P1
3112      5024 5334 JMP XCER
3113      5025 4322 XOK1, JMS XGET
3114      5026 5334 JMP XCER
3115      5027 5334 JMP XCER
3116      5030 5234 JMP XINT01
3117      5031 4322 XL00P2, JMS XGET
3120      5032 5334 JMP XCER
3121      5033 5244 JMP XDONE1
3122      5034 1051 XINT01, TAD ITEM
3123      5035 4765 JMS I XCON
3124      5036 3764 DCA I XXXP
3125      5037 2364 ISZ XXXP
3126      5040 1051 TAD ITEM
3127      5041 4264 JMS XPUT
3130      5042 5231 JMP XL00P2
3131      5043 5334 JMP XCER
3132      /
3133      5044 7240 XDONE1, CLA CMA
3134      5045 4264 JMS XPUT
3135      5046 7000 NOP
3136      5047 4322 JMS XGET

```

|      |      |      |         |            |
|------|------|------|---------|------------|
| 3137 | 5050 | 5743 | JMP I   | XN         |
| 3140 | 5051 | 5334 | JMP     | XCER       |
| 3141 | 5052 | 3033 | DCA     | EVAL       |
| 3142 | 5053 | 3020 | DCA     | ANY        |
| 3143 | 5054 | 4477 | JMS I   | EXPR1      |
| 3144 | 5055 | 1020 | TAD     | ANY        |
| 3145 | 5056 | 7650 | SNA CLA |            |
| 3146 | 5057 | 5743 | JMP I   | XN         |
| 3147 | 5060 | 1033 | TAD     | EVAL       |
| 3150 | 5061 | 0553 | AND     | XL77       |
| 3151 | 5062 | 3366 | DCA     | XTHISBLOCK |
| 3152 | 5063 | 5743 | JMP I   | XN         |
| 3153 |      |      | /       |            |
| 3154 |      |      | /       |            |
| 3155 |      |      | /       |            |
| 3156 |      |      | /       |            |
| 3157 |      |      | /       |            |
| 3160 |      |      | /       |            |
| 3161 | 5064 | 0000 | XPUT,   | 0          |
| 3162 | 5065 | 2017 | ISZ     | 17         |
| 3163 | 5066 | 5272 | JMP     | ,+4        |
| 3164 | 5067 | 7200 | CLA     |            |
| 3165 | 5070 | 2264 | ISZ     | XPUT       |
| 3166 | 5071 | 5664 | JMP I   | XPUT       |
| 3167 | 5072 | 0353 | AND     | XL77       |
| 3170 | 5073 | 3322 | DCA     | XGET       |
| 3171 | 5074 | 2355 | ISZ     | XHALF      |
| 3172 | 5075 | 5304 | JMP     | XRIGHT     |
| 3173 | 5076 | 1322 | TAD     | XGET       |
| 3174 | 5077 | 7106 | CLL RTL |            |
| 3175 | 5100 | 7006 | RTL     |            |
| 3176 | 5101 | 7006 | RTL     |            |
| 3177 | 5102 | 3356 | DCA     | XT         |
| 3200 | 5103 | 5664 | JMP I   | XPUT       |
| 3201 |      |      | /       |            |
| 3202 | 5104 | 7240 | XRIGHT, | CLA CMA    |
| 3203 | 5105 | 3355 | DCA     | XHALF      |
| 3204 | 5106 | 1356 | TAD     | XT         |
| 3205 | 5107 | 1322 | TAD     | XGET       |
| 3206 | 5110 | 3416 | DCA I   | 16         |
| 3207 | 5111 | 5664 | JMP I   | XPUT       |
| 3210 |      |      | /       |            |
| 3211 |      |      | /       |            |
| 3212 |      |      | /       |            |
| 3213 |      |      | /       |            |
| 3214 | 5112 | 0000 | XRESET, | 0          |
| 3215 | 5113 | 7240 | CLA CMA |            |
| 3216 | 5114 | 3355 | DCA     | XHALF      |
| 3217 | 5115 | 1347 | TAD     | XM9        |
| 3220 | 5116 | 3017 | DCA     | 17         |
| 3221 | 5117 | 1357 | TAD     | XP         |
| 3222 | 5120 | 3016 | DCA     | 16         |
| 3223 | 5121 | 5712 | JMP I   | XRESET     |
| 3224 |      |      | /       |            |
| 3225 |      |      | /       |            |
| 3226 |      |      | /       |            |
| 3227 | 5122 | 0000 | XGET,   | 0          |
| 3230 | 5123 | 4751 | JMS I   | XGETIN     |
| 3231 | 5124 | 1352 | TAD     | XM43       |
| 3232 | 5125 | 7450 | SNA     |            |
| 3233 | 5126 | 5722 | JMP I   | XGET       |
| 3234 | 5127 | 2322 | ISZ     | XGET       |
| 3235 | 5130 | 7001 | IAC     |            |

|      |      |      |             |        |     |        |
|------|------|------|-------------|--------|-----|--------|
| 3236 | 5131 | 7640 |             | SZA    | CLA |        |
| 3237 | 5132 | 2322 |             | ISZ    |     | XGET   |
| 3240 | 5133 | 5722 |             | JMP    | I   | XGET   |
| 3241 |      |      | /           |        |     |        |
| 3242 |      |      | /           |        |     |        |
| 3243 |      |      | /           |        |     |        |
| 3244 |      |      | /           |        |     |        |
| 3245 |      |      | /           |        |     |        |
| 3246 |      |      | /           |        |     |        |
| 3247 | 5134 | 1344 | XCER,       | TAO    |     | XL6    |
| 3250 | 5135 | 4476 |             | JMS    | I   | ERR1   |
| 3251 | 5136 | 4465 |             | JMS    | I   | PASSEK |
| 3252 | 5137 | 4466 |             | JMS    | I   | CURSKA |
| 3253 | 5140 | 4532 |             | JMS    | I   | SCLEAR |
| 3254 | 5141 | 4530 |             | JMS    | I   | LINENO |
| 3255 | 5142 | 5517 |             | JMP    | I   | MON    |
| 3256 |      |      | /           |        |     |        |
| 3257 | 5143 | 5202 | XN,         | XXN    |     |        |
| 3260 | 5144 | 0006 | XL6,        | 6      |     |        |
| 3261 | 5145 | 0017 | XL17,       | 17     |     |        |
| 3262 | 5146 | 7740 | XH40,       | -40    |     |        |
| 3263 | 5147 | 7767 | XH9,        | -9     |     |        |
| 3264 | 5150 | 7731 | XH47,       | -47    |     |        |
| 3265 | 5151 | 3670 | XGETIN,     | GETIN  |     |        |
| 3266 | 5152 | 7735 | XH43,       | -43    |     |        |
| 3267 | 5153 | 0077 | XL77,       | 77     |     |        |
| 3270 | 5154 | 4175 | XBBLOCK,    | BBLOCK |     |        |
| 3271 | 5155 | 0000 | XHALF,      | 0      |     |        |
| 3272 | 5156 | 0000 | XT,         | 0      |     |        |
| 3273 | 5157 | 5157 | XP,         |        |     |        |
| 3274 | 5160 | 0000 | XXN1,       | 0      |     |        |
| 3275 | 5161 | 0000 | XXN2,       | 0      |     |        |
| 3276 | 5162 | 0000 | XXN3,       | 0      |     |        |
| 3277 | 5163 | 0000 | XXN4,       | 0      |     |        |
| 3300 | 5164 | 5310 | XXXP,       | XNM1   |     |        |
| 3301 | 5165 | 3600 | XCON,       | CONVRT |     |        |
| 3302 |      |      | /           |        |     |        |
| 3303 |      |      | /           |        |     |        |
| 3304 |      |      | /           |        |     |        |
| 3305 |      |      | /           |        |     |        |
| 3306 |      |      | /           |        |     |        |
| 3307 | 5166 | 7777 | XTHISBLOCK, |        | -1  |        |
| 3310 | 5167 | 0036 |             | 36     |     |        |
| 3311 | 5170 | 0346 |             | 346    |     |        |
| 3312 | 5171 | 0002 |             | 2      |     |        |
| 3313 |      |      | /           |        |     |        |
| 3314 |      |      | /           |        |     |        |
| 3315 |      |      | /           |        |     |        |
| 3316 |      |      | /           |        |     |        |
| 3317 |      |      | /           |        |     |        |
| 3320 |      |      | /           |        |     |        |
| 3321 |      |      | /           |        |     |        |
| 3322 | 5172 | 5600 | XXRET,      | JMP    | I   | XXXX   |
| 3323 |      |      | /           |        |     |        |
| 3324 |      |      | /           |        |     |        |
| 3325 |      |      | /           |        |     |        |
| 3326 |      |      | /           |        |     |        |
| 3327 |      |      | /           |        |     |        |
| 3330 |      |      | /           |        |     |        |
| 3331 |      |      | /           |        |     |        |
| 3332 |      |      |             | PAGE   |     |        |
| 3333 |      |      | /           |        |     |        |
| 3334 |      |      | /           |        |     |        |

|      |      |      |         |            |        |
|------|------|------|---------|------------|--------|
| 3335 |      | /    |         |            |        |
| 3336 |      | /    |         |            |        |
| 3337 |      | /    |         |            |        |
| 3340 | 5200 | 4754 | XXXR,   | JMS I      | XXGET  |
| 3341 | 5201 | 7410 |         | SKP        |        |
| 3342 | 5202 | 1051 | XXN,    | TAD        | ITEM   |
| 3343 | 5203 | 1341 |         | TAD        | XXM43  |
| 3344 | 5204 | 7640 |         | SZA CLA    |        |
| 3345 | 5205 | 5200 |         | JMP        | XXXR   |
| 3346 |      |      | /       |            |        |
| 3347 | 5206 | 4736 |         | JMS I      | XREAD  |
| 3350 | 5207 | 5166 |         | XTMISBLOCK |        |
| 3351 | 5210 | 1337 |         | TAD        | XL6777 |
| 3352 | 5211 | 3016 |         | DCA        | 16     |
| 3353 | 5212 | 1340 |         | TAD        | XM100  |
| 3354 | 5213 | 3202 |         | DCA        | XXN    |
| 3355 | 5214 | 1016 | XXLOOP, | TAD        | 16     |
| 3356 | 5215 | 7040 |         | CMA        |        |
| 3357 | 5216 | 0342 |         | AND        | XL7770 |
| 3360 | 5217 | 7040 |         | CMA        |        |
| 3361 | 5220 | 3016 |         | DCA        | 16     |
| 3362 | 5221 | 1343 |         | TAD        | XXP    |
| 3363 | 5222 | 3017 |         | DCA        | 17     |
| 3364 | 5223 | 1344 |         | TAD        | XM4    |
| 3365 | 5224 | 3203 |         | DCA        | XXN+1  |
| 3366 | 5225 | 6211 | XXLP,   | CDF        | 10     |
| 3367 | 5226 | 1416 |         | TAD I      | 16     |
| 3370 | 5227 | 6201 |         | CDF        | 0      |
| 3371 | 5230 | 7041 |         | CIA        |        |
| 3372 | 5231 | 1417 |         | TAD I      | 17     |
| 3373 | 5232 | 7640 |         | SZA CLA    |        |
| 3374 | 5233 | 5327 |         | JMP        | XXY    |
| 3375 | 5234 | 2203 |         | ISE        | XXN+1  |
| 3376 | 5235 | 5225 |         | JMP        | XXLP   |
| 3377 | 5236 | 6211 |         | CDF        | 10     |
| 3400 | 5237 | 1416 |         | TAD I      | 16     |
| 3401 | 5240 | 3204 |         | DCA        | XXN+2  |
| 3402 | 5241 | 1416 |         | TAD I      | 16     |
| 3403 | 5242 | 6201 |         | CDF        | 0      |
| 3404 | 5243 | 7710 |         | SPA CLA    |        |
| 3405 | 5244 | 5327 |         | JMP        | XXY    |
| 3406 | 5245 | 4465 |         | JMS I      | PASSEK |
| 3407 | 5246 | 4466 |         | JMS I      | CURSKA |
| 3410 | 5247 | 4532 |         | JMS I      | SCLEAR |
| 3411 | 5250 | 4530 |         | JMS I      | LINENO |
| 3412 | 5251 | 1204 |         | TAD        | XXN+2  |
| 3413 | 5252 | 3745 |         | DCA I      | XXL    |
| 3414 | 5253 | 1346 |         | TAD        | XL7400 |
| 3415 | 5254 | 3750 |         | DCA I      | XX1    |
| 3416 | 5255 | 3751 |         | DCA I      | XX2    |
| 3417 | 5256 | 1347 |         | TAD        | XL4400 |
| 3420 | 5257 | 3752 |         | DCA I      | XX3    |
| 3421 | 5260 | 3037 |         | DCA        | MODE   |
| 3422 | 5261 | 1733 |         | TAD I      | XTH    |
| 3423 | 5262 | 3734 |         | DCA I      | XHT    |
| 3424 | 5263 | 4736 |         | JMS I      | XREAD  |
| 3425 | 5264 | 4170 |         | LTAB       |        |
| 3426 | 5265 | 4465 |         | JMS I      | PASSEK |
| 3427 | 5266 | 5321 |         | JMP        | XXFH   |
| 3430 | 5267 | 1335 |         | TAD        | XM26   |
| 3431 | 5270 | 4475 |         | JMS I      | SKIP   |
| 3432 | 5271 | 1277 | XXT,    | TAD        | XXMESS |
| 3433 | 5272 | 7450 |         | SNA        |        |

|      |      |      |         |              |
|------|------|------|---------|--------------|
| 3434 | 5273 | 5321 | JMP     | XXFH         |
| 3435 | 5274 | 4401 | JMS I   | DECOU        |
| 3436 | 5275 | 2271 | ISZ     | XXT          |
| 3437 | 5276 | 5271 | JMP     | XXT          |
| 3440 |      |      | /       |              |
| 3441 |      |      | /       |              |
| 3442 | 5277 | 0303 | XXMESS, | 303          |
| 3443 | 5300 | 0310 |         | 310          |
| 3444 | 5301 | 0301 |         | 301          |
| 3445 | 5302 | 0311 |         | 311          |
| 3446 | 5303 | 0316 |         | 316          |
| 3447 | 5304 | 0311 |         | 311          |
| 3450 | 5305 | 0316 |         | 316          |
| 3451 | 5306 | 0307 |         | 307          |
| 3452 |      |      | /       |              |
| 3453 | 5307 | 0240 |         | 240          |
| 3454 |      |      | /       |              |
| 3455 | 5310 | 0000 | XNM1,   | 0            |
| 3456 | 5311 | 0000 |         | 0            |
| 3457 | 5312 | 0000 |         | 0            |
| 3460 | 5313 | 0000 |         | 0            |
| 3461 | 5314 | 0000 |         | 0            |
| 3462 | 5315 | 0000 |         | 0            |
| 3463 | 5316 | 0000 |         | 0            |
| 3464 | 5317 | 0000 |         | 0            |
| 3465 | 5320 | 0000 |         | 0            |
| 3466 |      |      | /       |              |
| 3467 |      |      | /       |              |
| 3470 | 5321 | 4500 | XXFH,   | JMS I TYCARI |
| 3471 | 5322 | 7240 |         | CLA CMA      |
| 3472 | 5323 | 3012 |         | OCA CURLIN   |
| 3473 | 5324 | 4753 | JMS I   | XBUFIN       |
| 3474 | 5325 | 5726 | JMP I   | ,+1          |
| 3475 | 5326 | 5172 | XXRET   |              |
| 3476 |      |      | /       |              |
| 3477 |      |      | /       |              |
| 3500 |      |      | /       |              |
| 3501 |      |      | /       |              |
| 3502 |      |      | /       |              |
| 3503 |      |      | /       |              |
| 3504 |      |      | /       |              |
| 3505 | 5327 | 2202 | XXY,    | ISZ XXN      |
| 3506 | 5330 | 5214 | JMP     | XXLOOP       |
| 3507 | 5331 | 5732 | JMP I   | ,+1          |
| 3510 | 5332 | 5134 | XCER    |              |
| 3511 |      |      | /       |              |
| 3512 |      |      | /       |              |
| 3513 |      |      | /       |              |
| 3514 |      |      | /       |              |
| 3515 |      |      | /       |              |
| 3516 |      |      | /       |              |
| 3517 |      |      | /       |              |
| 3520 |      |      | /       |              |
| 3521 |      |      | /       |              |
| 3522 |      |      | /       |              |
| 3523 | 5333 | 5166 | XTH,    | XTHISBLOCK   |
| 3524 | 5334 | 4175 | XHT,    | BBLOCK       |
| 3525 | 5335 | 7752 | XN26,   | -26          |
| 3526 | 5336 | 7774 | XREAD,  | READ         |
| 3527 | 5337 | 6777 | XL6777, | 6777         |
| 3530 | 5340 | 7700 | XM100,  | -100         |
| 3531 | 5341 | 7735 | XXM43,  | -43          |
| 3532 | 5342 | 7770 | XL7770, | 7770         |

|      |      |      |         |        |
|------|------|------|---------|--------|
| 3533 | 5343 | 5157 | XXP,    | XP     |
| 3534 | 5344 | 7774 | XM4,    | -4     |
| 3535 | 5345 | 4177 | XXL,    | BUFINZ |
| 3536 | 5346 | 7400 | XL7400, | 7400   |
| 3537 | 5347 | 4400 | XL4400, | 4400   |
| 3540 | 5350 | 3354 | XX1,    | CHRC   |
| 3541 | 5351 | 3351 | XX2,    | WDHALF |
| 3542 | 5352 | 3352 | XX3,    | PTBUFP |
| 3543 | 5353 | 4006 | XBUFIN, | BUFIN  |
| 3544 | 5354 | 3670 | XXGET,  | GETIN  |
| 3545 |      |      | /       |        |
| 3546 |      |      | /       |        |
| 3547 |      |      | /       |        |
| 3550 |      |      | /       |        |
| 3551 |      |      | /       |        |
| 3552 |      |      | /       |        |
| 3553 |      |      | /       |        |
| 3554 |      |      | /       |        |
| 3555 |      |      | /       |        |
| 3556 |      |      | /       |        |
| 3557 |      |      | /       |        |
| 3560 |      |      | /       |        |
| 3561 |      |      | /       |        |
| 3562 |      |      | /       |        |
| 3563 |      |      | /       |        |
| 3564 |      |      |         | EJECT  |

|      |   |                                             |                                           |
|------|---|---------------------------------------------|-------------------------------------------|
| 3565 | / | M A J O R   T A P E   D E F I N I T I O N S |                                           |
| 3566 | / |                                             |                                           |
| 3567 | / |                                             |                                           |
| 3570 | / |                                             |                                           |
| 3571 | / |                                             |                                           |
| 3572 |   | FUDGE1=326-300                              |                                           |
| 3573 |   | FUDGE2=467-370                              |                                           |
| 3574 |   | SCRATCH=325-300                             | /DEFINES A SCRATCH AREA FOR SAVSYM&LOOSYM |
| 3575 |   | BCHAIN=324-300                              |                                           |
| 3576 |   | BOOUT=450-370                               |                                           |
| 3577 | / |                                             |                                           |
| 3600 | / |                                             |                                           |
| 3601 | / |                                             |                                           |
| 3602 | / |                                             |                                           |
| 3603 | / |                                             |                                           |
| 3604 |   | DIALUNIT=100                                |                                           |
| 3605 |   | DIALBINARY=111                              |                                           |
| 3606 | / |                                             |                                           |
| 3607 | / |                                             |                                           |
| 3610 | / |                                             |                                           |
| 3611 | / |                                             |                                           |
| 3612 |   | READ=7774                                   |                                           |
| 3613 |   | WRITE=7775                                  |                                           |
| 3614 | / |                                             |                                           |
| 3615 | / |                                             |                                           |
| 3616 | / |                                             |                                           |
| 3617 | / |                                             |                                           |
| 3620 | / |                                             |                                           |
| 3621 | / |                                             |                                           |
| 3622 | / |                                             |                                           |
| 3623 | / |                                             |                                           |
| 3624 | / |                                             |                                           |
| 3625 |   | EJECT                                       |                                           |

```
3626      /      E N D   A S S E M T H R E E ! ! ! ! !
3627      /
3630      /
3631      /
3632      /
3633      /
3634      /
3635      /
3636      /
3637      /
3640      EJECT
-
```

|      |   |
|------|---|
| 3641 | / |
| 3642 | / |
| 3643 | / |
| 3644 | / |
| 3645 |   |

NO ERRORS

| SYMBOL  | VALUE | DEF  | REFERENCES                                                                                                    |
|---------|-------|------|---------------------------------------------------------------------------------------------------------------|
| A       | 1024  | 1515 | 1521 1545 1565 3003                                                                                           |
| AADR    | 0030  | 0331 | 0517 0522 0525 0553 0563 1025 1030 1032 2051 2061 2071 0115 0121 0177 0227 1022 1534 1557 1602 1613 1621 1624 |
| AAORL   | 3052  | 1603 | 1623 1626                                                                                                     |
| AALPHA  | 4004  | 2602 | 2671                                                                                                          |
| AAS     | 1332  | 2030 | 2041 1717                                                                                                     |
| ABANK   | 0046  | 0347 | 2070 0205 1616 2432 2773                                                                                      |
| ABNOP1  | 3042  | 1573 | 0067                                                                                                          |
| ABNOP2  | 3043  | 1574 | 0070                                                                                                          |
| ABNOP3  | 3044  | 1575 | 0071                                                                                                          |
| ABNOP4  | 3045  | 1576 | 0072                                                                                                          |
| ACHECK  | 5460  | 0357 | 0424                                                                                                          |
| ACOUNT  | 5400  | 0250 | 0340 0345 0346 0366                                                                                           |
| ADDCON  | 4254  | 3120 | 3103                                                                                                          |
| ADR     | 0165  | 0467 | 1602 1733                                                                                                     |
| AEJECT  | 5516  | 0434 | 0310                                                                                                          |
| AEPOINT | 5532  | 0452 | 0445                                                                                                          |
| AEF62   | 5531  | 0451 | 0441                                                                                                          |
| AFORT   | 5457  | 0356 | 0411                                                                                                          |
| ALL     | 7600  | 1467 | 1527                                                                                                          |
| ALPHA   | 2400  | 1116 | 1215 2602                                                                                                     |
| ALPHAB  | 3445  | 2153 | 2133                                                                                                          |
| AMAINP  | 5507  | 0421 | 0353                                                                                                          |
| AMM10   | 5505  | 0417 | 0323                                                                                                          |
| ANDS    | 3400  | 1463 | 1526                                                                                                          |
| ANLXC   | 6054  | 1240 | 1213 1261                                                                                                     |
| ANLXE   | 6057  | 1244 | 1212                                                                                                          |
| ANY     | 0020  | 0321 | 0513 0714 1021 0077 0103 0706 1106 1111 1505 2760 0261 3142 3144                                              |
| ANY     | 2367  | 1102 | 0704 1112                                                                                                     |
| APAEND  | 5477  | 0376 | 0420                                                                                                          |
| APAGE   | 0031  | 0332 | 1725 2055 0201 0231                                                                                           |
| ASEND   | 5506  | 0420 | 0351                                                                                                          |
| ASPAST  | 5467  | 0366 | 0363                                                                                                          |
| ASSNO   | 5503  | 0411 | 0263 0331 0337 0437 0443 0446 0465 0564 1002                                                                  |
| ASSRST  | 5471  | 0370 | 0401                                                                                                          |
| ASSYIN  | 5441  | 0340 | 0333                                                                                                          |
| AWAY    | 1241  | 1737 | 1723 1727                                                                                                     |
| AW      | 1313  | 2011 | 2027 2031 2033 2035                                                                                           |
| B       | 1076  | 1567 | 1517 1520                                                                                                     |
| BAD     | 6400  | 1326 | 1371 1372 1275                                                                                                |
| BAND    | 5031  | 0074 | 0115                                                                                                          |
| BANKCH  | 0200  | 0512 | 2677                                                                                                          |
| BANKEM  | 0220  | 0532 | 0515                                                                                                          |
| BANKHR  | 0211  | 0523 | 0540                                                                                                          |
| BANOP1  | 5024  | 0067 | 0106                                                                                                          |
| BANOP2  | 5025  | 0070 | 0110                                                                                                          |
| BANOP3  | 5026  | 0071 | 0112                                                                                                          |
| BANOP4  | 5027  | 0072 | 0114                                                                                                          |
| BANOP5  | 5030  | 0073 | 0116                                                                                                          |
| BBLOCK  | 4175  | 3021 | 2606 2617 3270 3524                                                                                           |
| BCHAIN  | 0024  | 3575 | 1315                                                                                                          |
| BEGNAS  | 5002  | 0045 |                                                                                                               |
| BEGNLI  | 5000  | 0043 |                                                                                                               |
| BINBLK  | 0011  | 0311 | 1553 2363 2423 2647 3212                                                                                      |
| BINCK   | 3020  | 1551 | 1545                                                                                                          |
| BINHOR  | 4164  | 3004 | 2664                                                                                                          |
| BINPTR  | 0010  | 0310 | 1542 1546 1550 1551 2420 2422 2621                                                                            |
| BINTAB  | 0176  | 0500 | 1541                                                                                                          |
| BIT     | 2047  | 0554 | 2456                                                                                                          |
| BITT    | 3712  | 2456 | 2455                                                                                                          |
| BIT6    | 3400  | 2102 | 0366 2113 2126 2141 2150 2152 2155                                                                            |

| SYMBOL  | VALUE | DEF  | REFERENCES                                                                           |
|---------|-------|------|--------------------------------------------------------------------------------------|
| BIT6A   | 0064  | 0366 | 0554                                                                                 |
| BIT6M3  | 3462  | 2170 | 2134                                                                                 |
| BLKUSE  | 4340  | 3301 | 3222                                                                                 |
| BLM10   | 5035  | 0100 | 0123                                                                                 |
| BLPM22  | 5036  | 0101 | 0124                                                                                 |
| BLTAD   | 5032  | 0075 | 0117 0121                                                                            |
| BLTAD1  | 5033  | 0076 | 0120                                                                                 |
| BLTAD2  | 5034  | 0077 | 0122                                                                                 |
| BOOUT   | 0060  | 3576 | 0604                                                                                 |
| BUFI    | 3350  | 2027 | 2014                                                                                 |
| BUFIN   | 4006  | 2604 | 2027 2610 2630 3543                                                                  |
| BUFINZ  | 4177  | 3023 | 2607 2615 3535                                                                       |
| BUMP    | 0726  | 1323 | 1270 1312 1324 1325 1331 1332 1333 1334                                              |
| B1      | 1041  | 1532 | 0755 1542 1543                                                                       |
| B11     | 0434  | 0755 | 0734 0744                                                                            |
| B2      | 1046  | 1537 | 0405 1550 1574                                                                       |
| B21     | 0103  | 0405 | 1111 0733 0735 1774 2767                                                             |
| C       | 1055  | 1546 | 1522 1523 1524 1526 1540 1552 1554 1556 2325                                         |
| CARET   | 3443  | 2151 | 2124                                                                                 |
| CCDF    | 0753  | 1355 | 1353                                                                                 |
| CCNOP   | 6304  | 1456 | 1453                                                                                 |
| CHARAC  | 0057  | 0360 | 0147 0567 0573 0616 0633 0640                                                        |
| CHARX   | 0044  | 0345 | 0722 1773 2007 0555 0556                                                             |
| CHARX2  | 0045  | 0346 | 0737 0557 0711 0720                                                                  |
| CHECK   | 0740  | 1342 | 0443 1171 1345 1347 1351 1361                                                        |
| CHECKO  | 5737  | 1032 | 2646 1047                                                                            |
| CHEKER  | 0070  | 0372 | 1243                                                                                 |
| CHKL    | 1264  | 1762 | 1750                                                                                 |
| CHKLA   | 1222  | 1720 | 1710                                                                                 |
| CHO     | 4045  | 2643 | 2624                                                                                 |
| CHRC    | 3354  | 2033 | 2006 2011 2643 3540                                                                  |
| CJMP1   | 1547  | 0224 | 0375 0062 0222                                                                       |
| CKUN    | 3262  | 1740 | 1731 1744                                                                            |
| CLEARRR | 2132  | 0637 | 0434 0127 0650                                                                       |
| CLEUP   | 1453  | 0127 | 0071                                                                                 |
| CNOP    | 0747  | 1351 | 1456                                                                                 |
| CNTLP1  | 0004  | 0302 | 0070 1244 1404 2725                                                                  |
| CNTMON  | 2363  | 1075 | 1072                                                                                 |
| CNTRLP  | 2353  | 1064 | 0302 1066 1073                                                                       |
| COMMA   | 0600  | 1456 | 1541                                                                                 |
| COMMAN  | 0260  | 0574 | 0544 0254                                                                            |
| COMMEN  | 3701  | 2445 | 0540 2452                                                                            |
| COMS    | 2036  | 0540 | 0546                                                                                 |
| CON     | 0164  | 0466 | 1472 1474 1606 1627 1711 0265 0270 0271 2323 2462 2463 3054 3063 3064 3102 3110 3111 |
| CONREC  | 3725  | 2502 | 2453                                                                                 |
| CONSKP  | 3674  | 2440 | 2446 2454                                                                            |
| CONVRT  | 3600  | 2340 | 0062 2346 3301                                                                       |
| CONYES  | 3700  | 2444 | 2445 0073                                                                            |
| COSA    | 3541  | 2270 | 2255                                                                                 |
| COUNT   | 0760  | 1370 | 1167 1264 1265 1304 1314 1343 2746 1272                                              |
| CRFOUN  | 6010  | 1151 | 1143                                                                                 |
| CRIOR   | 1165  | 1660 | 1505 1546 1551 1553 1555 1603 1636 1644                                              |
| CTYPO   | 0763  | 1373 | 1357                                                                                 |
| CURLIN  | 0012  | 0312 | 0406 0414 0425 0433 1247 1573 1665 2635 3472                                         |
| CURR    | 0043  | 0344 | 0771 0774 0775 1002 1010 1015                                                        |
| CURREN  | 0227  | 0542 | 1667                                                                                 |
| CURSKA  | 0066  | 0370 | 0530 1035 1655 2064 0173 0341 1511 3252 3407                                         |
| CURSKP  | 2520  | 1246 | 0370 1254                                                                            |
| C1A     | 3576  | 2325 | 2324                                                                                 |
| DECIM   | 2225  | 0732 | 2717                                                                                 |
| DECIMS  | 2231  | 0736 | 0732 0741                                                                            |
| DECOUT  | 0001  | 0277 | 1407 0657 1240 1242 1451 1471 1674 1703 2211 2262 2264 2721 2723 2724 0657 1042 3435 |

| SYMBOL | VALUE | DEF  | REFERENCES                                                                                                    |
|--------|-------|------|---------------------------------------------------------------------------------------------------------------|
| DEFIN1 | 1174  | 1667 | 1600                                                                                                          |
| DIALBI | 0111  | 3605 | 3004 3176 0520 0602                                                                                           |
| DIALUN | 0100  | 3604 | 1313 1321 2060 3011 3243 0207                                                                                 |
| DISRE  | 4001  | 2576 | 2665                                                                                                          |
| DISRET | 3463  | 2171 | 2216 2576                                                                                                     |
| DLFT   | 2126  | 0633 | 0673                                                                                                          |
| DOCAL  | 1426  | 0102 | 0252 0425                                                                                                     |
| DOLEFT | 1474  | 0151 | 0143                                                                                                          |
| DOLIST | 0554  | 1076 | 2733                                                                                                          |
| DOPOUT | 0567  | 1111 | 1103                                                                                                          |
| DPCS   | 2212  | 0717 | 0714 0731                                                                                                     |
| DPCV   | 2200  | 0705 | 0736 0723 0726                                                                                                |
| DPCVM2 | 2242  | 0747 | 0715                                                                                                          |
| DPCVOP | 0415  | 0736 | 0733                                                                                                          |
| DRHT   | 2111  | 0616 | 0674                                                                                                          |
| DV3    | 2640  | 1433 | 1430                                                                                                          |
| D6     | 0300  | 0246 | 1775 3013                                                                                                     |
| D7     | 0370  | 0247 | 0310 0500 2600 2637 3006 3222                                                                                 |
| D8     | 0410  | 0250 |                                                                                                               |
| D9     | 0446  | 0251 |                                                                                                               |
| EDEFIN | 3771  | 2561 | 2556                                                                                                          |
| EMLEVE | 2130  | 0635 | 0571 0620                                                                                                     |
| EMPTRA | 0172  | 0474 | 0576 0600 0601 0622                                                                                           |
| EMPTY  | 2054  | 0561 | 0363 0566 0636                                                                                                |
| EMPTY1 | 0061  | 0363 | 0110 1253                                                                                                     |
| ENDMS  | 3717  | 2471 | 0700 0427                                                                                                     |
| ENTS   | 2000  | 0501 | 0525 0531 0533 1745 0075                                                                                      |
| ENTS1  | 3267  | 1745 | 1733                                                                                                          |
| EQCHK  | 2332  | 1043 | 1032 1050                                                                                                     |
| EQCHKI | 2340  | 1051 | 1045                                                                                                          |
| EQRETI | 0006  | 0304 | 1637 1643 1047 2046                                                                                           |
| EQU    | 1142  | 1635 | 1572                                                                                                          |
| EQUALS | 1400  | 1460 | 1544 1572                                                                                                     |
| EGUCHK | 3366  | 2045 | 1051 2050                                                                                                     |
| EQUERR | 3766  | 2555 | 1633                                                                                                          |
| ERONE  | 1140  | 1632 | 1544                                                                                                          |
| ERRCNT | 0167  | 0471 | 0167 0210 0642 0645 1570 1662 1706                                                                            |
| ERROR  | 3201  | 1656 | 0400 0400 1713                                                                                                |
| ERR1   | 0076  | 0400 | 0757 1671 1026 1037 1100 2047 2157 2555 3250                                                                  |
| ERR12  | 2364  | 1077 | 1057                                                                                                          |
| ERR2   | 2275  | 1006 | 0576 1012 1024 1027 1053 1060                                                                                 |
| ERSY   | 0674  | 1245 | 1156                                                                                                          |
| ERTOT  | 0002  | 0300 | 0206 0643 0644 2173                                                                                           |
| EVAL   | 0033  | 0334 | 0526 0533 0535 1033 1037 1607 1646 1720 1741 1751 1754 1755 1767 2050 2052 2062 2066 0106 0113 0203 0305 0316 |
|        |       |      | 0323 0324 0360 1513 1517 1522 1563 1606 0264 3141 3147                                                        |
| EXCLAM | 2200  | 1462 | 1524                                                                                                          |
| EXPR   | 1000  | 1471 | 0401 1512 1605 1630                                                                                           |
| EXPR1  | 0077  | 0401 | 0512 1020 1641 1702 2044 0102 1501 0260 0434 3143                                                             |
| E7     | 0000  | 0245 |                                                                                                               |
| FACTOR | 4330  | 3222 | 3213                                                                                                          |
| FORT   | 2714  | 1510 | 0377 1504 1515 1521 1524 0356                                                                                 |
| FIELDM | 2717  | 1513 | 1507                                                                                                          |
| FINEND | 2173  | 0700 | 0553 0426                                                                                                     |
| FIRST  | 6110  | 1325 | 1271 1313 1371 1274 1275                                                                                      |
| FLOUP  | 2705  | 1501 | 2663                                                                                                          |
| FLOWD  | 0036  | 0337 | 0534 0207 1523 1543                                                                                           |
| FOUND  | 2507  | 1231 | 1120 1205 1213 1221                                                                                           |
| FSBLK  | 4002  | 2600 | 2613 0147 0152                                                                                                |
| FSUNIT | 4003  | 2601 | 2616 0145                                                                                                     |
| FUDGE1 | 0026  | 3572 | 0211                                                                                                          |
| FUDGE2 | 0077  | 3573 | 0522                                                                                                          |
| G      | 1113  | 1604 | 1525 1527                                                                                                     |

| SYMBOL | VALUE | DEF  | REFERENCES                                                                                               |
|--------|-------|------|----------------------------------------------------------------------------------------------------------|
| GENINI | 1703  | 0401 | 0275 0330 0370                                                                                           |
| GET    | 1110  | 1601 | 1541 1573                                                                                                |
| GETADR | 1200  | 1676 | 1665 1700 1735 1736                                                                                      |
| GETAD1 | 1172  | 1665 | 1601                                                                                                     |
| GETASY | 3242  | 1720 | 0412 1723 1726 1736 1737 1743                                                                            |
| GETAS1 | 0110  | 0412 | 0762                                                                                                     |
| GETCHR | 0067  | 0371 | 2435                                                                                                     |
| GETIN  | 3670  | 2434 | 0401 0537 2443 2447 3265 3544                                                                            |
| GETLET | 0441  | 0762 | 0727                                                                                                     |
| GETS   | 2035  | 0537 | 0543                                                                                                     |
| GETSYM | 0400  | 0721 | 0411 0761 0767 0772 0776 1003 1011                                                                       |
| GETSY1 | 0107  | 0411 | 1016 1516 1537 1571                                                                                      |
| GETVAL | 5401  | 0251 | 0265 0327 0334 0461 1001                                                                                 |
| GETYE  | 0365  | 0703 | 0675                                                                                                     |
| GETYPE | 3623  | 2367 | 0703 2316 2416                                                                                           |
| GGGET  | 0676  | 1262 | 1273 1356                                                                                                |
| GSTART | 0446  | 0767 | 0753 1004 1017                                                                                           |
| GTYPOT | 0464  | 1005 | 1001                                                                                                     |
| GYP    | 3567  | 2316 | 2302                                                                                                     |
| H      | 1066  | 1557 | 1577 3007                                                                                                |
| HIC    | 0052  | 0353 | 0707 1422 1431 1435                                                                                      |
| H6     | 1075  | 1566 | 1507 1511 1563 1564 1610 1613                                                                            |
| IDX    | 2037  | 0542 | 0406 0560 2161                                                                                           |
| IDX1   | 0104  | 0406 | 0760 1515 1536 1640 1677 2004 2043 0072 0717 2766 0364                                                   |
| IMINIT | 5132  | 0203 | 0172                                                                                                     |
| IMMOVE | 5130  | 0201 | 0156                                                                                                     |
| IMREAD | 5131  | 0202 | 0164                                                                                                     |
| IMTYPO | 5127  | 0200 | 0171                                                                                                     |
| IM215  | 5126  | 0177 | 0170                                                                                                     |
| IN     | 0050  | 0351 | 2003 1722 1724                                                                                           |
| INAS   | 0073  | 0375 | 0125                                                                                                     |
| IND    | 0166  | 0470 | 1611 1744 1753 1770 0100                                                                                 |
| IND1   | 0370  | 0706 | 0622 0630 0636                                                                                           |
| IND1B  | 0364  | 0702 | 0674                                                                                                     |
| IND1C  | 0366  | 0704 | 0677                                                                                                     |
| INIT   | 4013  | 2611 | 0361 2636 0203                                                                                           |
| INIT1  | 0060  | 0361 | 0221 0167                                                                                                |
| INP    | 0722  | 1313 | 1311 2744                                                                                                |
| INST   | 0055  | 0356 | 0650 1475 1477 1622 1625 0236                                                                            |
| INSYM  | 3375  | 2062 | 1767                                                                                                     |
| ITABIN | 5133  | 0207 | 0165                                                                                                     |
| ITEM   | 0051  | 0352 | 0135 0141 0311 0321 0335 0347 0364 0547 2103 2130 2137 2154 2436 2442 0341 0360 3102 3106 3122 3126 3342 |
| JBHSTR | 1707  | 0406 | 0413 1356                                                                                                |
| JBNOP  | 5014  | 0057 | 0105 0107 0111 0113                                                                                      |
| JBSMF  | 1732  | 0433 | 0412                                                                                                     |
| JBTEST | 1706  | 0405 | 0432 0422 0430                                                                                           |
| JEND   | 1734  | 0435 | 1113 0423 0062                                                                                           |
| JJEND  | 0571  | 1113 | 1110                                                                                                     |
| JJSTAR | 0570  | 1112 | 1105                                                                                                     |
| JNOP   | 1730  | 0427 | 0102                                                                                                     |
| JSTART | 1733  | 0434 | 1112 0420 0063                                                                                           |
| JUDGE  | 2452  | 1170 | 1152 1162                                                                                                |
| KILBUF | 4075  | 2701 | 1075                                                                                                     |
| KLOOP  | 4114  | 2725 | 2704 2731                                                                                                |
| KLUG2  | 2170  | 0675 | 0667                                                                                                     |
| LBA0   | 6103  | 1275 | 1227                                                                                                     |
| LCB    | 6662  | 1552 | 1260 1266 1420 1435                                                                                      |
| LCF    | 6652  | 1550 | 1240 1250 1344 1417 1506                                                                                 |
| LCOUNT | 6100  | 1272 | 1220                                                                                                     |
| LDT    | 1274  | 1772 | 1775 2006 2010 2012 2025 2037                                                                            |
| LEAVE  | 1236  | 1734 | 1732 1745 1756                                                                                           |
| LEFTHF | 3340  | 2017 | 2001                                                                                                     |

| SYMBOL | VALUE | DEF  | REFERENCES                                                  |
|--------|-------|------|-------------------------------------------------------------|
| LEVEL  | 1256  | 1754 | 1717 1766 1771                                              |
| LFF    | 6032  | 1174 | 1161                                                        |
| LFF2   | 6033  | 1175 | 1165                                                        |
| LFIRST | 6102  | 1274 | 1224                                                        |
| LFTEST | 6014  | 1155 | 1202                                                        |
| LINENO | 0130  | 0432 | 0065 0345 3254 3411                                         |
| LINKDT | 4152  | 2770 | 2763                                                        |
| LINKI  | 1104  | 1575 | 1561                                                        |
| LINKIC | 1370  | 2066 | 2047                                                        |
| LINKB  | 2624  | 1417 | 1400                                                        |
| LINKBA | 3551  | 2300 | 1417 2307 2315                                              |
| LISREG | 5062  | 0125 | 0104                                                        |
| LISTA  | 2731  | 1525 | 0234                                                        |
| LISTWD | 0003  | 0301 | 0132 0164 0411 0427 0562 1565 2035 2501 0046 0053           |
| LIST1  | 1557  | 0234 | 0075                                                        |
| LLB    | 6654  | 1547 | 1240                                                        |
| LNCK   | 3737  | 2517 | 0653 2523                                                   |
| LNCKA  | 2146  | 0653 | 0621 0651                                                   |
| LNINST | 1560  | 0235 | 0702 0251                                                   |
| LNKA   | 3062  | 1613 | 1601                                                        |
| LNKADD | 4250  | 3113 | 3106                                                        |
| LNKMOD | 3303  | 1762 | 2673                                                        |
| LNKOR  | 1250  | 1746 | 1705                                                        |
| LNKREG | 1452  | 0126 | 0120                                                        |
| LNKSUB | 4225  | 3066 | 3060                                                        |
| LNRT   | 1365  | 2063 | 2057 2072                                                   |
| LOK    | 6200  | 1343 | 1400 1403 1406 1410 1423 1527                               |
| LOKA2  | 6314  | 1504 | 1441                                                        |
| LOK0   | 6210  | 1354 | 1520                                                        |
| LOK1   | 6212  | 1357 | 1447 1450 1451 1452 1454                                    |
| LOUTP  | 6101  | 1273 | 1222 1223 1225 1230 1231                                    |
| LOWTMP | 0140  | 0442 | 1267 1272 1562 1564                                         |
| LPBACK | 6041  | 1220 | 1146                                                        |
| LPR    | 6064  | 1553 | 2733 1251 1260 1266 1345 1435 1506                          |
| LPRENT | 6065  | 1252 | 1267                                                        |
| LPTEST | 6244  | 1413 | 0137                                                        |
| LP00C  | 6071  | 1260 | 1523                                                        |
| LP00E  | 6073  | 1263 | 1524                                                        |
| LSD    | 6661  | 1551 | 1350 1145 1421 1437                                         |
| LSRTE  | 1472  | 0147 | 0157                                                        |
| LSTSYM | 1454  | 0131 | 0373 0134 0140 0150                                         |
| LTAB   | 4170  | 3011 | 2633 3425                                                   |
| LTREAD | 4005  | 2603 | 2605 2632                                                   |
| LWC    | 0053  | 0354 | 1533 0710 0752 0770 0772 0777 1001 1421 1423 1434 1437 2320 |
| LWD    | 3477  | 2206 | 2175 2203 2204 2212                                         |
| LWDE   | 3504  | 2214 | 2210 1530                                                   |
| LWDEFD | 3507  | 2224 | 1527                                                        |
| L10    | 6104  | 1300 | 1247                                                        |
| L17    | 6305  | 1466 | 1343                                                        |
| L7710  | 5520  | 0436 | 0325                                                        |
| L777   | 4174  | 3017 | 2614                                                        |
| MAIN   | 1400  | 0054 | 0374 0063                                                   |
| MAINEX | 0072  | 0374 | 2474 0126 0166                                              |
| MAINS  | 1504  | 0161 | 0105 0421 0422                                              |
| MAINSE | 1411  | 0065 | 0403 0111 0163 0166 0174                                    |
| MAINSF | 1416  | 0072 | 0600 2561                                                   |
| MAIN1  | 0101  | 0403 | 0531 1036 1656 2065 1303 1512                               |
| MAIN1A | 0264  | 0600 | 0572                                                        |
| MASSIF | 5426  | 0325 | 0306                                                        |
| MESS   | 3510  | 2232 | 2206                                                        |
| MESSS  | 3520  | 2242 | 2200                                                        |
| MICRO  | 0056  | 0357 | 1623 1706                                                   |

| SYMBOL | VALUE | DEF | REFERENCES |
|--------|-------|-----|------------|
|--------|-------|-----|------------|

|         |      |      |                                                                                 |
|---------|------|------|---------------------------------------------------------------------------------|
| MICROB  | 0005 | 0303 | 1617 1746                                                                       |
| MICROL  | 0054 | 0355 | 1500 1502 1616 1621 0246                                                        |
| MINUS   | 5200 | 1465 | 1523                                                                            |
| MMHOVE  | 4136 | 2751 | 2712                                                                            |
| MODE    | 0037 | 0340 | 0550 0672 1557 1703 2045 0060 0116 1502 1577 1764 1765 2631 2761 3056 3104 3421 |
| MON     | 0117 | 0421 | 1672 0356 1101 2673 3255                                                        |
| MONIT   | 4101 | 2712 | 0421                                                                            |
| MOVA    | 0545 | 1066 | 1056                                                                            |
| MOVB    | 0540 | 1061 | 1074                                                                            |
| MOVEAA  | 0262 | 0576 | 0547                                                                            |
| MOVE1   | 0074 | 0376 | 1176                                                                            |
| MOVE8   | 0524 | 1045 | 0376 1051 1053 1054 1055 1065 1100 1104 1106                                    |
| MTAD    | 2265 | 0775 | 0757 0763 1002                                                                  |
| MTDG    | 2273 | 1004 | 0727 0761                                                                       |
| MTHREE  | 7346 | 0256 | 0656                                                                            |
| MTRL    | 2257 | 0766 | 0755 0756 0760 0773                                                             |
| MTSW    | 2251 | 0757 | 0737 0740 0744                                                                  |
| MTWO    | 7344 | 0254 | 1234 0520 1516 1604 1610 1667 1704 2413                                         |
| MT10    | 2243 | 0751 | 0730 0764                                                                       |
| M1      | 0523 | 1044 | 1046                                                                            |
| M1A     | 1735 | 0436 |                                                                                 |
| M10     | 2171 | 0676 | 0664                                                                            |
| M1000   | 0121 | 0423 | 0216 0743                                                                       |
| M12A    | 0414 | 0735 | 0730                                                                            |
| M140    | 6060 | 1245 |                                                                                 |
| M154    | 6311 | 1472 | 1347                                                                            |
| M20     | 3454 | 2162 | 2107                                                                            |
| M212    | 6105 | 1301 | 1157                                                                            |
| M215    | 0151 | 0453 | 1070 1141 1163 1203                                                             |
| M22     | 2535 | 1263 | 1251 0101                                                                       |
| M26     | 3661 | 2425 | 2355                                                                            |
| M33     | 0123 | 0425 | 0725 1445 2131                                                                  |
| M4      | 0135 | 0437 | 1043 1457 2256 1414 1431                                                        |
| M40     | 0013 | 0313 | 2104 2342 2653 0324                                                             |
| M43     | 0122 | 0424 | 0136 0334 0365 2450 0342 0361                                                   |
| M45     | 0125 | 0427 | 2000 0721 1426                                                                  |
| M47     | 0133 | 0435 | 0612 0627                                                                       |
| M57     | 2034 | 0536 | 0544                                                                            |
| M6      | 3101 | 1632 | 1575                                                                            |
| M6A     | 1261 | 1757 | 1715                                                                            |
| M60     | 1576 | 0253 | 0073 0403 1467                                                                  |
| M60A    | 3662 | 2426 | 0441 2357 1470                                                                  |
| M67     | 0126 | 0430 | 0712                                                                            |
| M7      | 2702 | 1476 | 1376                                                                            |
| M7 TWO  | 2174 | 0701 | 0075                                                                            |
| M72     | 6310 | 1471 | 1510 1512                                                                       |
| M77     | 3756 | 2537 | 2505                                                                            |
| NASS IF | 5424 | 0323 | 0304 0504                                                                       |
| NONER   | 3046 | 1577 | 1572                                                                            |
| NOS     | 4137 | 2755 | 2775                                                                            |
| NOTAB1  | 2107 | 0614 | 0076                                                                            |
| NOTAB2  | 2124 | 0631 | 0077                                                                            |
| NOTBLK  | 3616 | 2362 | 1627 2366                                                                       |
| NOTBL1  | 3076 | 1627 | 1555                                                                            |
| NOTLDT  | 1311 | 2007 | 2002                                                                            |
| NUM     | 6106 | 1313 | 1153 1171 1174 1253                                                             |
| NUPAGE  | 1262 | 1760 | 1731 1776 1777 2005 2013 2014 2023                                              |
| NXCAND  | 2413 | 1131 | 1211                                                                            |
| OASSNO  | 5725 | 1002 | 0624 0655 0660 0744                                                             |
| OBIN    | 5720 | 0756 | 0735                                                                            |
| OBLOCK  | 5735 | 1013 | 0632 0722                                                                       |
| OBSET   | 5706 | 0733 | 0627 0736                                                                       |

| SYMBOL  | VALUE | DEF  | REFERENCES                                                                 |
|---------|-------|------|----------------------------------------------------------------------------|
| OCCRLF  | 5751  | 1046 | 1035 1041                                                                  |
| OCDOIT  | 5743  | 1037 | 1043 1044                                                                  |
| OCHECK  | 4050  | 2646 | 2666                                                                       |
| OCPNT   | 5753  | 1054 | 1037                                                                       |
| OCT     | 2227  | 0734 | 2713                                                                       |
| OCTS    | 2235  | 0742 | 0212 0734 0745                                                             |
| OCTS1   | 1535  | 0212 | 0064 0217                                                                  |
| OCUSES  | 5776  | 1102 | 1033                                                                       |
| OEND    | 5661  | 0673 | 0667 0700                                                                  |
| OGTVAL  | 5724  | 1001 | 0620                                                                       |
| OLOC    | 5730  | 1005 | 0634 0674 0706 0707 0713 0714                                              |
| OL17    | 5726  | 1003 | 0630                                                                       |
| OL377   | 5732  | 1007 | 0675 0711 0715                                                             |
| OL6400  | 5731  | 1006 | 0633 0712                                                                  |
| ONEREG  | 1436  | 0112 | 0107 0124 0126 0402                                                        |
| ONEREI  | 1704  | 0402 | 0306 0361                                                                  |
| ONVRT   | 2155  | 0662 | 0656                                                                       |
| OOPUT   | 5667  | 0705 | 0670 0673 0717 0723                                                        |
| OPOINT  | 5727  | 1004 | 0656                                                                       |
| OPS     | 2661  | 1455 | 0364 1402 1473 1474 1531                                                   |
| OPUT    | 5654  | 0665 | 0621 0625 0671 0677 1004                                                   |
| OP1     | 4273  | 3150 | 3134 3135 3136 3137 3140 3141 3143 3146                                    |
| ORGIC   | 1345  | 2043 | 2753                                                                       |
| OSANDV  | 2600  | 1373 | 1230 1405                                                                  |
| OSYM    | 2613  | 1406 | 1374 1416 1425 1444 1456 1461 1464 1465                                    |
| OUNIT   | 5733  | 1011 | 0631 0721                                                                  |
| OUTP    | 0705  | 1271 | 1266 2745 1273                                                             |
| OUTPUT  | 5615  | 0620 | 2627                                                                       |
| OUTTAB  | 2156  | 0663 | 0614 0631                                                                  |
| OUTTP   | 2147  | 0654 | 0615 0632 0661                                                             |
| OX      | 2625  | 1420 | 1411 1413 1415 1441                                                        |
| OXREAD  | 2546  | 1310 | 1276 1301                                                                  |
| OXWRIT  | 2547  | 1311 | 1274                                                                       |
| PACHECK | 5512  | 0424 | 0347                                                                       |
| PAFTER  | 4270  | 3145 | 0277                                                                       |
| PAGEC   | 0477  | 1020 | 2737                                                                       |
| PAGEM   | 0516  | 1037 | 1023                                                                       |
| PAGE1   | 0102  | 0404 | 0524 1031 2060 0125 1773                                                   |
| PAGHER  | 0510  | 1031 | 1043                                                                       |
| PAGSET  | 3524  | 2253 | 0372 2267 2274 2277                                                        |
| PASS    | 0035  | 0336 | 1101 0057 0161 0215 0564 1031 1035 1041 2040 2437 2471 0462 0502 0622      |
| PASSEK  | 0065  | 0367 | 0527 1034 1694 2063 0172 0307 0340 0362 1510 1661 1710 0435 3251 3406 3426 |
| PASSER  | 3355  | 2034 | 0367 2042 2043 2044                                                        |
| PASS1   | 1401  | 0055 | 0054 0224                                                                  |
| PASS1P  | 1407  | 0063 | 0223                                                                       |
| PASS2   | 1537  | 0214 | 0225                                                                       |
| PCFIG   | 6037  | 1212 | 1172 1526                                                                  |
| PCHECK  | 0141  | 0443 | 1303 1315 2726                                                             |
| PCLP    | 6040  | 1213 | 1147 1525                                                                  |
| PCONT   | 3522  | 2251 | 2214 2215 2254 2270 2276 2513 0452                                         |
| PCOT    | 3736  | 2513 | 2504                                                                       |
| PCOUNT  | 4133  | 2746 | 2701 2727                                                                  |
| PC1     | 6306  | 1467 | 1350 1351 1511                                                             |
| PC2     | 6307  | 1470 | 1352 1513                                                                  |
| PDLK    | 4147  | 2765 | 2757 3000                                                                  |
| PDOCAL  | 5513  | 0425 | 0370                                                                       |
| PDOLLA  | 5514  | 0426 | 0352 0373                                                                  |
| PDPBER  | 0253  | 0566 | 0562                                                                       |
| PDPBMO  | 3304  | 1763 | 2667                                                                       |
| PENDMS  | 5515  | 0427 | 0372 0377                                                                  |
| PERIOD  | 4142  | 2760 | 2757                                                                       |
| PERMA   | 0032  | 0333 | 0644 0055 1121 2550                                                        |

| SYMBOL | VALUE | DEF  | REFERENCES                              |
|--------|-------|------|-----------------------------------------|
| PGJS   | 3723  | 2500 | 2503 2512 2674                          |
| PINP   | 4131  | 2744 | 2702                                    |
| PJEND  | 5017  | 0062 | 0052                                    |
| PJMP   | 5414  | 0303 | 0300                                    |
| PJNOP  | 5037  | 0102 | 0060 0061                               |
| PJSTAR | 5020  | 0063 | 0050                                    |
| PLIST  | 0555  | 1077 | 2727                                    |
| PLQVR  | 0762  | 1372 | 1326                                    |
| PLP08C | 6331  | 1523 | 1514                                    |
| PLP08E | 6332  | 1524 | 1516                                    |
| PLTST  | 0761  | 1371 | 1305 1330                               |
| PLUS   | 4600  | 1464 | 1522                                    |
| PL17   | 4130  | 2741 | 2732                                    |
| PL212  | 4135  | 2750 | 2722                                    |
| PL215  | 4134  | 2747 | 2720                                    |
| PMAINS | 5510  | 0422 | 0354 0367 0374 0400                     |
| PM60A  | 0137  | 0441 | 0440                                    |
| POINT  | 0047  | 0350 | 0076 1055 1061 1062 3171 3172 3173      |
| POPUP1 | 4307  | 3167 | 0410 3174                               |
| POPUI  | 0106  | 0410 | 1604 1612 1614 1620 1624 1626 1734      |
| POSAND | 2506  | 1230 | 1223                                    |
| POUTP  | 4132  | 2745 | 2703                                    |
| PPSEUD | 4274  | 3152 | 3144                                    |
| PRINT  | 0107  | 1314 |                                         |
| PRINT  | 0000  | 1140 | 1152 1154 1167 1175 1232 1242 1255 1404 |
| PRSYMB | 2474  | 1213 | 1135                                    |
| PR1    | 6226  | 1373 | 1360                                    |
| PR2    | 6227  | 1374 | 1362                                    |
| PR3    | 6230  | 1375 | 1364                                    |
| PR4    | 6231  | 1376 | 1366                                    |
| PR5    | 6232  | 1377 | 1370                                    |
| PSEUDO | 5410  | 0277 | 3152 0375 0412                          |
| PSINTE | 1575  | 0252 | 0101 0423                               |
| PSRET  | 4201  | 3151 | 3145 3147                               |
| PSTCHK | 0261  | 0575 | 0546                                    |
| PSTS   | 4074  | 2674 | 2672                                    |
| PSUD   | 0334  | 0652 | 0643                                    |
| PSUD01 | 4265  | 3142 | 2637                                    |
| PSUD02 | 4264  | 3141 | 2643                                    |
| PSUD03 | 4263  | 3140 | 2647                                    |
| PSUD04 | 4262  | 3137 | 2703                                    |
| PSUD05 | 4261  | 3136 | 2633                                    |
| PSUD06 | 4260  | 3135 | 2657                                    |
| PSUD07 | 4257  | 3134 | 2653                                    |
| PSWITC | 5511  | 0423 | 0350 0371                               |
| PTB    | 4047  | 2645 | 2627                                    |
| PTBUFF | 3353  | 2032 | 2012                                    |
| PTBUFP | 3352  | 2031 | 2002 2005 2013 2021 2645 3542           |
| PTHREE | 7325  | 0252 | 0756 1164 2156                          |
| PTOCHC | 1577  | 0254 | 0067                                    |
| PT1    | 6312  | 1473 | 1416 1424 1433 1442                     |
| PT2    | 6313  | 1474 | 1415 1426 1432 1444                     |
| PUNONE | 3001  | 1532 | 0422 0213 1533 1567 1574 1603 1607 1612 |
| PUNON1 | 0120  | 0422 | 0114                                    |
| PUSH1  | 0105  | 0407 | 1473 1476 1501 1504 1510 1513 1701      |
| PUSH2  | 2341  | 1052 | 0407 1063                               |
| PUT    | 0710  | 1301 | 0277 1307 1316                          |
| PV8MOV | 5552  | 0501 | 0474                                    |
| PXL1   | 6333  | 1525 | 1515                                    |
| PXL2   | 6334  | 1526 | 1517                                    |
| P1T037 | 3425  | 2127 | 2106                                    |
| P10    | 1173  | 1666 | 1575 1576                               |

| SYMBOL | VALUE | DEF | REFERENCES |
|--------|-------|-----|------------|
|--------|-------|-----|------------|

|        |      |      |                                                                                           |
|--------|------|------|-------------------------------------------------------------------------------------------|
| P100   | 3751 | 2531 | 2344                                                                                      |
| P11    | 0015 | 0315 | 1077 2114                                                                                 |
| P12    | 2241 | 0746 | 0724                                                                                      |
| P177   | 0145 | 0447 | 0770 1742 0122                                                                            |
| P1777  | 0144 | 0446 | 0555 0616 1203 1752 2067 1625 2305 2777                                                   |
| P2     | 3456 | 2164 | 2117 2122                                                                                 |
| P20    | 3100 | 1631 | 1547                                                                                      |
| P200   | 0147 | 0451 | 1013 1026 1737                                                                            |
| P2000  | 0152 | 0454 | 0520 0746 0764                                                                            |
| P212   | 0153 | 0455 | 1241 2261 1265 1505                                                                       |
| P215   | 0154 | 0456 | 1237 2263 1144 1371                                                                       |
| P2270  | 4041 | 2637 | 2620                                                                                      |
| P240   | 0134 | 0436 | 1406 2345 1434                                                                            |
| P260   | 2704 | 1500 | 1450 1470                                                                                 |
| P300   | 0143 | 0445 | 1673 1702                                                                                 |
| P367A  | 3450 | 2156 | 2121 2136                                                                                 |
| P377   | 3077 | 1630 | 1560                                                                                      |
| P3777  | 0131 | 0433 | 0624 0632 1071 1213 1224 1622 2756                                                        |
| P40T57 | 3412 | 2114 | 2111                                                                                      |
| P400   | 0124 | 0426 | 1562 2405                                                                                 |
| P4000  | 0150 | 0452 | 0566 0662 0665 1217 1231 0176 0200 0202 0204 2375                                         |
| P41    | 3460 | 2166 | 2140                                                                                      |
| P43    | 3523 | 2252 | 2260                                                                                      |
| P4400  | 4046 | 2644 | 2626                                                                                      |
| P45    | 2701 | 1475 | 1433                                                                                      |
| P46    | 3461 | 2167 | 2151                                                                                      |
| P4774  | 5023 | 0066 | 0103                                                                                      |
| P4775  | 5021 | 0064 | 0047                                                                                      |
| P4776  | 5022 | 0065 | 0051                                                                                      |
| P5     | 3770 | 2560 | 2553                                                                                      |
| P5000  | 0142 | 0444 | 1054 1561                                                                                 |
| P53    | 2703 | 1477 | 1447                                                                                      |
| P54    | 3455 | 2163 | 2147                                                                                      |
| P57    | 3457 | 2165 | 2125                                                                                      |
| P6000  | 0136 | 0440 | 0521 0564 1154 1206 1240 1764 0242 1010 1015 1104 1107 1614 1741 2312 2371 2411 2431 2771 |
| P6321  | 3316 | 1775 | 1766                                                                                      |
| P7     | 0157 | 0461 | 1670 1714 1467 1244 1246                                                                  |
| P7400  | 0155 | 0457 | 1535 2010 2623                                                                            |
| P76    | 0127 | 0431 | 2272                                                                                      |
| P7600  | 0146 | 0450 | 0773 0777 1012 1027 1721 2053 0230                                                        |
| P77    | 0160 | 0462 | 0312 0322 0605 0623 1672 1701 2003 2025 2341                                              |
| P777   | 0156 | 0460 | 1057 0753                                                                                 |
| RADO   | 4235 | 3100 | 1661 3112                                                                                 |
| RADDRT | 4246 | 3111 | 3123                                                                                      |
| RAND   | 3713 | 2461 | 1663 2464                                                                                 |
| READ   | 7774 | 3612 | 1134 2051 2603 3166 0202 0477 0576 3526                                                   |
| REJECT | 2470 | 1206 | 1145 1171                                                                                 |
| REPUN1 | 1171 | 1664 | 1635                                                                                      |
| RESET  | 1551 | 0226 | 0404 0232                                                                                 |
| RESETL | 3663 | 2427 | 0365 2433                                                                                 |
| RESET1 | 0063 | 0365 | 0523 1772                                                                                 |
| RESTOR | 1751 | 0462 | 0577 1664 0451 0453 0455 0463 0465 0467 0471                                              |
| RESTO1 | 0263 | 0577 | 0543                                                                                      |
| RETURN | 4051 | 2647 | 0420                                                                                      |
| RET1   | 0116 | 0420 | 2473                                                                                      |
| R10R   | 1600 | 0263 | 1660 0272                                                                                 |
| RSUB   | 4211 | 3052 | 1662 3065                                                                                 |
| RSUBRT | 4223 | 3064 | 3077                                                                                      |
| RTHFT  | 3336 | 2015 | 2007                                                                                      |
| RWRITE | 4163 | 3002 | 2663                                                                                      |
| R1     | 6233 | 1400 | 1357                                                                                      |
| R2     | 6235 | 1402 | 1361                                                                                      |

| SYMBOL | VALUE | DEF  | REFERENCES                                                                                                                                                                                                           |
|--------|-------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R3     | 6237  | 1404 | 1363                                                                                                                                                                                                                 |
| R4     | 6241  | 1406 | 1365                                                                                                                                                                                                                 |
| R5     | 6243  | 1410 | 1367                                                                                                                                                                                                                 |
| SADR   | 0023  | 0324 | 0613 0641 1163 1165 1202 1204 1207 1210 1211 1212 1214 1220 1221 1222 1223 1225 1232 1233 1235 1236 0510 0511<br>0513 0514 0516 0521 0524 1141 1153 1155 1163 1165 1175 1204 1216 2370 2373 2374 2401 2402 2414 2415 |
| SADROV | 2033  | 0535 | 0523                                                                                                                                                                                                                 |
| SAMES  | 3073  | 1624 | 1620                                                                                                                                                                                                                 |
| SAVE   | 1736  | 0447 | 0602 0456                                                                                                                                                                                                            |
| SAVE1  | 0265  | 0602 | 0605                                                                                                                                                                                                                 |
| SCH1   | 0433  | 0754 | 0743                                                                                                                                                                                                                 |
| SCLEAR | 0132  | 0434 | 0342 3253 3410                                                                                                                                                                                                       |
| SCOLON | 0200  | 1455 | 1525                                                                                                                                                                                                                 |
| SCRATC | 0025  | 3574 | 1323 3245                                                                                                                                                                                                            |
| SCURR  | 0026  | 0327 | 0606 0610 0646 0706 0707 0450 1122 1131 1136 1173 1210                                                                                                                                                               |
| SDEC01 | 0112  | 0414 | 0611 1645 0464 0503 1137 1174                                                                                                                                                                                        |
| SDEC08 | 0600  | 1151 | 0414 1172 1405 1410                                                                                                                                                                                                  |
| SEARCH | 0266  | 0604 | 0676 0701 0754 1746                                                                                                                                                                                                  |
| SEARC1 | 0272  | 0610 | 0713                                                                                                                                                                                                                 |
| SEAR1  | 3270  | 1746 | 1727                                                                                                                                                                                                                 |
| SEMF0G | 2573  | 1355 | 1346                                                                                                                                                                                                                 |
| SEMITX | 2561  | 1340 | 1345 1351 2144                                                                                                                                                                                                       |
| SEMITZ | 3440  | 2144 | 2112                                                                                                                                                                                                                 |
| SEMPYF | 2574  | 1356 | 1347                                                                                                                                                                                                                 |
| SEND   | 0027  | 0330 | 0711 0056 0502 0532 1133 0467 0511                                                                                                                                                                                   |
| SETINU | 0177  | 0501 | 1556                                                                                                                                                                                                                 |
| SETORG | 1520  | 0175 | 0061 0211 0220                                                                                                                                                                                                       |
| SETUSE | 3653  | 2417 | 2365 2424 2641                                                                                                                                                                                                       |
| SGN    | 0034  | 0335 | 1503 1506 1534 1547 1570 1615 2765                                                                                                                                                                                   |
| SIT    | 4315  | 3176 | 3046 3160                                                                                                                                                                                                            |
| SKIP   | 0075  | 0377 | 0670 1252 1377 1576 1605 1611 1670 1705 2356 3431                                                                                                                                                                    |
| SKIPAS | 5435  | 0334 | 0312                                                                                                                                                                                                                 |
| SKIP2  | 0764  | 1404 | 0377 1412                                                                                                                                                                                                            |
| SL777  | 4304  | 3163 | 3043 3155                                                                                                                                                                                                            |
| SM370  | 4305  | 3164 | 3042 3154                                                                                                                                                                                                            |
| SNUM   | 0171  | 0473 | 0545 1152 1153 1157 2546                                                                                                                                                                                             |
| SOMEW  | 4227  | 3071 | 3053 3061                                                                                                                                                                                                            |
| SOPS   | 0062  | 0364 | 1250 1666 2202                                                                                                                                                                                                       |
| SPACE  | 5600  | 1466 | 1521 1540 1573                                                                                                                                                                                                       |
| SPACK  | 3744  | 2524 | 0675 2535                                                                                                                                                                                                            |
| SPCUNT | 0173  | 0475 | 0074 0344 2360 2520 2530 2533                                                                                                                                                                                        |
| SPECAL | 0416  | 0737 | 0724 0732                                                                                                                                                                                                            |
| SPT060 | 1705  | 0403 | 0343                                                                                                                                                                                                                 |
| SQERR  | 1702  | 0400 | 0354 0355                                                                                                                                                                                                            |
| SQF0RT | 1701  | 0377 | 0367                                                                                                                                                                                                                 |
| SRD1   | 0115  | 0417 | 2421                                                                                                                                                                                                                 |
| SREAD1 | 4275  | 3153 | 0417 3161                                                                                                                                                                                                            |
| SRITIC | 0113  | 0415 | 2364 2650                                                                                                                                                                                                            |
| STAB   | 0675  | 1246 | 1162                                                                                                                                                                                                                 |
| STDUMP | 1667  | 0360 | 0304 0320                                                                                                                                                                                                            |
| STEXD  | 1175  | 1670 | 1245 0535                                                                                                                                                                                                            |
| STGET  | 1641  | 0327 | 0300 0303 0317 0333 0346 0353                                                                                                                                                                                        |
| STINTO | 1624  | 0311 | 0302                                                                                                                                                                                                                 |
| STLOOP | 1616  | 0303 | 0325                                                                                                                                                                                                                 |
| STNODU | 1673  | 0364 | 0301 0371                                                                                                                                                                                                            |
| STNOT  | 1661  | 0347 | 0337                                                                                                                                                                                                                 |
| STOCHK | 3757  | 2545 | 0575 2552                                                                                                                                                                                                            |
| STRING | 1610  | 0275 | 2723                                                                                                                                                                                                                 |
| SUBCON | 4232  | 3074 | 3055                                                                                                                                                                                                                 |
| SWITCH | 0170  | 0472 | 0610 0625 0672                                                                                                                                                                                                       |
| SWOT   | 1556  | 0233 | 0142 0156 0677                                                                                                                                                                                                       |
| SWOTA  | 2172  | 0677 | 0641                                                                                                                                                                                                                 |
| SWOUT  | 4317  | 3200 | 3044 3156                                                                                                                                                                                                            |

| SYMBOL | VALUE | DEF  | REFERENCES                                                                                                    |
|--------|-------|------|---------------------------------------------------------------------------------------------------------------|
| SWRC   | 4201  | 3041 | 0415 3047 3151                                                                                                |
| SW1    | 6035  | 1202 | 1151                                                                                                          |
| SW2    | 6036  | 1203 | 1166                                                                                                          |
| SY     | 0174  | 0476 | 0575 0646                                                                                                     |
| SYB    | 0114  | 0416 | 0647                                                                                                          |
| SYCRAP | 3373  | 2060 | 1771                                                                                                          |
| SYMB   | 4000  | 1457 | 1520 1542 1653                                                                                                |
| SYMBLP | 2402  | 1120 | 1224                                                                                                          |
| SYMBU  | 1503  | 0160 | 0416 0144 0145 0146 0154                                                                                      |
| SYMBUF | 3102  | 1633 | 0476 0160                                                                                                     |
| SYMFIX | 6335  | 1527 | 1354                                                                                                          |
| SYML0D | 5553  | 0502 | 0307                                                                                                          |
| SYMLST | 0071  | 0373 | 2441                                                                                                          |
| SYMOVR | 6336  | 1530 | 1355                                                                                                          |
| SYMSAV | 5533  | 0461 | 0311                                                                                                          |
| SYREAD | 3372  | 2051 | 1770                                                                                                          |
| SYSIN  | 3317  | 1776 | 0371 2016 2026                                                                                                |
| TAABA  | 0362  | 0700 | 0715                                                                                                          |
| TABA   | 3441  | 2147 | 2116                                                                                                          |
| TABLE  | 3271  | 1750 | 1654                                                                                                          |
| TACK   | 1434  | 0110 | 0171                                                                                                          |
| TATA   | 3532  | 2261 | 2266                                                                                                          |
| TBCONT | 0175  | 0477 | 0574 0660 0663 0671 1375 1407 1452 1460 1472                                                                  |
| TEMP   | 0007  | 0305 | 0663 0667 1060 1061 1063 1070 1740 1743 0606 0611 0624 0626 0655 1033 1036 2004 2015 2525 2526 2534 3214 3216 |
| TEMPA  | 2510  | 1264 |                                                                                                               |
| TEMP1  | 6034  | 1201 | 1155 1156 1162 1254                                                                                           |
| TEM1   | 0161  | 0463 | 0620 0742 2032 0507                                                                                           |
| TEM1A  | 0040  | 0341 | 1124 1126 1130 1147 1157 1167 1177 1410 1412 1414                                                             |
| TEM2   | 0162  | 0464 | 0626 0741 2034 0512                                                                                           |
| TEM3   | 0163  | 0465 | 0634 0740 2036 0515                                                                                           |
| TERMC  | 0000  | 0266 | 0350 0375 0550                                                                                                |
| TEST   | 6252  | 1421 | 1425 1427                                                                                                     |
| TESTC2 | 6267  | 1437 | 1443 1445                                                                                                     |
| THOU   | 3570  | 2317 | 2763                                                                                                          |
| TIC    | 2274  | 1005 | 0754 0762 1000                                                                                                |
| TOP    | 0367  | 0705 | 0637                                                                                                          |
| TOUSE  | 4043  | 2641 | 2622                                                                                                          |
| TWOCH  | 2144  | 0651 | 0634                                                                                                          |
| TWOCHR | 2073  | 0600 | 0652                                                                                                          |
| TYBASE | 2400  | 1546 | 1400 1403 1406 1410                                                                                           |
| TYCAR  | 2510  | 1236 | 0402 1245 1264                                                                                                |
| TYCARI | 0100  | 0402 | 0310 0363 0635 1117 1403 1664 1712 2172 2354 2507 2612 1046 3470                                              |
| TYPADD | 0622  | 1173 | 0413 1175 1243 1244 1302 1310                                                                                 |
| TYPAD1 | 0111  | 0413 | 0570 1652 0527                                                                                                |
| TYPE   | 0021  | 0322 | 0745 0752 0763 1006 1200 1205 1215 1226 1237 1241 0454 0470 1007 1103 1740 2412                               |
| TYPEA  | 0014  | 0314 | 1176 1177 1762 0452 0466 1013 2304 2310 2372 2377 2400 2406 2407 2410                                         |
| TYPO   | 2527  | 1255 | 1373 1262 0133 0135 0141 0200 1373 1374 1375 1376 1377 1400 1403 1406 1410                                    |
| UBITS  | 4321  | 3211 | 0501 3217                                                                                                     |
| UN     | 1105  | 1576 | 2767                                                                                                          |
| UNDERR | 2317  | 1030 | 1760 1042 1716                                                                                                |
| UNITAS | 4777  | 0244 |                                                                                                               |
| UNREF  | 3240  | 1716 | 1660 1671 1675                                                                                                |
| UNTRAN | 2647  | 1442 | 1436 1440 1453                                                                                                |
| UPLN   | 3607  | 2353 | 2361 2522                                                                                                     |
| USEDNO | 4337  | 3277 | 2642 1102                                                                                                     |
| USES   | 4044  | 2642 | 2651 2655 2660                                                                                                |
| USYMB  | 2000  | 1461 | 1517 1543 1734                                                                                                |
| VADR   | 0022  | 0323 | 0557 0655 1166 1650 0506 1017                                                                                 |
| VAL    | 0024  | 0325 | 0671 0750 0751 0766 1530 1567 2016 2017 2022 2024 2026 0241 0250 0264 0267 0277 0331 0505 1203 1401 1732 2313 |
| VALK   | 0025  | 0326 | 0653 0670 0247                                                                                                |
| VASSNO | 5603  | 0564 | 0562                                                                                                          |

| SYMBOL | VALUE | DEF  | REFERENCES                                             |
|--------|-------|------|--------------------------------------------------------|
| VCLEAN | 5602  | 0562 | 0571                                                   |
| VERSIO | 0017  | 0235 | 0251                                                   |
| VIN    | 5573  | 0537 | 0473 0512                                              |
| VINIT  | 5565  | 0514 | 0513                                                   |
| VL6777 | 5551  | 0500 | 0470 0510                                              |
| VOUT   | 5572  | 0536 | 0466 0505                                              |
| VPRSRV | 0773  | 1427 | 1432 0536                                              |
| VPRSRW | 0777  | 1437 | 1430                                                   |
| VREAD  | 5610  | 0576 | 0567 0734                                              |
| VRSTR  | 0572  | 1124 | 1127 0537                                              |
| VRSTRR | 0576  | 1134 | 1125                                                   |
| VSAVE  | 4331  | 3243 | 1126 1431                                              |
| VTAB   | 5611  | 0602 | 0560 0570                                              |
| VVINIT | 5604  | 0567 | 0514                                                   |
| VWRITE | 5607  | 0575 | 0557 0720                                              |
| V8MOVE | 5600  | 0557 | 0501                                                   |
| WDH    | 4042  | 2640 | 2625                                                   |
| WDHALF | 3351  | 2030 | 2000 2020 2640 3541                                    |
| WHERE  | 4252  | 3116 | 3101 3107                                              |
| WREAD  | 4306  | 3166 | 3157                                                   |
| WRITE  | 7775  | 3613 | 1437 3002 3051 0476 0575                               |
| WWRITE | 4210  | 3051 | 3045                                                   |
| XBBL0C | 5154  | 3270 | 3067                                                   |
| XBUFIN | 5353  | 3543 | 3473                                                   |
| XCER   | 5134  | 3247 | 3100 3112 3114 3115 3120 3131 3140 3510                |
| XCHAIN | 2536  | 1274 | 2623                                                   |
| XCON   | 5165  | 3301 | 3123                                                   |
| XDONE1 | 5044  | 3133 | 3121                                                   |
| XGET   | 5122  | 3227 | 3077 3113 3117 3136 3170 3173 3205 3233 3234 3237 3240 |
| XGETIN | 5151  | 3265 | 3230                                                   |
| XHALF  | 5155  | 3271 | 3171 3203 3216                                         |
| XHT    | 5334  | 3524 | 3423                                                   |
| XINIT  | 2551  | 1313 | 1277                                                   |
| XINT01 | 5034  | 3122 | 3116                                                   |
| XL00P1 | 5011  | 3077 | 3105 3111                                              |
| XL00P2 | 5031  | 3117 | 3130                                                   |
| XL17   | 5145  | 3261 |                                                        |
| XL4400 | 5347  | 3537 | 3417                                                   |
| XL6    | 5144  | 3260 | 3247                                                   |
| XL6777 | 5337  | 3527 | 3351                                                   |
| XL7400 | 5346  | 3536 | 3414                                                   |
| XL77   | 5153  | 3267 | 3070 3150 3167                                         |
| XL7770 | 5342  | 3532 | 3357                                                   |
| XM100  | 5340  | 3530 | 3353                                                   |
| XM13   | 2575  | 1357 | 1341                                                   |
| XM26   | 5335  | 3525 | 3430                                                   |
| XM4    | 5344  | 3534 | 3364                                                   |
| XM40   | 5146  | 3262 | 3107                                                   |
| XM43   | 5152  | 3266 | 3231                                                   |
| XM47   | 5150  | 3264 | 3103                                                   |
| XM9    | 5147  | 3263 | 3217                                                   |
| XN     | 5143  | 3257 | 3137 3146 3152                                         |
| XNM1   | 5310  | 3455 | 3300                                                   |
| XOK1   | 5025  | 3113 | 3101                                                   |
| XP     | 5157  | 3273 | 3221 3533                                              |
| XPP    | 2550  | 1312 | 1300                                                   |
| XPSEU  | 2555  | 1321 | 1275 1302                                              |
| XPUT   | 5064  | 3161 | 3074 3127 3134 3165 3166 3200 3207                     |
| XP46   | 2576  | 1360 | 1344 1350                                              |
| XREAD  | 5336  | 3526 | 3347 3424                                              |
| XRESET | 5112  | 3214 | 3072 3076 3223                                         |
| XRIGHT | 5104  | 3202 | 3172                                                   |

| SYMBOL | VALUE | DEF  | REFERENCES                         |
|--------|-------|------|------------------------------------|
| XT     | 5156  | 3272 | 3177 3204                          |
| XTH    | 5333  | 3523 | 3422                               |
| XTHISB | 5166  | 3307 | 3071 3151 3350 3523                |
| XXFH   | 5321  | 3470 | 3427 3434                          |
| XXGET  | 5354  | 3544 | 3340                               |
| XXL    | 5345  | 3535 | 3413                               |
| XXLOOP | 5214  | 3355 | 3506                               |
| XXLP   | 5225  | 3366 | 3376                               |
| XXMESS | 5277  | 3442 | 3432                               |
| XXM43  | 5341  | 3531 | 3343                               |
| XXN    | 5202  | 3342 | 3257 3354 3365 3375 3401 3412 3505 |
| XXN1   | 5160  | 3274 |                                    |
| XXN2   | 5161  | 3275 |                                    |
| XXN3   | 5162  | 3276 |                                    |
| XXN4   | 5163  | 3277 |                                    |
| XXP    | 5343  | 3533 | 3362                               |
| XXRET  | 5172  | 3322 | 3475                               |
| XXT    | 5271  | 3432 | 3436 3437                          |
| XXXP   | 5164  | 3300 | 3124 3125                          |
| XXXR   | 5200  | 3340 | 3345                               |
| XXXX   | 5000  | 3066 | 1312 3322                          |
| XXY    | 5327  | 3505 | 3374 3405                          |
| XX1    | 5350  | 3540 | 3415                               |
| XX2    | 5351  | 3541 | 3416                               |
| XX3    | 5352  | 3542 | 3420                               |
| YP1    | 5566  | 0520 | 0472 0507                          |
| YREAD  | 5550  | 0477 | 0506                               |
| YWRITE | 5547  | 0476 | 0471                               |
| ZASSIF | 5425  | 0324 | 0305                               |